Foundations of Econometrics

Estimands, identification, and inference: what a regression can tell you, and how sure to be

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Abstract

A regression measures how two variables move together. Reading a coefficient as a causal effect requires an assumption that the data cannot test. This chapter identifies that assumption, shows how to argue for it, and shows how much confidence the result then deserves.

Every empirical study answers three questions in order. The first asks what quantity is the target. The second asks whether the data can identify it. The third asks how precisely the data estimate it. Errors pass downward only. Precise estimation does not help when the data cannot identify the quantity. Most disputes concern the first two questions, but most effort goes into the third. Regression residuals are uncorrelated with the regressors by construction. No diagnostic computed from the fit can therefore detect the bias that matters most, and the case for a causal reading must come from outside the regression. A wrong identifying assumption gives a wrong answer. A wrong assumption about dependence gives a right answer with a wrong standard error. Robust and clustered standard errors fix only the second problem.

Precision depends on the number of independent observations, not the number of rows. A persistent series of 1,000 observations can carry the information of about 53 independent ones. Forty years of overlapping 12-month returns carry about 40. With 50 clusters of 100 and an intra-cluster correlation of 0.05, the naive standard error is 2.4 times too small. An underpowered study misleads, because the only estimates that reach significance are overestimates. Controls can add bias as easily as remove it. Conditioning on a mediator removes part of the effect, and conditioning on a collider can reverse its sign. Survivorship and look-ahead filters act as colliders, and financial data contain many of them. Regressing one random walk on an unrelated one usually yields a "significant" fit, and more data makes the problem worse. Granger causality measures predictive timing, not causation.

Machine learning predicts well, but it is not a research design. Its feature importances describe the model and not the world. Standard cross-validation leaks information on financial data. The chapter concludes that most failures come from data handling: joins, units, timing, corporate actions and delistings. Published return predictors lose about a quarter of their effect out of sample and more than half after publication. The recommended defences are procedural. State the target before estimating, count independent observations, log every specification tried, and report the range of estimates across all reasonable choices.

What this is. This chapter is a first-principles tutorial on econometrics. Econometrics is the set of methods for learning about economic and financial relationships from data that nobody designed as an experiment. The chapter covers three areas. The first is the foundations: what a regression estimates and what probability theory contributes. The second is the methods: the estimator family, inference, the research designs that turn correlations into causal claims, and time-series econometrics. The third is practice: how empirical work goes wrong and the working habits that prevent it. The emphasis throughout is on intuition and use. Each formula comes with an explanation of its meaning before it and a number after it. Each method comes with an account of how it works and how it breaks.

Who it is for. The chapter is for a reader who is mathematically comfortable and technically strong, who has no training as an econometrician, and who wants to *do* something with the material. The reader might want to run a regression and know what it means. The reader might want to evaluate an empirical claim. The reader might want to build a forecasting or signal-research pipeline that does not fool its builder, or to read the empirical finance literature critically. The chapter assumes calculus, linear algebra, and basic probability, and nothing more. Appendix A explains from first principles any concept that the text uses before it defines the concept fully.

How to read it. The chapter has six parts.

- **Part I (§1–§3): what econometrics is for.** §1 states the idea that organises everything else, and no reader should skip it. §2 is the minimum probability needed. Readers who know the material can skim it, but should read §2.2 and §2.5. §3 builds regression from the conditional expectation function, and its §3.4 and §3.5 are the two results the rest of the chapter relies on most.
- **Part II (§4–§5): estimation and inference.** §4 covers standard errors, tests, and the ways they mislead. For someone who already runs regressions, it is the most practically important section. §5 shows that OLS, instrumental variables, GMM, and maximum likelihood are one estimator with different inputs.

- **Part III (§6–§7): identification.** §6 explains endogeneity, the reason a regression coefficient differs from a causal effect. §7 surveys the research designs that address it: experiments, instruments, panels, difference-in-differences, regression discontinuity, synthetic control, and structural models.
- **Part IV (§8): time series.** §8 covers what changes when observations have a time order, from autoregressions to unit roots, cointegration, VARs, volatility models, breaks, and forecast evaluation.
- **Part V (§9–§11): practice.** §9 covers the econometrics specific to finance: predictive regressions, event studies, asset-pricing tests, panel standard errors, and the replication debate. §10 covers prediction, model selection, and machine learning. §11 covers workflow, reporting, and a catalogue of failure modes.
- **Part VI (§12–§15): synthesis.** §12 collects the taxonomy and the equivalences. §13 gives the history. §14 gives the synthesis and a decision tree. §15 lists the references.

Readers short of time should read four parts. **§1.5–§1.6** cover the master form and the sandwich. **§3.4–§3.5** explain what "controlling for" does and what leaving something out costs. **§4.4** shows why 5,000 observations can carry the information of 840. **§6.5** covers the controls that create bias instead of removing it. Readers who work on return prediction should add §2.5, §8.3, §9.2, and §10.6.

Objectives. After this chapter, you should be able to:

- ask three questions of any empirical claim and answer them in order: what the estimand is, whether it is identified, and how precisely the data estimate it.
- write an estimator as a moment condition, and derive its standard error from the sandwich formula.
- explain what "controlling for" a variable does, and compute the bias from omitting one.
- choose a standard error that matches the dependence in the data, and count the independent observations behind it.
- pick a research design for a causal question, and name the way that design typically fails.
- recognise the failures specific to time series and financial data: spurious regression, persistent predictors, overlapping horizons, and multiple testing.

Relationship to the other notes. Simple and Log Returns §6 treats the time-series econometrics of returns from the side of the return convention. It covers unit roots, volatility models, cointegration, and long-horizon predictive regressions. This chapter supplies the general methods behind it. Market Regimes and Hidden Markov Models goes deep on structural breaks, regime-switching models, and time-series validation. This chapter outlines those topics only (§8.7, §10.6). Portfolio Construction and the Covariance Matrix covers covariance estimation, shrinkage, and the statistics of Sharpe ratios. The momentum and trend-following notes apply the tools of this chapter to specific signals. Each note stands alone.

A warning about scope. Nothing here is investment advice. The numerical illustrations are simulations built to isolate one mechanism at a time, not evidence about any market. Where a number comes from a simulation run for this chapter rather than from a published study, the text says so.

Epistemic tags. The chapters in this collection tag claims by status where the status changes what a reader should do:

- **[Fact]** — a mathematical result, or an empirical finding that others have replicated or independently reanalysed, with broad agreement among those who have examined it.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner or professional convention. The evidence may be informal or absent. A [Practice] tag does not mean the claim has been debunked.

A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations, or arithmetic, true by construction rather than by evidence. A fifth label, **[Simulated]**, marks results from simulations run for this chapter. It records provenance and not epistemic status. The generating code is in `figures/em_*.py`, committed alongside this chapter. Each script prints the numbers quoted in the text. Readers can change the parameters and rerun the scripts.

Notation. The table lists the symbols that recur in the chapter, with the section that defines or first uses each. The text defines symbols used in a single derivation or appendix entry where they appear.

Symbol	Meaning	Defined in
$i = 1, \dots, n; t = 1, \dots, T$	Units in a cross-section; dates in a time series	§1.5, §2.5
$N; T$	In a panel, the number of units and of periods	§7.4

Symbol	Meaning	Defined in
G; g; n_g, $\bar{n}_g$	Number of **clusters**; cluster index; observations in cluster g, and the average cluster size	§2.5, §4.4
Y_i	Outcome	§1.5
$X_i \in \mathbb{R}^k$	Column vector of k regressors, usually including a constant, so $X_i'\beta$ is a scalar; a prime denotes transpose	§1.5
$\mathbf{X}$; $\mathbf{Y}$	The $n \times k$ matrix whose rows are X_i'; the n-vector of outcomes	§3.3
$Z_i \in \mathbb{R}^\ell$	Vector of $\ell \geq k$ **instruments**	§1.5, §5.2
C_i	A control variable	§3.5
θ; θ_0	Generic parameter vector; its true value	§1.5
$\hat{\cdot}$; $\bar{\cdot}$	Estimate ($\hat{\beta}$); sample average ($\bar{Y}$)	—
β; β_s	Linear projection coefficient; structural (causal) coefficient	§3.2
e_i; $\hat{e}_i$	**Projection error**, uncorrelated with X_i by construction; regression residual	§3.2, §3.3
u_i	**Structural error**: the unobserved part of an equation meant to describe how Y_i is generated	§3.2
ε_t	White-noise innovation in a time-series model	§8.2
$m(x)$; ϵ_i	**Conditional expectation function** (CEF) $\mathbb{E}[Y_i \mid X_i = x]$; CEF error $Y_i - m(X_i)$, mean-independent of X_i by construction	§3.1
D_i; $Y_i(1)$, $Y_i(0)$; τ_i	Binary treatment; **potential outcomes** with and without treatment; unit-level effect $Y_i(1) - Y_i(0)$	§6.1
$\psi_i(\theta)$	Moment function: $\mathbb{E}[\psi_i(\theta_0)] = 0$ defines the estimator	§1.5
Q; Ω	**Bread**, $\mathbb{E}[\partial\psi_i(\theta_0)/\partial\theta']$; **meat**, the long-run variance of $n^{-1/2}\sum_i \psi_i(\theta_0)$	§1.6
Q_{XX}	$\mathbb{E}[X_i X_i']$	§1.6
W	GMM weighting matrix	§5.3
$\mathcal{F}_t$	Information available at time t	§1.4
α; α_i; λ_t	Intercept; unit fixed effect; time fixed effect	§3.5, §7.4
γ; δ	In the omitted-variable formula: effect of the omitted variable; its projection coefficient on the included regressor	§3.5
π; F	First-stage coefficient in IV; first-stage F-statistic	§7.3
κ	Regularisation penalty	§5.6, §10.4
ϕ; ρ_j	Autoregressive coefficient; autocorrelation at lag j	§2.3, §2.5
h; L	Forecast or return horizon; HAC bandwidth (number of lags)	§2.5, §4.5
ρ_u; ρ_x	**Intra-cluster** correlations of the error and of the regressor	§4.4
n_{eff}	Effective sample size	§2.5
σ^2; σ_t^2	Error variance; conditional variance	§2.4, §8.6
ω, a, b	GARCH parameters	§8.6
R^2; R^2_{OS}	In-sample fit; out-of-sample R^2 against the historical mean	§3.8

Symbol	Meaning	Defined in
Φ; φ	Standard normal distribution function; standard normal density	§12.2, A.11
$\mathbb{1}\{\cdot\}$	Indicator of the condition in braces	§1.5
$\xrightarrow{p}$; $\xrightarrow{d}$; plim	Convergence in probability; convergence in distribution; probability limit	§2.3, §3.5
Avar; se$(\cdot)$	Asymptotic variance; standard error	§1.6, §4.1
SR	Sharpe ratio	§2.6

Keep two error terms strictly apart. The distinction between them carries much of §1 and §3. The projection error e_i is the part of Y_i that remains after the best linear prediction from X_i. It is uncorrelated with X_i *by construction* (§3.2). Whether the structural error u_i is uncorrelated with X_i is an *assumption*. The CEF error ϵ_i has a stronger property than e_i. It is mean-independent of X_i and not merely uncorrelated with it, per the hierarchy of §2.2.

Three notational conventions need explanation. In §8.5 only, bold lowercase letters are vectors of several time series at one date, and plain capitals are coefficient matrices. This is standard for vector autoregressions. §8.6 writes the GARCH parameters as ω, a, b instead of the literature's ω, α, β. This keeps the regression meanings of α and β. The chapter writes significance levels as percentages, with no symbol.

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Part I — What econometrics is for

1. What econometrics is

1.1 The wrong intuition

Most readers arrive with a view like this one:

> *Econometrics is statistics applied to economic data. You regress an outcome on the variables you think matter, check which coefficients are significant, and read each significant coefficient as the effect of its variable.*

Each clause is defensible on its own. The sentence as a whole is wrong in three ways. Each way corresponds to a part of this chapter.

A regression coefficient is not an effect. A coefficient summarises how the average outcome differs between units whose regressor differs, after a linear adjustment for the other regressors. It becomes an *effect* only under an assumption about *why* the regressor differs across units. An effect is what would happen if someone changed the regressor. Consider firms that buy back their own shares. A regression of subsequent returns on a buyback indicator compares firms that chose to repurchase with firms that did not. Managers tend to repurchase when they believe their stock is cheap. The coefficient therefore mixes any effect of the buyback itself with the private information that prompted it. Both components are economically interesting, and they call for opposite conclusions. Part III separates them.

Significance is neither truth nor importance. A t-statistic is an estimate divided by its standard error. The analyst computes the standard error under an assumed model of how the sample could have come out differently. If that model is wrong, the t-statistic is off by *multiples*, not by decimals (§4). The model goes wrong in common ways. Analysts ignore that observations share shocks, ignore that returns overlap, or ignore that they tried twenty specifications before this one. Also, a precise estimate can be of an economically trivial effect, and an important effect can have an imprecise estimate.

The hard part is the argument, not the fitting. A regression takes microseconds to compute. The intellectual content of econometrics is the argument that connects the computed number to the question the analyst asks. That argument concerns the process that *generated* the data, not the data themselves. So two competent econometricians can run the same regression on the same data and disagree completely about what it means. They agree about the arithmetic and disagree about the data-generating process.

The field's own evidence against the naive view came from a clean test. In the 1970s the National Supported Work Demonstration randomly assigned disadvantaged workers to a job-training programme. The experiment therefore revealed the programme's true effect on earnings. LaLonde (1986) set aside the experimental control group and substituted comparison samples drawn from standard household surveys. LaLonde then applied the non-experimental estimators that economists used at the time to evaluate training programmes. [Fact] The resulting estimates varied widely around the experimental benchmark. Many were far off. Some had the wrong sign. The specification tests of the day did not reliably separate the good estimates from the bad ones. That paper is a large part of why the field now discusses research design before it discusses estimators (§13).

A better definition:

> **Econometrics is the discipline of stating, and defending, the conditions under which a number computed from data answers a question about the world. The computation is easy, and it is almost always a valid answer to *some* question. The hard step is making it an answer to the question actually asked.**

1.2 The problem: data that nobody randomised

Almost all economic and financial data are **observational**. They record the outcomes of decisions made by people and firms pursuing their own goals. A physicist who wants to know how a spring responds to force can apply the force. An economist who wants to know how earnings respond to schooling cannot assign schooling. A financial economist who wants to know how prices respond to index inclusion cannot assign stocks to indices.

This fact makes econometrics a discipline and not a branch of statistics. **Agents who optimise generated the variation in the data, using information the econometrician does not see.** People choose schooling partly on ability, which also affects earnings. Firms choose leverage partly on the stability of their cash flows, which also affects their risk. Investors buy funds after good performance, and the resulting inflows affect subsequent performance. Central banks raise rates when they expect inflation, so a naive regression can make rate rises appear to cause the inflation they were responding to.

In 1944 Trygve Haavelmo gave the field its framework. He treated observed economic data as a single draw from a joint probability distribution. A system of behavioural relationships generates that distribution. Examples are demand curves,

decision rules, and pricing equations. The analyst wants to learn the relationships but can see only the distribution. **Identification** is the question of whether the relationships can be recovered from the distribution. The analyst must answer it before any estimation starts.

Three consequences recur throughout the chapter:

1. **Correlation mixes effects with selection.** When a regressor is a choice, it is correlated with the reasons for the choice, and those reasons usually affect the outcome too (§6).
2. **Jointly determined variables cannot be regressed on each other and interpreted.** Supply and demand set price and quantity together, so a scatter of price against quantity shows neither the supply curve nor the demand curve (Working, 1927). The same holds for returns and order flow, volume and volatility, and fund flows and fund performance (§6.4).
3. **Relationships estimated under one regime can break when the regime changes.** Agents re-optimise when policy or market structure changes, so a relationship that was stable in the past need not survive an intervention (Lucas, 1976). This is why models can work in backtests and fail in deployment (§7.8, §8.7).

Much of finance appears to escape these problems. A trader who wants tomorrow's expected return asks a *prediction* question, not a causal one, and prediction does not need identification (§1.4). Two facts keep finance within the scope of the problem. First, a great deal of finance *is* causal. Examples are whether index inclusion moves prices, whether buybacks create value, and whether a regulation changes liquidity. Second, prediction has its own version of the problem. A predictive relationship is useful only if it holds in data the model has not seen. Re-optimisation and selection, described above, make such relationships unstable. The inference problems of §4 apply everywhere in full. They include dependent data, overlapping horizons, and many tested hypotheses.

1.3 Three questions, in order

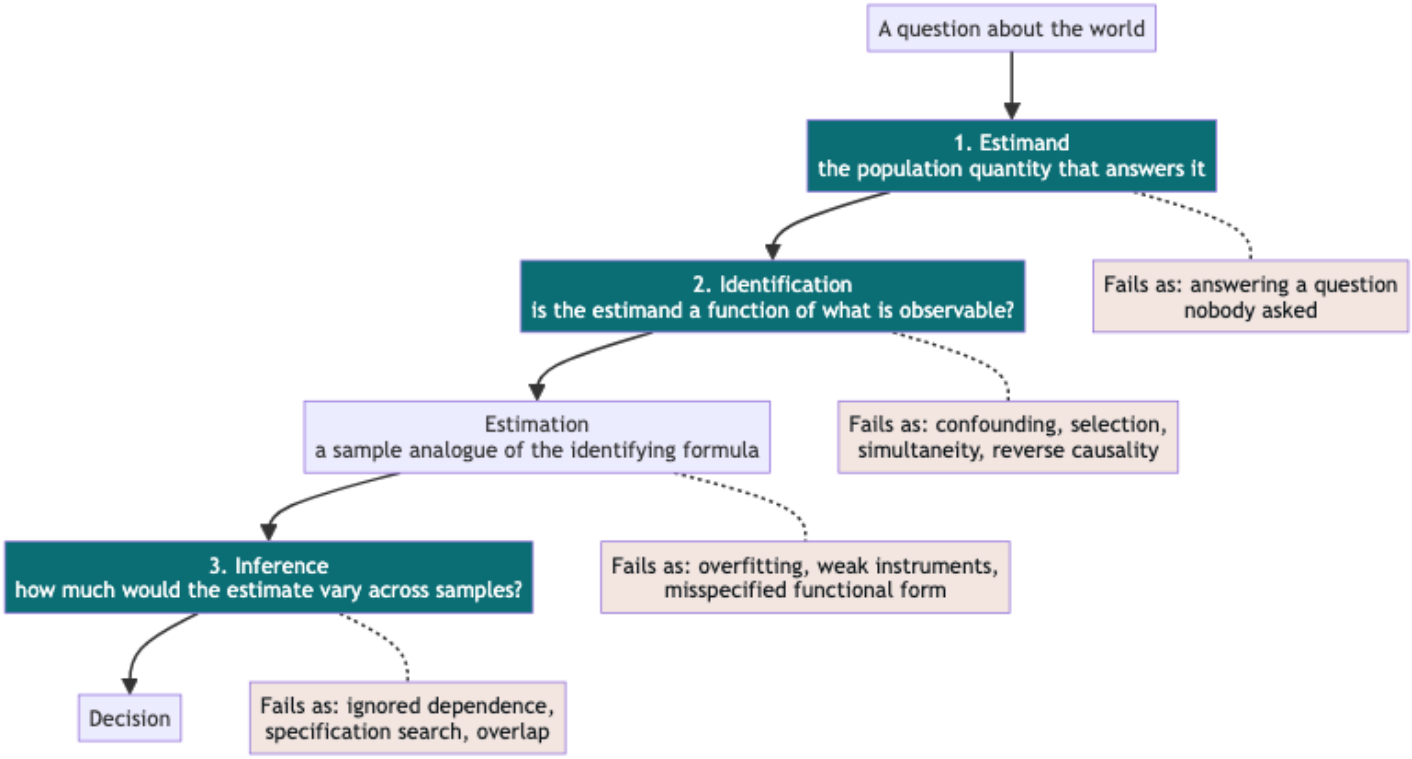
Every empirical claim answers three questions. This holds whether the claim is a published paper, a backtest, or a regression in a notebook. The analyst must answer the questions in order.

1. The estimand. The first question is what the estimand is. The estimand is the precise population quantity the analyst wants to learn. "The effect of index inclusion on prices" is not an estimand. This is an estimand: the average change in a stock's price over the month after it joins the S&P 500. The comparison is to what its price would have been without the addition. The average runs over stocks of the kind that get added. A thought experiment defines the estimand, not the data. The data never say which question to ask.

2. Identification. The second question is whether one could recover the estimand from the population distribution of the observable variables, that is, from an infinite amount of data. Identification is a property of the assumptions and the data-generating process, not of the sample. If the data cannot identify the estimand, more data only give a more precise estimate of the wrong quantity. The economics enters here. Every identifying assumption is a claim about behaviour. One example is "the lottery number affected earnings only through military service." Another is "firms just above and just below the index cutoff are otherwise comparable."

3. Precision. The third question is how much the estimate would vary across the other samples that the data-generating process could have produced, given identification and an estimator. This question is the subject of **inference**, and its central quantity is the standard error (§4).

The order matters because errors propagate downwards and never upwards. A perfect standard error around an unidentified parameter is worthless. An elegant estimator cannot fix a badly chosen estimand. Practice nonetheless often inverts the order. Analysts choose the estimator first, because the software offers it. They compute standard errors second. They state the identifying assumption last or not at all. The diagram shows the correct order, with the typical failure at each step.



1.4 Four kinds of question

The question "what does the data say about X and Y?" can mean four different questions. The four need different assumptions and different evaluation. Confusing them is one of the most common errors in empirical work. The table sets them side by side.

Question	Estimand	What must be true	How it is checked	Typical tools
Description	A conditional mean or linear projection	Only sampling assumptions	Inference alone	Regression read as a projection; portfolio sorts
Prediction	$\mathbb{E}[Y_{t+h} \mid \mathcal{F}_t]$, or a forecast that minimises a loss	The relationship is stable out of sample	Out-of-sample evaluation	Forecasting models; machine learning (§8.8, §10)
Causal effect	An average effect of a well-defined intervention	An identifying assumption about why treatment varies	Design diagnostics, placebo tests, sensitivity analysis	Experiments, IV, panels, DiD, RD (§7)
Structural parameter	A deep parameter of preferences or technology	The economic model is correctly specified	Fit to features of the data the model was not tuned to	GMM, maximum likelihood, simulated moments (§5, §7.8)

$\mathcal{F}_t$ in the table is the information available at time t.

An analyst can read the same regression as any of the four. The analyst's assumptions determine the reading, not the output. A regression of next month's return on this month's book-to-market ratio is a valid *description*, and possibly a useful *prediction*. It does not estimate what would happen to a firm's return if someone changed its book value. Nothing in the output warns against that reading. §10.1 returns to this point as the distinction between problems about $\hat{Y}$ and problems about $\hat{\beta}$.

Most machine learning in finance is prediction, and most empirical corporate finance is causal. Asset-pricing tests are closer to structural questions, because the moment conditions come from an economic model of prices. Knowing which kind of question a project asks shows which of the later sections apply to it.

1.5 The master form: one equation behind every estimator

This subsection states the idea that organises the rest of the chapter. Almost every estimator in econometrics rests on a **moment condition**. A moment condition is a function of the data and the parameters whose population expectation is zero at the true parameter value:

$$\mathbb{E}[\psi_i(\theta_0)] = 0.$$

The estimator is the **sample analogue**. It chooses $\hat{\theta}$ so that the sample average of ψ_i is zero, or as close to zero as possible when the equations outnumber the unknowns:

$$\frac{1}{n}\sum_{i=1}^{n} \psi_i(\hat{\theta}) = 0.$$

Methods differ only in the choice of ψ. The table lists the common choices. The last column gives the assumption behind "$= 0$" for each one.

Estimator	Moment function $\psi_i(\theta)$	The assumption behind "$= 0$"
Sample mean	$Y_i - \mu$	None. This is the definition of μ
OLS	$X_i\,(Y_i - X_i'\beta)$	The error is uncorrelated with X_i. This is automatic for a projection and an assumption for a causal β
Instrumental variables	$Z_i\,(Y_i - X_i'\beta)$	The instruments are uncorrelated with the structural error
Maximum likelihood	$\partial \log f(Y_i \mid X_i;\theta)/\partial\theta$	The model's density is correct, or at least its score has mean zero
Fixed effects	$(X_{it}-\bar{X}_i)(Y_{it}-\bar{Y}_i-(X_{it}-\bar{X}_i)'\beta)$	Regressors are uncorrelated with the error at every date: strict exogeneity
Quantile regression	$X_i\,(q - \mathbb{1}\{Y_i < X_i'\beta\})$	$X_i'\beta$ is the conditional q-quantile
Asset-pricing Euler equation	$Z_t\,(M_{t+1}(\theta)\,R_{t+1} - 1)$	Investors' first-order condition holds and Z_t is known at t

In the last two rows, $q \in (0,1)$ is the quantile of interest and R_{t+1} is a gross asset return. $M_{t+1}(\theta)$ is a **stochastic discount factor**, the model's valuation of a unit payoff at $t+1$ (§5.3). In the fixed-effects row, $\bar{X}_i$ and $\bar{Y}_i$ are unit i's averages over time (§7.4).

Writing estimators this way yields three facts. The chapter uses each of them repeatedly.

Identification is uniqueness of the solution. A parameter is identified when the population moment condition holds at θ_0 and nowhere else. For linear moment conditions this reduces to a rank condition. Instrumental variables, for example, needs $\mathbb{E}[Z_i X_i']$ to have full column rank. This is the formal version of "the instruments must actually move the regressors" (§7.3).

The substantive assumption is always an orthogonality condition. Some observed variable is uncorrelated with some unobserved one. In the just-identified case, *the data cannot check this assumption*. A just-identified model has exactly as many equations as unknowns. The estimator forces the sample analogue to hold exactly. OLS residuals are orthogonal to the regressors by construction, so no plot of residuals against X can reveal that X is correlated with the structural error. [Fact] The residuals of a regression show no sign of the endogeneity that contaminates it. Over-identified models permit a partial check. The check tests only whether the instruments agree with each other (§5.3).

A new method changes only what must be orthogonal to the error. Adding controls requires orthogonality conditional on the controls. Fixed effects require orthogonality with within-unit variation only. Instrumental variables require orthogonality with the instrument instead of the regressor. Regression discontinuity requires it only in a small window around a cutoff. The first question to ask of a new design is therefore *which orthogonality condition it requires, and what that requirement costs*.

1.6 The sandwich: one formula behind every standard error

The master form also gives every standard error in econometrics, through a single derivation. Start from the sample moment condition at the estimate and expand it around the true value. To first order,

$$0 = \frac{1}{n}\sum_{i=1}^{n}\psi_i(\hat{\theta}) \;\approx\; \frac{1}{n}\sum_{i=1}^{n}\psi_i(\theta_0) \;+\; \underbrace{\Big[\frac{1}{n}\sum_{i=1}^{n}\frac{\partial\psi_i(\theta_0)}{\partial\theta'}\Big]}_{\to Q}(\hat{\theta}-\theta_0).$$

Solving for the estimation error and scaling by $\sqrt{n}$ gives

$$\sqrt{n}\,(\hat{\theta}-\theta_0) \;\approx\; -Q^{-1}\,\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i(\theta_0).$$

The estimation error is a fixed matrix times a scaled sum of the moment contributions. A central limit theorem (§2.4) makes that sum approximately normal, with variance Ω. The asymptotic variance is therefore

$$\text{Avar}\left(\sqrt{n}\,(\hat{\theta}-\theta_0)\right) \;=\; \underbrace{Q^{-1}}_{\text{bread}}\;\underbrace{\Omega}_{\text{meat}}\;\underbrace{Q^{-1\prime}}_{\text{bread}}, \qquad \Omega = \lim_{n\to\infty}\text{Var}\Big(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_i(\theta_0)\Big).$$

Dividing by n gives the approximate variance of $\hat{\theta}$ itself. The square roots of its diagonal are the standard errors.

The two ingredients do different jobs, so keep them apart.

- **The model sets the bread, Q.** The bread measures how sharply the moment condition responds when the parameter moves. For OLS, $\psi_i = X_i(Y_i - X_i'\beta)$, so $Q = -\mathbb{E}[X_iX_i'] = -Q_{XX}$. More variation in the regressors gives a larger bread and a smaller variance. So an experiment with a wide spread of treatment doses is more informative than one with a narrow spread.
- **The dependence in the data sets the meat, Ω.** The meat is the variance of a sum. If the moment contributions are independent across observations, it is the sum of their variances. If they are correlated, the covariances enter too. With many positively correlated pairs, the covariances dominate.

Every standard-error "correction" in applied work is a different estimate of the meat. Each makes a different assumption about which observations are independent of which. The table lists the common estimates.

Name	Estimate of the meat Ω	Independence it assumes
Classical	$\hat{\sigma}^2\hat{Q}_{XX}$	All observations independent, with equal error variance
Heteroskedasticity-robust (White)	$\frac{1}{n}\sum_i \hat{e}_i^2 X_iX_i'$	All observations independent, variances free
Cluster-robust	$\frac{1}{n}\sum_g \big(\sum_{i\in g} X_i\hat{e}_i\big)\big(\sum_{i\in g} X_i\hat{e}_i\big)'$	Independent across clusters. Arbitrary dependence within clusters
Newey–West (HAC)	Weighted sum of sample autocovariances of $X_t\hat{e}_t$	Serial dependence that fades with the lag
Driscoll–Kraay	Newey–West applied to cross-sectional sums	Arbitrary cross-sectional dependence, fading serial dependence
Fama–MacBeth	Variance of the period-by-period estimates	Independence over time. Arbitrary dependence within a period
Two-way clustered	Firm-clustered + time-clustered − heteroskedasticity-robust	Independence across both firms and dates, except through those two groupings

All of them use the same bread. §4 derives the entries and says when each is right. In a few cases the equivalence between rows is exact (§12.2).

The table implies the most important practical fact about inference. This fact is the second organising idea of the chapter:

> **The number of independent pieces of information in the meat sets the precision of an estimate, not the number of rows in the data.**

Two examples show this. First, take 5,000 observations in 50 groups of 100. The regressor takes one value per group, and groups share 5% of the error variance internally. [Simulated] The effective sample size is about 840. Classical standard errors reject a true null hypothesis 42% of the time at a nominal 5% (§4.4). Second, take 40 years of monthly data on 12-month returns. The data have 480 rows. For estimating an average 12-month return, they carry the information of roughly 40 non-overlapping observations (§4.5).

The two failures, stated once. Every estimate can go wrong in two ways, and the two come from different sources:

> **Bias comes from a false orthogonality condition. False precision comes from a wrong meat.**

Almost every method in the rest of this chapter does one of three things. It makes an orthogonality condition more credible (Part III). It estimates the meat correctly (§4, §8, §9). Or it trades a little bias for a large reduction in variance (§5.6, §10). Placing a new technique in one of these three groups says most of what matters about it.

1.7 What econometrics is not

Several distinctions come up often enough to fix at the start. The table lists the pairs most often confused.

Often conflated	The difference
Econometrics and statistics	The mathematics is the same. Econometrics adds two things: data generated by optimising agents, which makes identification the central problem, and economic models that supply moment conditions. Other fields have reached the same problems under other names. Epidemiology's "confounding" is econometrics' "omitted variable bias."
Econometrics and machine learning	Machine learning optimises out-of-sample prediction of Y. Much of econometrics estimates a parameter with a causal or structural meaning and a correct measure of its uncertainty. The tools now overlap (§10). The goals do not.
A good fit and a good model	A high R^2 does not indicate identification, and a low one is compatible with a precisely estimated causal effect (§3.8).
Statistically significant and important	Significance measures signal relative to noise. Importance depends on magnitude (§4.10).
Unbiased and consistent	Unbiased means right on average at a fixed sample size. Consistent means converging to the truth as the sample grows. Neither implies the other (§2.3).
Granger causality and causality	Granger causality means that the past of X improves forecasts of Y. It measures predictive precedence. It does not show what an intervention on X would do (§8.5).
The error term and the residual	The error (u_i or e_i) is a population quantity. The residual $\hat{e}_i$ comes from a sample and is orthogonal to the regressors by construction (§3.3).
A backtest and an econometric study	They are the same thing. A backtest estimates a mean from dependent data after a search over specifications. It has every problem described in §4 and §9, whether or not anyone acknowledges it (§9.7).

§1 Key takeaways

1. A regression coefficient is a ratio of covariance to variance, not an effect. It becomes an effect only under an assumption about why the regressor varies across units.
2. Econometrics exists because agents generate economic data and choose their regressors using information the analyst does not have.
3. Answer three questions in order: what the estimand is, whether it is identified, and how precisely the data estimate it. Errors propagate downwards and never upwards.
4. Description, prediction, causal effects, and structural parameters are four different questions. The same regression can serve any of them, and the output never says which.

5. Almost every estimator solves a sample moment condition. The substantive assumption is always an orthogonality condition, and in the just-identified case the data cannot test it.
6. Every standard error is a sandwich: bread from the model, meat from the dependence in the data. Every standard-error correction is a different estimate of the meat.
7. Bias comes from a false orthogonality condition, and false precision comes from a wrong meat. Precision depends on the number of independent pieces of information, not the number of rows.

2. The probability you actually need

This section covers the minimum probability theory that the rest of the chapter uses. A reader who has seen the definitions before can read it quickly. Two subsections go beyond standard textbook material, and no reader should skip them. §2.2 covers three strengths of "unrelated." §2.5 covers how dependence reduces the effective number of observations.

2.1 Populations, samples, and the data-generating process

A random variable is a model of what the data *could have been*. The **data-generating process** (DGP) is the joint probability distribution from which the observed data are one draw. In a cross-section, it is the distribution of $(Y_i, X_i, Z_i, \dots)$ for a randomly drawn unit. In a time series, it is the stochastic process that produced the whole path.

An estimate is uncertain because a different draw would have produced a different estimate, and a standard error quantifies that variation. A standard error is therefore meaningful only once the analyst states *what is being redrawn*. Two answers are standard.

- **Sampling-based uncertainty.** The data are a random sample from a larger population, and the uncertainty concerns which units the sample happened to include. This is the textbook case, and it fits survey data.
- **Design-based uncertainty.** The data cover the entire population of interest, such as all fifty US states or every stock in an index, so sampling variation in the usual sense does not arise. The random element is which units received the treatment, and therefore which potential outcome of each unit the analyst observes. Abadie, Athey, Imbens & Wooldridge (2020) show that conventional standard errors remain sensible for causal questions under this reading, though they can be conservative. For purely descriptive questions about a fully observed population, conventional standard errors measure sampling variation that does not exist in the data.

Finance has a third answer, which matters for everything that follows. A study of every listed stock does not sample the cross-section. **The sample is the history.** The relevant thought experiment is that the same economy could have produced a different sequence of shocks. The uncertainty concerns which history occurred. So the number of independent time periods usually governs precision in empirical finance, not the number of stocks (§2.5, §9.1).

2.2 Conditional expectation, and three strengths of "unrelated"

Conditional expectation. $\mathbb{E}[Y \mid X]$ is the average of Y among units with a given value of X, viewed as a function of X. Three properties make it central to the field.

1. **It is the best predictor.** Among all functions $g(X)$, the one that minimises the mean squared error $\mathbb{E}[(Y - g(X))^2]$ is $g(X) = \mathbb{E}[Y \mid X]$. To see why, add and subtract the conditional mean:

$$\mathbb{E}\big[(Y - g(X))^2\big] = \mathbb{E}\big[(Y - \mathbb{E}[Y \mid X])^2\big] + \mathbb{E}\big[(\mathbb{E}[Y \mid X] - g(X))^2\big].$$

   The cross term vanishes because the prediction error $Y - \mathbb{E}[Y \mid X]$ has mean zero at every value of X. The first term does not depend on g, and the second is zero only when g equals the conditional mean.
2. **The law of iterated expectations.** $\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big]$. The overall average is the average of the group averages, weighted by group size.
3. **The law of total variance.** $\mathrm{Var}(Y) = \mathrm{Var}\left(\mathbb{E}[Y \mid X]\right) + \mathbb{E}\big[\mathrm{Var}(Y \mid X)\big]$. Total variation splits into the part explained by X and the part left over. R^2 is a sample version of the first share (§3.8).

Three strengths of "unrelated." Econometric assumptions often require an unobserved U (with $\mathbb{E}[U] = 0$) to be "unrelated" to an observed X. The table gives three distinct ways to state this, in increasing strength.

Condition	Statement	What it rules out	What needs it
Uncorrelated	$\mathbb{E}[XU] = 0$	A *linear* relationship between X and U	Consistency of OLS and IV
Mean-independent	$\mathbb{E}[U \mid X] = 0$	*Any* relationship between X and the average of U	Unbiasedness; treating any function of X as a valid regressor or instrument
Independent	$U \perp X$	Any relationship between X and the whole distribution of U	Quantile and distributional claims; randomisation

Each condition implies the one above it. Neither converse holds. The standard counterexample takes $X \sim N(0,1)$ and $U = X^2 - 1$. Then $\mathbb{E}[XU] = \mathbb{E}[X^3] - \mathbb{E}[X] = 0$. The two are uncorrelated, and a regression of U on X has a slope of exactly zero, even though U is a deterministic function of X.

Financial returns show the same gap in practice. [Fact] Daily returns on broad equity indices are close to uncorrelated over time but far from independent. Large moves cluster, and squared returns are strongly autocorrelated. The **martingale difference** property lies between the two conditions: $\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] = 0$ for a demeaned return r_{t+1}. It means that the past cannot predict the *level* of future returns. It allows their variance to be predictable. A test of "no autocorrelation" whose standard errors assume independence is mis-sized on returns. Independence is false even when the hypothesis under test is true (§8.1).

2.3 Estimators are random variables

An estimator $\hat{\theta}$ is a function of the sample, so it has a distribution. This **sampling distribution** is the distribution of $\hat{\theta}$ across all the samples the DGP could have produced. Three summaries of it matter.

- **Bias:** $\mathbb{E}[\hat{\theta}] - \theta_0$, whether the estimator is right on average.
- **Variance:** how much it moves from sample to sample. The **mean squared error** combines the two: MSE $=$ bias$^2 +$ variance.
- **Consistency:** whether $\hat{\theta} \xrightarrow{p} \theta_0$ as the sample grows, that is, whether the estimator converges to the truth.

Unbiasedness and consistency are logically independent. Using only the first observation, $\hat{\mu} = Y_1$, gives an unbiased estimator of a mean, but its variance never shrinks, so it is inconsistent. The OLS estimate of an autoregressive coefficient ϕ is biased downward in finite samples, by roughly $-(1+3\phi)/T$ (Kendall, 1954), but it is consistent. The bias matters when T is small and ϕ is near one, as §9.2 shows.

Econometrics relies on consistency more than on unbiasedness, because almost no estimator used in practice is unbiased. Instrumental variables, maximum likelihood, and every nonlinear estimator are biased in finite samples. In practice the analyst asks two things. The first is whether the estimator converges to the right quantity. The second is whether the sample is large enough that the remaining bias is small relative to the standard error.

The MSE decomposition also previews a topic of §5.6 and §10. **An estimator with some bias can have a lower mean squared error than an unbiased one**, if the bias comes with a large enough reduction in variance. Shrinkage estimators use this deliberately.

2.4 The two limit theorems, and what they require

The law of large numbers (LLN) says that sample averages converge to population means. It needs a finite mean and either independence or dependence that fades with distance. For time series the relevant condition is **ergodicity**: averaging along one long path is equivalent to averaging across many independent paths.

The central limit theorem (CLT) says that sample averages are approximately normal, with a spread that shrinks at rate $\sqrt{n}$:

$$\sqrt{n}\,(\bar{Y} - \mu) \xrightarrow{d} N(0, \sigma^2).$$

It needs a finite variance and, again, independence or fading dependence. Under dependence, σ^2 becomes the *long-run* variance of §2.5.

The theorems have three consequences for practice:

1. **The $\sqrt{n}$ rule.** Standard errors fall in proportion to $1/\sqrt{n}$, so halving a standard error takes four times the data. Equivalently, detecting an effect half as large requires a sample four times as big. Precision is expensive, and its cost rises quickly.
2. **The approximation applies to the average, not to the data.** The CLT makes no claim about whether individual observations are normal. It justifies t-tests on regression coefficients, which are weighted averages, with non-normal data. It does not justify treating tail probabilities of the data themselves as normal.
3. **Heavy tails slow the approximation down.** When the variance exists but higher moments are large or infinite, the CLT still holds eventually, but the sample size it needs can be very large. [Fact] Daily equity returns have tails heavy enough that the fourth moment may not exist (Cont, 2001). Estimators of their means and variances work tolerably well. Any statistic that depends on fourth moments is far less precise than its standard formula implies. Examples are sample kurtosis, the standard error of a variance estimate, and the non-normal standard error of a Sharpe ratio.

2.5 Effective sample size: dependence reduces the number of observations

For the mean of a stationary series, the variance in the CLT is the **long-run variance**, which includes all the autocovariances, and not σ^2. The variance of a sample mean of T observations is approximately

$$\text{Var}(\bar{Y}) \approx \frac{\sigma^2}{T}\Big(1 + 2\sum_{j=1}^{\infty}\rho_j\Big),$$

where ρ_j is the autocorrelation at lag j. Positive autocorrelation inflates the variance, because neighbouring observations repeat part of the same information. Comparing this variance with the independent case defines the **effective sample size**:

$$n_{\text{eff}} = \frac{T}{1 + 2\sum_{j\geq 1}\rho_j}.$$

For a first-order autoregression, $\rho_j = \phi^j$ and the sum is geometric, which gives

$$n_{\text{eff}} = T\,\frac{1-\phi}{1+\phi}.$$

The table shows how quickly persistence reduces the effective number of observations.

Persistence ϕ	n_{eff}/T	1,000 observations are equivalent to
0	1.000	1,000
0.5	0.333	333
0.9	0.053	53
0.99	0.005	5

The same arithmetic holds across clusters. Suppose n observations fall into groups of size $\bar{n}_g$, and every pair within a group has correlation ρ. The effective sample size for a mean is then $n/(1+(\bar{n}_g-1)\rho)$. For a regression coefficient the formula also includes the within-group correlation of the regressor. This gives the Moulton factor of §4.4.

Overlapping observations are the most common case in finance. A series of overlapping h-period returns built from T one-period returns carries roughly the information of T/h non-overlapping ones, for the purpose of estimating a mean. Forty years of monthly data on 12-month returns have 480 rows and about 40 independent observations.

Merton's asymmetry. The precision of an estimated *mean* return depends on the calendar span of the data, not on how finely the data are sampled. The mean return over forty years is the total log price change divided by forty. Sampling daily instead of monthly does not change the endpoints. With an annual volatility of 16%, the standard error of a mean annual return estimated from 40 years is $16\%/\sqrt{40} = 2.5\%$. A 6% equity premium is therefore known only to within about ± 5 percentage points at 95% confidence. *Variances* behave the opposite way. Summing squared high-frequency returns estimates the variance more and more precisely as the sampling interval shrinks (Merton, 1980). This asymmetry explains a great deal of empirical finance. It explains three facts. Risk models work far better than return forecasts. The covariance matrix is estimable, but the mean vector is not. The evidence for any return anomaly is always weaker than its t-statistic implies.

2.6 The delta method, and other tools for approximations

Three results let the analyst manipulate estimates in the same way as the numbers they estimate.

- **The continuous mapping theorem.** If $\hat{\theta} \xrightarrow{p} \theta_0$ and g is continuous, then $g(\hat{\theta}) \xrightarrow{p} g(\theta_0)$.
- **Slutsky's theorem.** A statistic that converges in distribution, multiplied by something that converges in probability, converges to the obvious product. This allows an analyst to replace an unknown variance with a consistent estimate inside a t-statistic.

- **The delta method.** If $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V)$ and g is differentiable, then

$$\sqrt{n}\,(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0,\ g'(\theta_0)^2\, V).$$

  The standard error of a smooth function of an estimate is the standard error of the estimate times the slope of the function.

A worked example, and its failure. A shock to an AR(1) process decays as ϕ^h, so its **half-life**, the horizon at which half of the shock is gone, is $\ln(0.5)/\ln(\phi)$. Suppose $\hat{\phi} = 0.95$ with a standard error of 0.02. The point estimate of the half-life is $0.693/0.0513 = 13.5$ periods. Its derivative with respect to ϕ is $\ln(2)/(\phi\,(\ln\phi)^2) = 0.693/(0.95 \times 0.00263) = 277$, so the delta-method standard error is $277 \times 0.02 = 5.5$ periods. The resulting interval is roughly 13.5 ± 11.

That interval is badly wrong, and the failure generalises. Two standard errors above the estimate, $\phi = 0.99$ implies a half-life of 69 periods. The interval's upper limit is $13.5 + 11 = 24.5$. The half-life is so nonlinear near $\phi = 1$ that a linear approximation across the confidence region is meaningless. Delta-method intervals mislead whenever a parameter lies near a boundary, whether a unit root, a variance near zero, or a correlation near one. Transform the endpoints of an interval for ϕ instead, or bootstrap (§4.9).

A second example is the one that quants use most often. For independent, normally distributed returns, the delta method gives the standard error of an estimated per-period Sharpe ratio as approximately $\sqrt{(1 + \mathrm{SR}^2/2)/T}$ (Lo, 2002). In annual units, with a true Sharpe ratio of 0.5, that is about $1.06/\sqrt{\text{years}}$. The estimate needs around 18 years of data to reach two standard errors from zero.

2.7 When asymptotic approximations fail

Asymptotic approximations describe what happens as some quantity goes to infinity. The approximation fails when the quantity that governs it is not the quantity the analyst assumes is large. The table lists the common cases.

Situation	What is really small	Symptom	Where
Few clusters, or few *treated* clusters	The number of clusters, not observations	Tests reject true nulls far too often	§4.4
Weak instruments	The first-stage signal	IV biased toward OLS, oversized t-tests	§7.3
Near-unit-root regressors	The distance of ϕ from one, times T	Spurious significance, Stambaugh bias	§8.3, §9.2
Long overlapping horizons	The number of non-overlapping periods	Oversized Newey–West tests	§4.5, §9.2
Heavy tails	The effective number of tail observations	Unstable higher-moment estimates	§2.4
Many parameters	Observations per parameter	In-sample fit far above out-of-sample fit	§10.2
Estimation after model selection	Nothing: the approximation is invalid	Confidence intervals too narrow	§10.3
Maxima over many tests	Nothing: not a smooth function of means	Naive bootstrap fails	§4.8, §4.9

§2 Key takeaways

1. A standard error has no meaning until the analyst states what is being redrawn. In finance, the sample is usually the history, not the cross-section.
2. "Uncorrelated," "mean-independent," and "independent" are different assumptions. Returns are close to uncorrelated and far from independent, and tests that confuse the two are mis-sized.
3. Consistency matters more than unbiasedness in practice, because almost no estimator in use is unbiased. A little bias can buy a large reduction in variance.
4. Precision grows like $\sqrt{n}$: four times the data halves the standard error.

5. Dependence reduces the effective sample size. An AR(1) with $\phi = 0.9$ turns 1,000 observations into about 53, and 40 years of overlapping 12-month returns are worth about 40.
6. The precision of a mean return depends on calendar span, not on sampling frequency, while the precision of a variance improves with frequency. This asymmetry shapes most of empirical finance.
7. The delta method fails near boundaries such as unit roots. Near a boundary, transform the interval endpoints or bootstrap.
8. Asymptotic approximations fail when the quantity that governs them (clusters, instrument strength, distance from a unit root, independent periods) is small, however many rows the data have.

3. Regression from first principles

Textbooks usually teach regression as a model. They assume $Y = X'\beta + u$ with some list of properties for u and then estimate β. This section builds regression the other way round. It starts from objects that exist whether or not any model is true. The result is a clean answer to the question that the rest of the chapter returns to. The question is when a regression coefficient means what the analyst wants it to mean.

3.1 The conditional expectation function

The **conditional expectation function** (CEF) $m(x) = \mathbb{E}[Y_i \mid X_i = x]$ gives the average outcome at each value of the regressors. Define the CEF error as the remainder:

$$Y_i = m(X_i) + \epsilon_i, \qquad \epsilon_i = Y_i - \mathbb{E}[Y_i \mid X_i].$$

By the law of iterated expectations, $\mathbb{E}[\epsilon_i \mid X_i] = 0$ automatically. **This is a definition, not an assumption.** Every outcome splits into its conditional mean and a mean-independent remainder, whatever the data are and whatever caused them. So a statement such as "the error has mean zero given X" has no content until the analyst gives the error a meaning *beyond* "whatever is left over." §3.2 separates out that meaning.

By §2.2, the CEF is the best predictor of Y given X. The average log wage at each level of schooling is a CEF, and so is the average next-month return at each level of the dividend yield. A CEF describes the data. On its own, it does not show what would happen to a person's wage after more schooling, or to a market's return if its dividend yield changed.

3.2 Linear projection: the regression that always exists

The CEF can have any shape. The **linear projection** of Y on X is the best *linear* predictor. The vector β minimises the mean squared error of $X'\beta$:

$$\beta = \arg\min_b \mathbb{E}\big[(Y_i - X_i'b)^2\big] = \mathbb{E}[X_iX_i']^{-1}\,\mathbb{E}[X_iY_i].$$

The projection exists whenever second moments are finite and no regressor is an exact linear combination of the others. Its error $e_i = Y_i - X_i'\beta$ satisfies the first-order condition

$$\mathbb{E}[X_ie_i] = 0$$

by construction. With a single regressor and an intercept, the slope reduces to the familiar $\mathrm{Cov}(X,Y)/\,\mathrm{Var}(X)$.

Two facts connect the projection to the CEF. If the CEF happens to be linear, the two coincide. If it is not, the projection is still the best linear *approximation* to the CEF, in the sense that it minimises $\mathbb{E}[(m(X_i)-X_i'b)^2]$. A regression is therefore always a valid summary of how the average outcome varies with the regressors, to the extent that a straight line can summarise it.

This is the first half of the central point of this section:

> **A regression is never wrong. It always estimates the linear projection, a well-defined population object. Only the claim about what the projection measures can be wrong.**

The second half concerns causation. Suppose that a **structural equation** generates Y. A structural equation is meant to describe cause and effect:

$$Y_i = X_i'\beta_s + u_i,$$

Here a thought experiment defines β_s: change X, hold everything else fixed, and observe Y. The error u_i collects everything else that affects Y. Nothing guarantees $\mathbb{E}[X_iu_i] = 0$. Substituting the structural equation into the projection formula gives

$$\beta = \mathbb{E}[X_iX_i']^{-1}\,\mathbb{E}\big[X_i(X_i'\beta_s + u_i)\big] = \beta_s + \underbrace{\mathbb{E}[X_iX_i']^{-1}\,\mathbb{E}[X_iu_i]}_{\text{endogeneity bias}}.$$

This line states the whole problem of Part III. **The regression estimates β, while the question usually concerns β_s. The two differ by the projection of the structural error on the regressors.** That term is zero exactly when the regressors are

uncorrelated with everything else that moves the outcome. No amount of data shrinks it, because it is a difference between two population quantities. The projection error e_i is uncorrelated with X_i by definition. The structural error u_i is uncorrelated with X_i only if the data-generating process makes it so.

3.3 OLS: the sample moment condition, and its geometry

Ordinary least squares replaces population moments with sample moments:

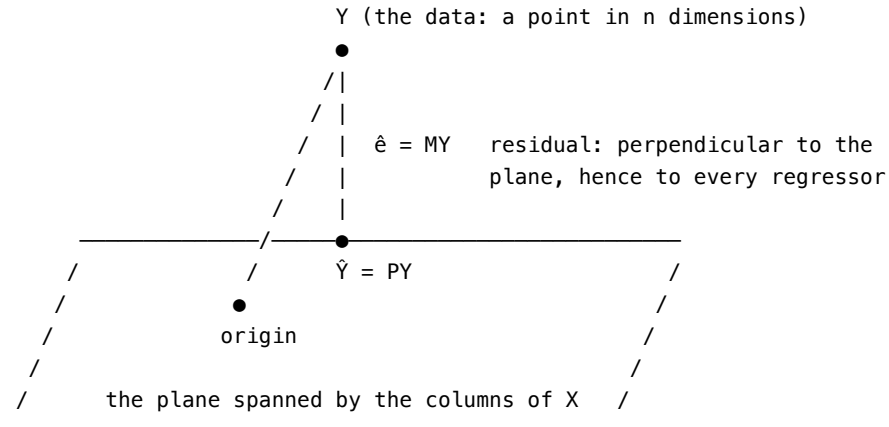
$$\hat{\beta} = \Big(\sum_{i=1}^{n} X_i X_i' \Big)^{-1} \sum_{i=1}^{n} X_i Y_i = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Its defining property is the **normal equations** $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\beta}) = 0$, the sample analogue of $\mathbb{E}[X_i e_i] = 0$. This is the master form of §1.5 with $\psi_i(\beta) = X_i(Y_i - X_i'\beta)$.

The geometry. Think of $\mathbf{Y}$ as a point in n-dimensional space, and of the k columns of $\mathbf{X}$ as spanning a k-dimensional flat subspace. OLS finds the point of that subspace closest to $\mathbf{Y}$, which is the orthogonal projection of $\mathbf{Y}$:

$$\hat{\mathbf{Y}} = \mathbf{P}\mathbf{Y}, \quad \mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \hat{\mathbf{e}} = \mathbf{M}\mathbf{Y}, \quad \mathbf{M} = \mathbf{I} - \mathbf{P}.$$

The residual vector is perpendicular to every column of $\mathbf{X}$, and therefore to the fitted values. The diagram shows this geometry.



The geometry has two consequences.

- **Pythagoras gives R^2.** With an intercept, the total sum of squares of Y around its mean splits exactly into the explained part, $\|\hat{\mathbf{Y}} - \bar{Y}\|^2$, and the residual part, $\|\hat{\mathbf{e}}\|^2$.
- **Residuals cannot reveal endogeneity.** $\mathbf{X}'\hat{\mathbf{e}} = 0$ holds in every sample by construction, whatever the true relationship between X and u (§1.5).

Leverage and influence. The diagonal element $\mathbf{P}_{ii}$ is observation i's **leverage**, a number between 0 and 1 that measures how far its regressors lie from the bulk of the data. The leverages sum to k. The fitted plane passes closer to high-leverage observations, so their residuals are *shrunk*. Under homoskedasticity, $\text{Var}(\hat{e}_i) = \sigma^2(1 - \mathbf{P}_{ii})$. Dropping observation i changes the estimate by

$$\hat{\beta} - \hat{\beta}_{(-i)} = (\mathbf{X}'\mathbf{X})^{-1} X_i \, \frac{\hat{e}_i}{1 - \mathbf{P}_{ii}},$$

An observation is therefore influential when it has both high leverage and a large residual. Financial data contain many such points. Examples are crash months, microcap stocks, and firms with extreme accounting ratios. [Practice] **Recommendation: before trusting a coefficient, drop the most influential 1% of observations and compute how much the coefficient moves.** If the coefficient moves "most of the way to zero," the finding comes from those observations and not from the sample as a whole.

3.4 Frisch–Waugh–Lovell: what "controlling for" means

Split the regressors into two groups, $\mathbf{Y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \mathbf{e}$. The **Frisch–Waugh–Lovell (FWL) theorem** says that two steps compute the multiple-regression coefficient $\hat{\beta}_1$:

1. Regress $\mathbf{X}_1$ on $\mathbf{X}_2$ and keep the residuals $\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1$, the part of $\mathbf{X}_1$ that $\mathbf{X}_2$ cannot linearly predict. Here $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$.
2. Regress $\mathbf{Y}$ on $\tilde{\mathbf{X}}_1$: $\hat{\beta}_1 = (\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\mathbf{Y}$.

Residualising $\mathbf{Y}$ on $\mathbf{X}_2$ as well gives the same coefficient. The theorem follows from the geometry of §3.3. The component of $\mathbf{X}_1$ that lies inside the space of $\mathbf{X}_2$ is already captured by $\mathbf{X}_2$'s coefficients, so only the orthogonal component can identify β_1.

The interpretation is the most useful single idea in applied regression:

> **"Controlling for X_2" means using only the variation in X_1 that X_2 cannot predict.** That residual variation is the only input to the coefficient estimate.

Four consequences follow. Each returns later in the chapter.

1. Controls reduce the identifying variation. The precision of $\hat{\beta}_1$ depends on how much residual variation remains. Let $R^2_{1\cdot 2}$ be the R^2 from regressing X_1 on the controls. Relative to a regression with no controls, the **variance inflation factor** $1/(1-R^2_{1\cdot 2})$ inflates the variance of $\hat{\beta}_1$. With $R^2_{1\cdot 2} = 0.9$, the standard error is $\sqrt{10} = 3.2$ times larger. This is equivalent to a sample ten times smaller. "Multicollinearity" violates no assumption. It is a shortage of independent variation. For this reason Goldberger (1991) suggested renaming it "micronumerosity."

2. The estimate is local to the residual variation. Consider a regression of stock returns on analyst coverage that controls for firm size. Size explains most of the cross-sectional variation in coverage, so the data identify the coefficient from firms with *unusually* high or low coverage for their size. Those firms are unusual in other ways too, such as recent listings, controversy, or index membership. The estimate describes them, and extrapolating it to a typical firm is an additional assumption.

3. Regression weights observations by the conditional variance of the regressor. With a binary treatment and a set of discrete controls, the OLS coefficient is a weighted average of the treatment–control differences within each cell of the controls. OLS weights each cell by its size times $p(1-p)$, where p is the share treated in that cell (Angrist, 1998). Cells where almost everyone or almost no one receives treatment get almost no weight. If the effect differs across cells, the regression coefficient is a variance-weighted average. It equals neither the average effect in the population nor the average effect on the treated. An unrepresentative subset can dominate this average (Aronow & Samii, 2016; Słoczyński, 2022).

4. Fixed effects and "partialling out" are FWL. Adding a dummy for every firm is algebraically identical to subtracting each firm's mean from both sides (§7.4). The double machine learning estimator of §10.5 uses FWL with flexible predictions in place of the linear first-step regressions.

3.5 The omitted variable bias formula

FWL shows what including a control does. The **omitted variable bias** (OVB) formula gives the cost of leaving one out. Suppose the desired regression is

$$Y_i = \alpha + \beta X_i + \gamma C_i + u_i,$$

and the feasible one omits C_i. Write the projection of the omitted variable on the included one as $C_i = \delta_0 + \delta X_i + v_i$. Substituting gives

$$Y_i = (\alpha + \gamma\delta_0) + (\beta + \gamma\delta)\, X_i + (\gamma v_i + u_i),$$

Because v_i is uncorrelated with X_i by construction, the short regression's slope converges to

$$\operatorname{plim} \hat{\beta}_{\text{short}} = \beta + \gamma\,\delta.$$

The bias equals the effect of the omitted variable on the outcome, times the coefficient of the omitted variable on the included one. Both links must be present. An omitted variable that affects Y but is uncorrelated with X reduces precision and causes no bias. One that is correlated with X but does not affect Y causes no bias either. The table gives the sign of the bias in each case.

	Omitted variable correlated positively with X ($\delta > 0$)	Correlated negatively ($\delta < 0$)
Omitted variable raises Y ($\gamma > 0$)	Upward bias	Downward bias
Omitted variable lowers Y ($\gamma < 0$)	Downward bias	Upward bias

A worked example, with illustrative numbers. Suppose a regression of log earnings on years of schooling gives $\hat{\beta}_{\text{short}} = 0.10$, a 10% earnings gain per year. Ability is unobserved. Suppose one standard deviation of ability raises log earnings by $\gamma = 0.05$ with schooling held fixed. Suppose also that each year of schooling goes with $\delta = 0.4$ standard deviations more ability. The bias is $0.05 \times 0.4 = 0.02$, so the causal return is about 0.08. A fifth of the naive estimate is selection. The formula does not supply γ and δ. It does state exactly what an analyst must believe about them to defend the estimate. This turns a vague worry into a specific argument.

The formula has three further uses.

- **Signing the bias without data.** An argument for the direction of both links shows whether the estimate is an upper or a lower bound.
- **Coefficient stability as evidence.** Suppose adding a set of observed controls barely moves the coefficient, and those controls are informative about the unobserved ones. Then the remaining bias may be small. §6.7 makes this argument formal and shows its limits.
- **Noisy controls control only partly.** A control measured with error is only partially correlated with the true confounder, so including it removes only part of the bias. [Simulated] In the simulation of §6.5, controlling for a proxy with a reliability of 0.5 removes under 40% of the bias. Authors who claim to have "controlled for" a confounder often controlled for a noisy version of it.

The multivariate generalisation is the endogeneity-bias formula at the end of §3.2, and OVB is its most common special case.

3.6 Reading coefficients

The analyst can interpret a coefficient only in the units of the model it came from. The table gives the interpretation of the slope in the common specifications.

Specification	Interpretation of the slope
$Y = \alpha + \beta X$	A one-unit difference in X goes with β units of Y
$\log Y = \alpha + \beta X$	A one-unit difference in X goes with about $100\beta\%$ of Y, or exactly $100(e^{\beta} - 1)\%$
$Y = \alpha + \beta \log X$	A 1% difference in X goes with $\beta/100$ units of Y
$\log Y = \alpha + \beta \log X$	An **elasticity**: a 1% difference in X goes with $\beta\%$ of Y
$\log Y = \alpha + \beta D$, D binary	A proportional difference of exactly $e^{\beta} - 1$
$Y = \alpha + \beta_1 X + \beta_2 X^2$	The slope $\beta_1 + 2\beta_2 X$ varies with X
$Y = \alpha + \beta_1 X + \beta_2 Z + \beta_3 XZ$	The slope of X is $\beta_1 + \beta_3 Z$. β_1 alone is the slope where $Z = 0$

Four errors are common.

- **Log approximations fail for large coefficients.** In a log regression, a dummy coefficient of 0.3 is a 35% difference, not 30%, and -0.3 is -26%, not -30%.
- **The main effect in an interaction model applies where the other variable is zero.** If Z is firm size in dollars, β_1 is the effect of X for a firm of size zero, which does not exist. Centre Z at a meaningful value before interacting.
- **Logs of variables with zeros are not unit-free.** Analysts often apply transforms such as $\log(1 + Y)$ or $\operatorname{arcsinh}(Y)$ to trading volume, counts, or R&D spending. They give coefficients that change when Y is rescaled, so the "percentage" reading is not well defined (Chen & Roth, 2024).
- **Predicting $\log Y$ is not predicting Y.** $\exp(\mathbb{E}[\log Y \mid X])$ is the conditional *geometric* mean, which is below $\mathbb{E}[Y \mid X]$. Retransforming a log-scale forecast needs a correction (Simple and Log Returns §11.5).

"Holding other variables constant" needs its own warning. The phrase describes the arithmetic of the linear projection, a partial derivative of $X'\beta$. It describes no feasible change in the world. Suppose analyst coverage causes firms to grow. Then "the effect of coverage holding size constant" is the effect of coverage through every channel *except* size. That may not be the effect that anyone wanted (§6.5).

3.7 The classical assumptions, ranked by what breaks

Textbooks list the assumptions of the linear model as if all were equally important. The assumptions differ in importance, according to what goes wrong when each one fails. The table ranks them by that consequence.

Assumption	What it buys	If it fails	The modern response
Exogeneity: $\mathbb{E}[u_i \mid X_i] = 0$	$\hat{\beta}$ estimates the structural β_s	**Wrong answer**, and more data do not help	Research design (Part III)
Independence across observations	The simple meat	**Wrong precision**, often by large factors	Cluster-robust or HAC standard errors (§4)
Homoskedasticity	Classical standard errors; OLS is efficient	Wrong precision, usually too small; some efficiency lost	Heteroskedasticity-robust standard errors (§4.3)
Linearity of the interpreted model	The coefficient has the meaning assigned to it	The regression estimates the best linear approximation, which may mislead	Flexible specifications; binned scatter plots (§11.4)
No perfect collinearity	A unique $\hat{\beta}$	Not identified	Reparametrise. Near-collinearity costs precision, not bias
Normal errors	Exact finite-sample t and F distributions	Almost nothing in moderate samples, by the CLT	Asymptotic inference; the bootstrap in small samples

[Fact] A failure of exogeneity gives the wrong answer. A failure of independence or homoskedasticity gives the wrong confidence in an answer. A failure of normality in a sample of reasonable size costs almost nothing.

The **Gauss–Markov theorem** says that under the first five assumptions, OLS has the smallest variance among linear unbiased estimators. The theorem receives more attention than it deserves, for three reasons. The class of "linear unbiased" estimators is narrow. Efficiency is a second-order concern next to bias. Since White (1980), valid inference has not needed homoskedasticity at all. The theorem's main modern use is to show what is lost by *not* weighting when the error variance is known.

3.8 R-squared, and what it is not

$R^2 = 1 - \sum_i \hat{e}_i^2 / \sum_i (Y_i - \bar{Y})^2$ is the share of the sample variance of Y that the linear projection explains. It also equals the squared correlation between Y and the fitted values $\hat{Y}$. It measures fit only.

- **R-squared does not test causal validity.** A randomised experiment with a noisy outcome can estimate a treatment effect precisely with an R^2 of 0.02. Two unrelated trending series produce an R^2 above 0.3 about a third of the time. Shared drift alone causes this (§8.3).
- **R-squared does not measure out-of-sample performance.** In-sample R^2 never falls when a regressor is added, whatever the regressor is (§10.2).
- **A small R^2 can be economically large.** This version matters in finance.

The last point has a short derivation. It changes intuition about return predictability. Consider an investor with mean-variance preferences and risk aversion γ_r. The investor holds a risky asset with excess-return mean μ, volatility σ, and Sharpe ratio $S = \mu/\sigma$. Without a forecast, the optimal weight is $\mu/(\gamma_r \sigma^2)$. The expected excess return on the investor's portfolio is then S^2/γ_r. Now give the investor a predictor that explains a fraction R^2 of the variance of the asset's return. The conditional mean μ_t then has variance $R^2\sigma^2$ around μ, and the unexplained variance is $(1 - R^2)\sigma^2$. The optimal weight becomes $\mu_t/(\gamma_r(1 - R^2)\sigma^2)$. The expected portfolio excess return is this weight times μ_t, averaged over the distribution of μ_t. This average is $\mathbb{E}[\mu_t^2]/(\gamma_r(1 - R^2)\sigma^2)$. Since μ_t has mean μ and variance $R^2\sigma^2$, $\mathbb{E}[\mu_t^2] = \mu^2 + R^2\sigma^2$. The expected portfolio excess return is therefore

$$\frac{\mathbb{E}[\mu_t^2]}{\gamma_r(1 - R^2)\sigma^2} = \frac{\mu^2 + R^2\sigma^2}{\gamma_r(1 - R^2)\sigma^2} = \frac{S^2 + R^2}{\gamma_r(1 - R^2)}.$$

Relative to no forecast, the expected return rises by a factor of $(1 + R^2/S^2)/(1 - R^2)$, which is approximately $1 + R^2/S^2$.

The right benchmark for a predictive R^2 is the squared Sharpe ratio, not 1. A monthly Sharpe ratio of 0.12 (about 0.42

annualised) gives $S^2 = 0.0144$. A monthly R^2 of 0.5% would then raise the expected return by about 35%. This is the logic of Campbell & Thompson (2008). It explains why an out-of-sample R^2 that looks negligible can have economic value.

The same logic has a second consequence. The benchmark is so small that an in-sample R^2 of that size is also easy to produce by accident. A search over predictors, overlapping observations, or a persistent regressor can produce it (§9.2). The relevant statistic for prediction is the **out-of-sample** R^2,

$$R^2_{\text{OS}} = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t (r_t - \bar{r}_t)^2},$$

where $\hat{r}_t$ is the forecast made with information available before t, and $\bar{r}_t$ is the historical average at that date. This statistic can be negative, and often is (§8.8).

§3 Key takeaways

1. The error of the conditional expectation function is mean-independent of the regressors by definition. The statement “the error has mean zero” carries no information until the analyst gives the error a structural meaning.
2. A regression always estimates the linear projection, a valid population object. The projection equals the causal coefficient only when the structural error is uncorrelated with the regressors. The gap does not shrink with sample size.
3. Residuals are orthogonal to the regressors by construction, so no residual diagnostic can detect endogeneity.
4. Controlling for a variable means using only the variation it cannot predict. Controls use up identifying variation and make the estimate local to the observations that retain it.
5. Omitted variable bias is the omitted variable’s effect times its coefficient on the included regressor. Use the formula to sign biases. A noisy control removes only part of a bias.
6. With heterogeneous effects, regression estimates a variance-weighted average. An unrepresentative subset of the data can dominate that average.
7. Rank the classical assumptions by consequence. Exogeneity failures give wrong answers, dependence and heteroskedasticity give wrong precision, and non-normality rarely matters.
8. Judge a predictive R^2 against the squared Sharpe ratio. A monthly R^2 of 0.5% can be economically large, and it is also easy to produce by accident in-sample.

Part II — Estimation and inference

4. Inference: how precise an estimate is

Identification decides whether an estimate measures the right parameter. Inference decides how far the estimate is likely to be from that parameter. This section covers the second question. Its central message is the one from §1.6. Every standard error rests on an assumption about which observations are independent of which. When that assumption is wrong, the standard error is wrong by multiples, not by small amounts.

4.1 What a standard error measures

A **standard error** is an estimate of the standard deviation of an estimator's sampling distribution. It measures how much the estimate would vary across the samples the data-generating process could have produced. A t**-statistic** divides the distance between the estimate and a hypothesised value by the standard error. A **95% confidence interval** is a procedure that covers the true value in 95% of repeated samples. It is roughly the estimate plus or minus two standard errors.

That last sentence is easy to misread. The correct and useful reading is that **the confidence interval is the set of parameter values that the data do not reject at the 5% level.** Values inside it are compatible with the data, and values outside it are not. A computed interval either contains the true value or does not, so in the frequentist sense it is not "95% likely" to contain it. On the correct reading, an interval of $[-0.1, 3.0]$ and one of $[-0.01, 0.02]$ are both "not significant," yet they say completely different things.

Every standard error is conditional on a model of how the sample could have differed. §4.2–§4.5 present four versions of that model.

4.2 The sandwich, derived for OLS

Substitute $Y_i = X_i'\beta + e_i$ into the OLS formula and rearrange:

$$\sqrt{n}\,(\hat{\beta} - \beta) = \Big(\frac{1}{n}\sum_{i=1}^{n} X_i X_i'\Big)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i e_i.$$

The first factor converges to Q_{XX}^{-1}. The second is a scaled sum of the moment contributions $X_i e_i$. A central limit theorem makes it approximately normal with variance $\Omega = \text{Var}\left(n^{-1/2}\sum_i X_i e_i\right)$. So

$$\text{Avar}\left(\sqrt{n}(\hat{\beta} - \beta)\right) = Q_{XX}^{-1}\,\Omega\,Q_{XX}^{-1}.$$

This is §1.6 with the OLS bread. The rest of this section concerns Ω.

- **If the observations are independent**, the cross-products between different observations have expectation zero and $\Omega = \mathbb{E}[e_i^2 X_i X_i']$.
- **If, in addition, the error variance does not depend on** X ($\mathbb{E}[e_i^2 \mid X_i] = \sigma^2$), then $\Omega = \sigma^2 Q_{XX}$. The sandwich then reduces to the classical formula $\sigma^2 Q_{XX}^{-1}$.

The classical standard error is therefore the *most* restrictive special case, not the neutral default. It assumes both independence and constant variance. It is correct only when both hold.

4.3 Heteroskedasticity

Replacing the population meat with its sample analogue gives the **heteroskedasticity-robust** variance estimator, also called the **White** estimator (White, 1980):

$$\hat{V}_{\text{HC}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{i=1}^{n} \hat{e}_i^2 X_i X_i'\Big)(\mathbf{X}'\mathbf{X})^{-1}.$$

The estimator is consistent whatever form the heteroskedasticity takes. Several small-sample variants exist. HC1 rescales by $n/(n-k)$. HC2 and HC3 divide each squared residual by $(1 - \mathbf{P}_{ii})$ and $(1 - \mathbf{P}_{ii})^2$ respectively. This undoes the

shrinkage of high-leverage residuals noted in §3.3. HC3 is close to a jackknife. It performs best in small samples with influential points (MacKinnon & White, 1985).

Three practical points follow:

- **Heteroskedasticity leaves $\hat{\beta}$ unbiased.** It distorts only the classical standard error, and it usually makes that standard error too small. This happens whenever the error variance is larger where the regressor is far from its mean. That pattern is typical, because small firms have both extreme characteristics and volatile returns.
- [Practice] **Recommendation: use robust standard errors by default.** In samples of reasonable size, they lose almost nothing when the errors happen to be homoskedastic. They remain valid when the errors are not.
- **Weighting recovers efficiency when the variance is predictable.** Weighted least squares divides each observation by its error standard deviation. Scaling returns by a volatility forecast before regressing is exactly this. It often improves the estimate more than any change of estimator does. A large gap between classical and robust standard errors also signals that the model is misspecified in some other way (King & Roberts, 2015).

4.4 Clustering

Now drop independence. Suppose observations come in G groups, such as states, industries, trading days, or firms. Suppose errors are correlated within a group and independent across groups. The task is to find how wrong the classical standard error becomes.

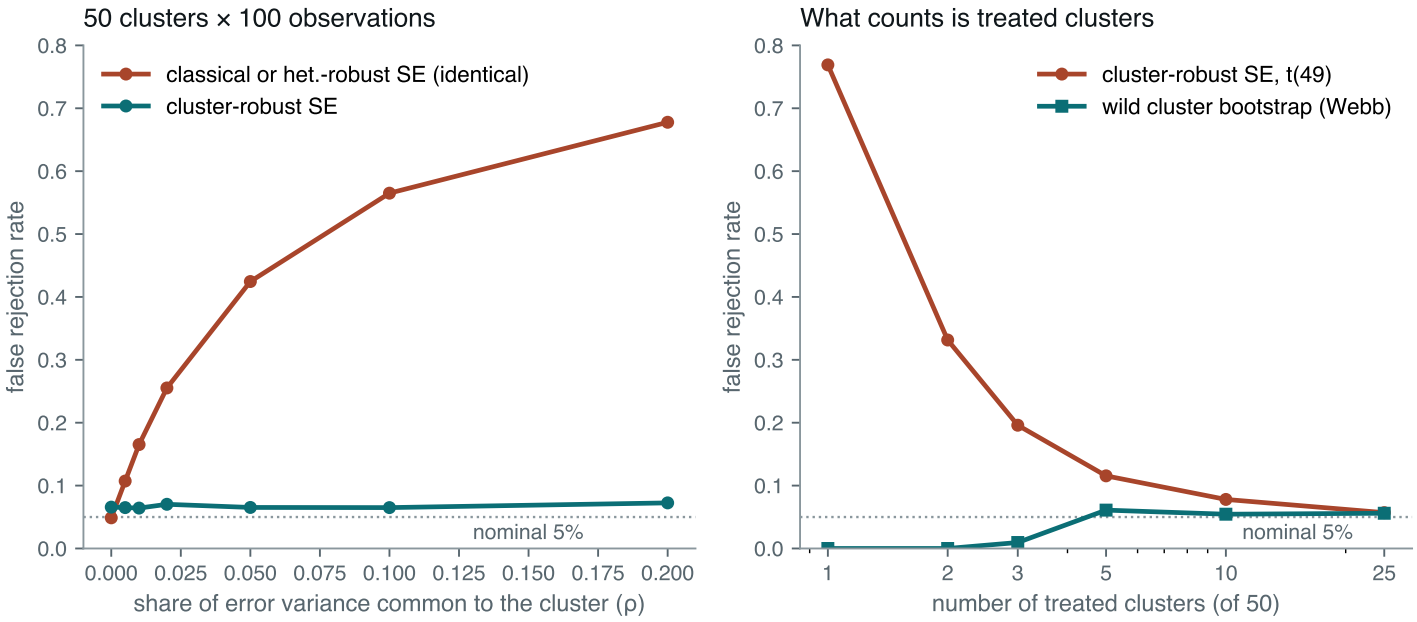
The Moulton factor. Take equal group sizes $\bar{n}_g$, an intra-cluster correlation of the error ρ_u, and an intra-cluster correlation of the regressor ρ_x. The true variance of the OLS slope then exceeds the classical formula by approximately (Moulton, 1990)

$$\frac{\mathsf{Var}_{\mathsf{true}}(\hat{\beta})}{\mathsf{Var}_{\mathsf{classical}}(\hat{\beta})} \approx 1 + (\bar{n}_g - 1)\,\rho_x\,\rho_u.$$

Read the formula term by term.

- **Both correlations must be present.** If the regressor varies independently within groups ($\rho_x = 0$), correlated errors do not change the variance. Clustering matters most when the regressor is *constant* within groups ($\rho_x = 1$). Examples are a state law, an industry shock, or a date-level event.
- **Large groups amplify small correlations.** The factor multiplies ρ_u by $\bar{n}_g - 1$. An error correlation of 0.05 is negligible with groups of three and enormous with groups of 1,000.

Take 50 groups of 100 observations, a group-level regressor, and $\rho_u = 0.05$. The factor is $1 + 99 \times 0.05 = 5.95$. The true standard error is therefore $\sqrt{5.95} = 2.44$ times the classical one. The 5,000 observations contain as much information as about 840 independent ones. A classical t-statistic of 2.4 equals about 1.0 once the analyst corrects the standard error. With groups of 1,000 instead, the same correlation gives a factor of $1 + 999 \times 0.05 = 51$. The standard errors are then 7.1 times too small. [Simulated] The simulation in the left panel below confirms the arithmetic. Classical standard errors reject a true null 42% of the time at a nominal 5%. They reject a true null 57% of the time when $\rho_u = 0.1$. Heteroskedasticity-robust errors do no better, because the problem is not heteroskedasticity.



The cluster-robust estimator allows arbitrary correlation among the errors within each group. It sums the moment contributions group by group before squaring:

$$\hat{V}_{\text{CR}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_{g=1}^{G}\mathbf{X}_g'\hat{\mathbf{e}}_g\hat{\mathbf{e}}_g'\mathbf{X}_g\Big)(\mathbf{X}'\mathbf{X})^{-1},$$

where $\mathbf{X}_g$ and $\hat{\mathbf{e}}_g$ stack the observations in group g. The usual small-sample adjustment multiplies by $\frac{G}{G-1}\cdot\frac{n-1}{n-k}$. Tests use the t distribution with $G-1$ degrees of freedom. The estimator now computes the meat from G group sums. **The effective number of observations for inference is the number of clusters, not the number of rows.**

Where to cluster. Two rules are in use, and they usually agree.

- **The design rule.** Cluster at the level where the treatment or regressor was assigned, or where the sample was drawn. If a policy varies by state, cluster by state. Abadie, Athey, Imbens & Wooldridge (2023) show that the design justifies clustering, not the presence of within-group residual correlation. They also show that clustering at a coarser level than the design requires can be unnecessarily conservative.
- **The conservative rule.** Cluster at the broadest level at which errors might be correlated, provided there are enough clusters (Cameron & Miller, 2015). Coarser clusters remove more dependence. They also leave fewer clusters from which to estimate the meat.

Few clusters, and few treated clusters. Cluster-robust inference is an asymptotic approximation in G. It degrades when clusters are few, when they are unequal in size, and above all when only a few of them are treated. A common rule of thumb calls fewer than 40 to 50 clusters few. The right panel above holds $G = 50$ and varies the number of treated clusters. [Simulated] With one treated cluster, the cluster-robust t-test rejects a true null 77% of the time. With two treated clusters it rejects 33% of the time, with five 12%, and with ten 8%. The reason is mechanical. A treated cluster's own dummy-like variation lets OLS fit that cluster's mean almost exactly. This forces that cluster's residual sum toward zero. That residual sum is precisely the piece of the meat that measures the treatment's noise. The **wild cluster bootstrap** is the standard remedy. It flips the signs of whole clusters' residuals at random and re-estimates (Cameron, Gelbach & Miller, 2008). As the panel shows, it errs in the opposite direction with one or two treated clusters, where it almost never rejects. It becomes reliable from about five (MacKinnon & Webb, 2017; MacKinnon, Nielsen & Webb, 2023). With a single treated unit, cluster-based inference is not available at all, so the design has to change. The alternatives are randomisation inference (§7.2), synthetic control (§7.7), or the approach of Conley & Taber (2011).

Two dimensions at once. Finance panels often have errors correlated both within firms over time and across firms at a date. **Two-way clustering** adds the firm-clustered and date-clustered meats. It subtracts the heteroskedasticity-robust meat that both double-count (Cameron, Gelbach & Miller, 2011; Thompson, 2011). It needs many clusters in *both* dimensions. So §9.5 recommends it for panels of thousands of firms over decades, and not for ten years of annual data.

The canonical warning. [Fact] Bertrand, Duflo & Mullainathan (2004) took about 20 years of state-level data on women's wages from the Current Population Survey. They assigned fake "laws" to randomly chosen states and years. They then ran

the standard difference-in-differences regression with conventional standard errors. The placebo laws were significant at 5% in up to 45% of simulations. Outcomes that are serially correlated within a state make the effective sample the number of states, not the number of state-years. Clustering by state fixes the problem when there are enough states.

4.5 Serial correlation and HAC standard errors

A time series is one ordered cluster. The moment contributions $\psi_t = X_t e_t$ are correlated with their neighbours. The meat is the **long-run variance**

$$\Omega = \sum_{j=-\infty}^{\infty} \Gamma_j, \qquad \Gamma_j = \mathbb{E}[\psi_t \psi_{t-j}'],$$

the sum of all the autocovariances of the moment contributions. It is the multivariate version of the variance inflation in §2.5.

The Newey–West estimator truncates the sum at a lag L and down-weights distant lags linearly:

$$\hat{\Omega}_{\text{NW}} = \hat{\Gamma}_0 + \sum_{j=1}^{L} \Big(1 - \frac{j}{L+1}\Big)(\hat{\Gamma}_j + \hat{\Gamma}_j').$$

The linear **Bartlett weights** guarantee that the estimate is positive semi-definite, so that no variance comes out negative. The unweighted sum does not give this guarantee (Newey & West, 1987). Such estimators are called **HAC**, for heteroskedasticity and autocorrelation consistent.

The bandwidth is a bias–variance choice. Too small an L misses autocorrelation and understates the variance. Too large an L makes the variance estimate noisy. This fattens the tails of the t-statistic. The textbook rule $L = 0.75\,T^{1/3}$ tends to err by choosing too small a bandwidth. Lazarus, Lewis, Stock & Watson (2018) simulate a regression with $T = 200$ and a moderately persistent regressor and error. [Fact] They find that this rule with normal critical values rejects a true null 9% to 18% of the time at a nominal 5%, depending on the persistence. They recommend a much larger bandwidth, $L = 1.3\,T^{1/2}$, combined with **fixed-b** critical values. These critical values account for the noise in the variance estimate. An alternative is an equal-weighted cosine estimator with $\nu = 0.4\,T^{2/3}$ terms and Student-t critical values with ν degrees of freedom. Both cut the distortions roughly in half.

HAC inference fails most often in finance with **overlapping observations**. In a regression of h-month returns on a predictor using monthly data, consecutive dependent variables share $h-1$ months. Under the null of no predictability, the error is a moving average of order $h-1$, so Newey–West with at least $h-1$ lags is consistent in principle. In a sample of realistic length, the test is far from the nominal size. [Simulated] With 40 years of monthly data and an exogenous, persistent predictor, Newey–West with h lags rejects a true null 15% of the time at $h = 12$. It rejects 28% of the time at $h = 60$. Classical standard errors reject 57% and 78% of the time. Hodrick (1992) proposed standard errors that impose the null of no predictability when estimating the meat. These keep the rejection rate at about 5% (§9.2).

The underlying rule comes from §2.5: **a long-horizon regression has about T/h independent observations, and no standard-error formula can create information that the data do not contain.** A formula can only fail to account for the missing information.

4.6 Tests, p-values, and confidence intervals

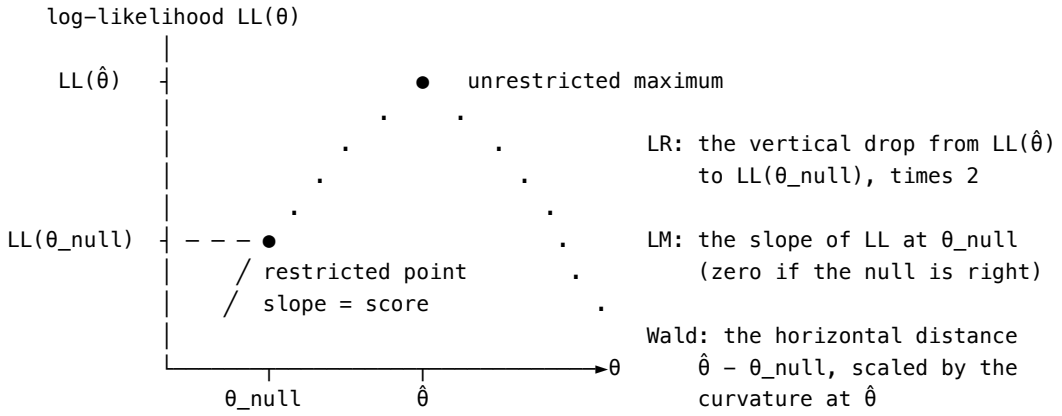
A **hypothesis test** is a rule for rejecting a null hypothesis. Its **size** is the probability of rejecting when the null is true. Its **power** is the probability of rejecting when a given alternative is true. Every section so far has aimed at keeping the actual size close to the nominal one.

The **p-value** is the probability of a test statistic at least as extreme as the one observed, computed as if the null hypothesis were true. The computation implicitly assumes the model. It also assumes that this was the only test that would be run. The American Statistical Association's statement on p-values (Wasserstein & Lazar, 2016) lists the misreadings to avoid:

- A p-value is **not** the probability that the null hypothesis is true.
- A p-value is **not** the probability that the result "is due to chance."
- A p-value does **not** measure the size or importance of an effect.
- A threshold such as 0.05 does **not** separate real effects from spurious ones.

Confidence intervals invert tests. The 95% interval is the set of null values that a 5% test would not reject (§4.1). Reporting the interval instead of the p-value shows the magnitude.

Three ways to test a restriction. A model estimated by maximum likelihood has three classical tests of a restriction such as $\theta = \theta_{\text{null}}$. They measure the same discrepancy along different axes of the log-likelihood curve. The diagram writes this curve as $LL(\theta)$.



- The **Wald** test estimates only the unrestricted model and asks whether $\widehat{\theta}$ is far from θ_{null} in standard-error units. Every t-test in a regression table is a Wald test.
- The **likelihood ratio (LR)** test estimates both models and compares their fit.
- The **Lagrange multiplier (LM)** or **score** test estimates only the restricted model and asks whether the log-likelihood is still rising there.

Under the null and a correctly specified likelihood, the three tests are asymptotically equivalent. In finite samples they differ. The Wald test has one known weakness. It is not invariant to how the restriction is written, so testing $\beta = 1$ and testing $1/\beta = 1$ can give different answers. The LM form is convenient when the unrestricted model is hard to estimate. Many specification tests are therefore LM tests. With one restriction, the F-statistic is the square of the t-statistic.

4.7 Power, and the winner's curse

A test with low power misses real effects. **When it does find one, it exaggerates it.** Estimates that pass a significance threshold are disproportionately those that overshot the true effect.

A two-sided 5% test has 80% power when the true effect is about $1.96 + 0.84 = 2.8$ standard errors. That is the **minimum detectable effect**. Compute it *before* an analysis, not after. The right panel of the figure in §4.8 plots the exaggeration of statistically significant estimates against power. [Simulated] At 80% power, significant estimates overstate the true effect by 12% on average. At 50% power they overstate it by 41%, and at 18% power by a factor of 2.4. At very low power, a meaningful share of significant estimates even have the wrong sign. The share is about one in five at 6% power. The framework of "Type M" (magnitude) and "Type S" (sign) errors comes from Gelman & Carlin (2014).

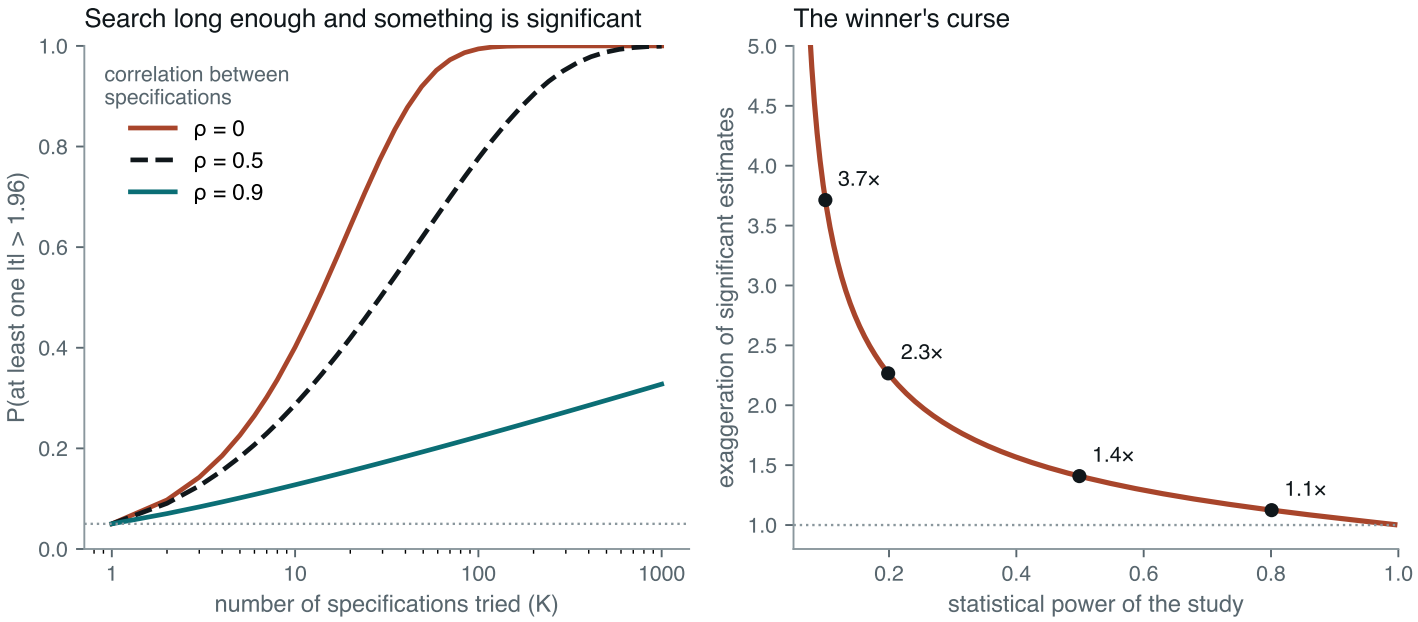
Low power is normal. Ioannidis, Stanley & Doucouliagos (2017) surveyed 159 empirical economics literatures containing about 64,000 estimates. They put median statistical power at 18% or less. They estimate that nearly 80% of reported effects are exaggerated, typically by a factor of two. [Contested] The magnitudes depend on their meta-analytic estimate of each literature's true effect, which is itself uncertain. The direction of the conclusion is not in serious dispute.

The finance version. Suppose a strategy has a true annual Sharpe ratio of 0.3 and is backtested over ten years. Its expected t-statistic is $0.3 \times \sqrt{10} = 0.95$. The power of a 5% test is about 16%. If the backtest is nonetheless significant, the Sharpe ratio it reports averages about 0.8, roughly 2.6 times the true value. The strategies whose backtests look good enough to deploy are systematically the lucky ones. This is one reason live performance disappoints even when nothing is wrong with the strategy.

4.8 Multiple testing and the garden of forking paths

A 5% test rejects a true null 5% of the time. Across 20 independent tests of true nulls, the chance that at least one rejects is $1 - 0.95^{20} = 64\%$.

Real specification searches test specifications that depend on each other. Specifications that share data and differ in a control or a sample period produce correlated test statistics. Correlation slows the growth of false positives with the number of tries but does not stop it. [Simulated] With a correlation of 0.5 between every pair of specifications, 20 tries find "significance" 42% of the time, and 100 tries find it 78% of the time. With a correlation of 0.9, the figures are 16% and 22%. The expected largest $|t|$ among K independent null statistics is about 1.9 for $K = 10$, 2.8 for $K = 100$, and 3.4 for $K = 1{,}000$.



Explicit multiplicity has standard corrections. They control different error rates, as the table shows.

Target	What it controls	Standard procedure	When it fits
Family-wise error rate	The probability of *any* false rejection	Bonferroni (test at $5\%/K$), or Holm's step-down version	A few tests, where any false claim is costly
False discovery rate	The expected *share* of rejections that are false	Benjamini & Hochberg (1995)	Screening many candidates, where some false positives are tolerable
Best of many strategies	Whether the best performer beats a benchmark by more than luck	White's reality check; Hansen's SPA test; Romano–Wolf stepdown	Backtests and forecast comparisons (§9.6)

For the cross-section of stock returns, Harvey, Liu & Zhu (2016) catalogue several hundred published return predictors. They argue that a new predictor should reach a t-statistic of about 3.0 rather than 2.0, given the search that produced the published predictors.

Implicit multiplicity is the harder problem. Gelman & Loken (2014) call it the **garden of forking paths**. A researcher who never runs more than one regression can still face a multiple-testing problem. The researcher chose the sample period, outlier rule, control set, horizon, and functional form after seeing the data. The researcher would have chosen differently had the data been different. A p-value is valid only if the analysis would have been the same for any data set. That premise is almost never true.

The remedies share one idea. Separate the choices from the data used to evaluate them.

- **Pre-specify** the analysis before seeing the outcome data, formally in a pre-analysis plan or informally in a dated note.
- **Hold out data** that the analysis uses only once, at the end.
- **Report every specification tried**, for example as a specification curve (§11.4), so a reader can see the whole distribution rather than the chosen tail.
- **Raise the threshold** in proportion to the search, as with the $t > 3$ hurdle or the deflated Sharpe ratio (Portfolio Construction, Appendix A.46).
- **Label exploratory findings as hypotheses** to be tested on new data.

4.9 The bootstrap

The **bootstrap** approximates an estimator's sampling distribution by treating the sample as if it were the population. It draws many samples from the data with replacement, recomputes the estimate in each, and uses the spread of the recomputed estimates (Efron, 1979). It replaces analytical derivations with computation. For complicated estimators it is often the only practical way to obtain a standard error.

The one rule: resample the independent units. The bootstrap is valid only if its assumption about which units are independent is right. The meat makes the same assumption. The table matches the resampling scheme to the dependence.

Dependence	Resample	Name
Independent observations	Individual observations, with Y and X together	Pairs bootstrap
Clusters	Whole clusters	Cluster (block) bootstrap
Independent observations, heteroskedastic errors	Flip residual signs at random, keeping X fixed	Wild bootstrap
Few clusters	Flip the signs of whole clusters' residuals	Wild cluster bootstrap (§4.4)
Time series	Contiguous blocks of observations	Moving block (Künsch, 1989) or stationary bootstrap (Politis & Romano, 1994)

Two refinements matter in practice. Bootstrapping the t-statistic (the **percentile-t** method) is more accurate than bootstrapping the estimate itself, because the distribution of the t-statistic depends less on unknown parameters. For time series, a block has to be longer than the dependence it preserves. Volatility clustering in daily returns decays over weeks to months, so blocks of a few days remove it.

When the bootstrap fails. The bootstrap fails in four situations.

- **The resampling scheme destroys the relevant dependence.** An independent bootstrap of autocorrelated or volatility-clustered returns understates uncertainty in the same way classical standard errors do.
- **The statistic is an extreme.** The maximum Sharpe ratio across K strategies is not a smooth function of means, and a naive bootstrap of it is inconsistent. White's reality check exists to handle this case.
- **The parameter is at or near a boundary**, as with unit roots or variances near zero.
- **The sample has too few independent units.** A few clusters or independent periods leave too little to resample.

4.10 Statistical significance and economic significance

Statistical significance measures how clearly an estimate differs from zero relative to its noise. Economic significance measures whether the effect is large enough to matter. The two measures are independent. McCloskey & Ziliak (1996) found that 70% of the full-length empirical papers in the *American Economic Review* in the 1980s did not distinguish between them.

The practical consequences:

- **Report magnitudes in meaningful units**: the effect of a one-standard-deviation change, annualised returns, basis points, or dollars. Put the confidence interval next to the point estimate.
- **A precisely estimated zero is a finding.** An interval of $[-0.02, 0.01]$ rules out every economically meaningful effect. An interval of $[-3, 5]$ rules out nothing, and the label "not significant" does not distinguish between the two.
- **With enough data, everything is significant.** With ten million observations, standard errors are tiny, and trivial effects clear any threshold. The questions become whether the effect is large enough to act on, and whether it survives the biases that no sample size removes (§3.2).

§4 Key takeaways

1. A confidence interval is the set of values the data do not reject. Read the interval, not the p-value.
2. The classical standard error is the most restrictive special case of the sandwich. It assumes independence and constant variance. It is correct only when both hold.
3. Heteroskedasticity leaves coefficients unbiased and distorts classical standard errors. Use robust standard errors by default.
4. Clustering matters when both the regressor and the error are correlated within groups. Large groups amplify small correlations. With 50 groups of 100 and an error correlation of 0.05, classical standard errors are 2.4 times too small.
5. The effective sample size for inference is the number of clusters, and with a binary treatment it is the number of treated clusters. With one or two treated clusters, conventional cluster-robust inference fails outright.
6. Newey–West with textbook bandwidths is oversized in realistic samples. Use larger bandwidths with fixed-b critical values, and do not trust any HAC correction to fix long overlapping horizons.

7. Low power misses effects and exaggerates the ones it finds. Compute the minimum detectable effect, about 2.8 standard errors, before looking at results.
8. Specification search inflates false positives even when the specifications are correlated. It does so even when no one runs more than one regression. Separate analysis choices from the data used to evaluate them.
9. Bootstrap the independent units. Resampling observations from dependent data reproduces the classical mistake.

5. One estimator family

Textbooks present OLS, instrumental variables, GMM, and maximum likelihood as four separate techniques, each with its own chapter. This section presents them as one technique, the master form of §1.5, with four ways of choosing the moment function. The unified view shows how the estimators relate to each other, gives every standard error from one formula, and shows which choice matters. That choice is the moment condition, not the estimator.

5.1 The method of moments

The oldest estimation idea is also the most general. A model implies that certain population averages take certain values. The method of moments replaces population averages with sample averages and solves for the parameters.

The simplest case is a mean and a variance. The model says $\mathbb{E}[Y_i - \mu] = 0$ and $\mathbb{E}[(Y_i - \mu)^2 - \sigma^2] = 0$. Setting the sample versions to zero gives $\hat{\mu} = \bar{Y}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_i (Y_i - \bar{Y})^2$.

OLS applies the same step to the projection's first-order condition, $\mathbb{E}[X_i(Y_i - X_i'\beta)] = 0$. When the model has as many moment conditions as parameters, it is **just-identified**. The sample equations then have an exact solution. The asymptotic variance is the sandwich $Q^{-1}\Omega Q^{-1\prime}$ from §1.6.

5.2 Instrumental variables as a moment condition

Suppose the structural equation is $Y_i = X_i'\beta + u_i$. Suppose some regressors are correlated with u_i, so that OLS converges to the wrong quantity (§3.2). Suppose also that a vector of **instruments** Z_i satisfies two conditions.

- **Relevance:** $\mathbb{E}[Z_i X_i']$ has full column rank. The instruments move the endogenous regressors.
- **Exogeneity:** $\mathbb{E}[Z_i u_i] = 0$. The instruments are uncorrelated with everything else that affects the outcome. This includes the **exclusion restriction**, the requirement that Z affects Y only through X.

The second condition is a moment condition, $\mathbb{E}[Z_i(Y_i - X_i'\beta)] = 0$. With as many instruments as regressors, its sample analogue solves to

$$\hat{\beta}_{\text{IV}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{Y}.$$

With one regressor, one instrument, and an intercept, this is a ratio of two regression slopes:

$$\hat{\beta}_{\text{IV}} = \frac{\widehat{\text{Cov}}(Z, Y)}{\widehat{\text{Cov}}(Z, X)} = \frac{\widehat{\text{Cov}}(Z, Y)/\widehat{\text{Var}}(Z)}{\widehat{\text{Cov}}(Z, X)/\widehat{\text{Var}}(Z)} = \frac{\text{reduced form: effect of } Z \text{ on } Y}{\text{first stage: effect of } Z \text{ on } X}.$$

The intuition: IV uses only the part of the variation in X that Z drives. It converts the effect of Z on Y into an effect of X on Y by dividing by how much Z moved X.

When Z is binary, this is the **Wald estimator**. It is the difference in average outcomes between the $Z = 1$ and $Z = 0$ groups, divided by the difference in their average treatment. The canonical example is Angrist (1990). The Vietnam-era draft lottery assigned men eligibility for conscription at random by birth date. Eligibility raised the probability of military service. Eligible men later earned less. The ratio of the earnings gap to the service gap estimates the effect of service itself. In the early 1980s, white veterans earned roughly 15% less than comparable non-veterans. Nobody chose a lottery number, which is the argument for exogeneity. The exclusion restriction is the assumption that the lottery affected earnings only by changing who served.

Two-stage least squares. With more instruments than endogenous regressors, 2SLS combines the instruments optimally under homoskedasticity. It projects X on Z and uses the fitted values:

$$\hat{\beta}_{\text{2SLS}} = (\mathbf{X}'\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_Z\mathbf{Y}, \qquad \mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'.$$

[Practice] **Never run the second stage by hand as an OLS regression of Y on $\hat{X}$ and report its standard errors.** The point estimate is right. The residuals are computed with $\hat{X}$ instead of X, so the standard errors are wrong. Use a routine that computes 2SLS as one estimator.

IV is consistent but biased in finite samples. The bias is toward the OLS estimate, and it grows as the first stage weakens. §7.3 covers weak or invalid instruments, which cause most of the practical difficulty.

5.3 GMM: more moments than parameters

With ℓ moment conditions and $k < \ell$ parameters, the model is **over-identified**. The sample moments cannot all be set to zero at once. The **generalised method of moments** (GMM) instead makes them jointly as small as possible in a weighted sense (Hansen, 1982):

$$\hat{\theta}_{\text{GMM}} = \arg\min_{\theta} \; \bar{\psi}(\theta)' \, W \, \bar{\psi}(\theta), \qquad \bar{\psi}(\theta) = \frac{1}{n} \sum_{i=1}^{n} \psi_i(\theta).$$

Any positive definite weighting matrix W gives a consistent estimator. The general asymptotic variance is

$$\text{Avar}\left(\sqrt{n}(\hat{\theta} - \theta_0)\right) = (Q'WQ)^{-1} \, Q'W \, \Omega \, WQ \, (Q'WQ)^{-1},$$

the same sandwich with the weighting matrix included in the bread. It is smallest when $W = \Omega^{-1}$, which gives $(Q'\Omega^{-1}Q)^{-1}$. **The efficient weighting matrix puts more weight on the moments that are estimated more precisely.** In practice, efficient GMM takes two steps. Estimate with a simple W, use the residuals to estimate Ω, and re-estimate with $\hat{\Omega}^{-1}$. 2SLS is GMM with $W = (\mathbf{Z}'\mathbf{Z}/n)^{-1}$. This choice is efficient when the errors are homoskedastic.

The J-test. Over-identification gives a test. If all ℓ moment conditions are valid, the minimised objective, scaled by n and computed with the efficient weighting matrix,

$$J = n \, \bar{\psi}(\hat{\theta})' \, \hat{\Omega}^{-1} \, \bar{\psi}(\hat{\theta}) \xrightarrow{d} \chi^2_{\ell - k},$$

is chi-squared with $\ell - k$ degrees of freedom. A large J indicates that the moment conditions cannot all hold at once.

- **The test can detect instruments that disagree with each other.** If one instrument implies $\beta = 0.1$ and another implies $\beta = 0.5$, they cannot both be valid.
- **The test cannot detect instruments that are wrong in the same way.** If every instrument is correlated with the error in the same direction, they agree on the same wrong answer and J is small.
- **In a just-identified model, the test statistic is identically zero.** Nothing remains to test. This repeats §1.5's point that the orthogonality condition cannot be tested when it exactly identifies the model.

A passed J-test is weak evidence of validity. A failed one is strong evidence of a problem.

GMM in asset pricing. GMM was built for moment conditions that come from economic theory, and asset pricing is its main application. An investor's first-order condition says that, for any gross return R_{t+1},

$$\mathbb{E}\big[M_{t+1}(\theta) \, R_{t+1} - 1 \,\big|\, \mathcal{F}_t\big] = 0, \qquad M_{t+1}(\theta) = d \left(\frac{C_{t+1}}{C_t}\right)^{-\eta},$$

where the **stochastic discount factor** M_{t+1} in the power-utility example depends on consumption growth, a subjective discount factor d, and relative risk aversion η. The condition holds conditional on information at t. Any variable Z_t known at t therefore gives an unconditional moment condition $\mathbb{E}[Z_t(M_{t+1}(\theta)R_{t+1} - 1)] = 0$. Hansen & Singleton (1982) estimated d and η this way. Almost all of empirical asset pricing can be written in this form. This form is the organising theme of Cochrane's *Asset Pricing* (2005).

Two practical warnings.

- [Fact] **Efficient GMM with many moments performs badly in finite samples.** The estimated weighting matrix is noisy. Its inverse overweights moments that happen to look precise in the sample. Altonji & Segal (1996) found large small-sample biases in efficient GMM for covariance structures. Equally weighted moments performed better. Iterated and continuously updated GMM help somewhat (Hansen, Heaton & Yaron, 1996).
- [Practice] **Statistical efficiency does not imply economic relevance.** The efficient weighting matrix may concentrate on portfolio combinations that are estimated precisely but are economically uninteresting, such as extreme long–short positions. Asset-pricing tests therefore often use a pre-specified weighting matrix, such as equal weights on test portfolios. This lets the test judge the model on the returns that matter to an investor (Hansen & Jagannathan, 1997).

5.4 Maximum likelihood, and quasi-likelihood

Suppose the model specifies the full conditional distribution of the data, $f(Y_i \mid X_i; \theta)$. The **maximum likelihood estimator** (MLE) then chooses the parameter under which the observed data were most probable:

$$\hat{\theta}_{\mathsf{ML}} = \arg\max_{\theta} \sum_{i=1}^{n} \log f(Y_i \mid X_i; \theta).$$

Its first-order condition sets the sample average of the **score** $s_i(\theta) = \partial \log f(Y_i \mid X_i; \theta)/\partial\theta$ to zero. So MLE is a method-of-moments estimator with $\psi_i = s_i$. The population moment condition $\mathbb{E}[s_i(\theta_0)] = 0$ holds whenever the model is correct. Differentiating $\int f(y \mid x; \theta)\, dy = 1$ with respect to θ gives $\int \partial f/\partial\theta \, dy = 0$. That integral is the expected score.

Why MLE is efficient, and when that stops being true. Under correct specification, the **information matrix equality** holds. The variance of the score equals minus the expected Hessian of the log-likelihood,

$$\mathbb{E}\big[s_i(\theta_0) s_i(\theta_0)'\big] = -\,\mathbb{E}\Big[\frac{\partial s_i(\theta_0)}{\partial \theta'}\Big] \equiv \mathcal{I}(\theta_0).$$

In sandwich terms, the meat equals minus the bread. Then $Q^{-1}\Omega Q^{-1\prime}$ reduces to $\mathcal{I}(\theta_0)^{-1}$, the inverse **Fisher information**. This is the Cramér–Rao lower bound, so no regular consistent estimator is more precise. For the linear model with normal errors, the MLE of β is exactly OLS.

The equality follows from the model being right. When the density is misspecified, the MLE converges to a **pseudo-true** value. This is the parameter whose implied distribution is closest to the truth in Kullback–Leibler divergence. The two sides of the equality then no longer match (White, 1982). The estimator is then a **quasi-maximum likelihood estimator** (QMLE). Its standard errors must use the full sandwich, with the Hessian for the bread and the outer product of the scores for the meat.

Some quasi-likelihoods are robust, and some are not. This is the practical content of QMLE theory.

- **Robust.** A Gaussian likelihood for a GARCH model gives consistent estimates of the mean and variance parameters even when returns are fat-tailed. This requires that the conditional mean and variance equations are right (Bollerslev & Wooldridge, 1992). The Poisson likelihood gives consistent estimates of an exponential conditional mean for any non-negative outcome, count or not. It also handles zeros that a log-linear regression cannot (Santos Silva & Tenreyro, 2006).
- **Not robust.** Probit and logit coefficients are inconsistent when the latent error is heteroskedastic or has the wrong distribution, although average marginal effects are often approximately right. Censored-regression (Tobit) estimates depend heavily on normality.

When to use likelihood. Likelihood is the right method in three situations. The first is when the distribution itself is the object of interest, as with volatility, tail risk, durations, and default probabilities. The second is when latent states must be filtered, as in the regime-switching models of Market Regimes and Hidden Markov Models. The third is when efficiency in a small sample is worth the risk of misspecification.

5.5 M-estimators beyond the mean

Everything so far targets conditional *means*. The same approach applies to any estimator defined by minimising a sum. Such estimators are known collectively as **M-estimators**.

Quantile regression (Koenker & Bassett, 1978) estimates how the conditional q-quantile of Y depends on X. It minimises $\sum_i \rho_q(Y_i - X_i'\beta)$ with the asymmetric "check" loss $\rho_q(v) = v\,(q - \mathbb{1}\{v < 0\})$. The median regression, $q = 0.5$, is least absolute deviations. Quantile regression is the natural method for questions about tails. The conditional 5% quantile of a portfolio's return is a value-at-risk model (Engle & Manganelli, 2004). It is also robust to outliers in Y, though not in X. A quantile coefficient describes how a quantile of the *distribution* shifts. It does not measure the effect on the particular unit that sits at that quantile, unless units keep their ranks.

Robust regression (Huber, 1964) replaces squared loss with a loss that grows only linearly for large residuals, so a few extreme observations cannot dominate the fit. It changes the estimand as well as the estimator. In finance the extreme observations are often the economically important ones, so being "robust to the crash" is a modelling decision, not a technical fix.

5.6 Shrinkage and the Bayesian reading

Every estimator so far is unbiased or consistent. §2.3 noted that a biased estimator can have a lower mean squared error. Shrinkage estimators make that trade deliberately.

Ridge regression adds a penalty on the size of the coefficients:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{b} \sum_{i=1}^{n} (Y_i - X_i' b)^2 + \kappa \|b\|^2 = (\mathbf{X}'\mathbf{X} + \kappa \mathbf{I})^{-1} \mathbf{X}'\mathbf{Y}.$$

The penalty shrinks the coefficients toward zero, adds bias, and reduces variance. It helps most when the model has many regressors, each with a small effect, and the regressors are correlated. Replacing $\|b\|^2$ with $\sum_j |b_j|$ gives the **LASSO**, which also sets some coefficients exactly to zero (§10.4).

The Bayesian reading. Suppose the errors are normal with variance σ^2. Suppose also that the prior is that each coefficient is normal around zero with variance τ^2. The posterior mean of β is then exactly the ridge estimator with $\kappa = \sigma^2/\tau^2$. The LASSO is the posterior mode under a Laplace (double-exponential) prior. **A penalty is a prior, and a prior is a penalty.** Choosing κ by cross-validation sets how strongly the analyst assumes that effects are small.

Shrinkage can provably dominate the obvious estimator. [Fact] Suppose three or more means are estimated at once. Shrinking all the sample means toward a common value then gives a lower total mean squared error than the sample means themselves. This holds whatever the true means are. This is Stein's paradox. Efron & Morris (1977) explain it accessibly. In that setting, unbiasedness is not a virtue.

Shrinkage is useful in econometrics in three recurring settings.

- **Many related effects.** Shrink estimates of skill for thousands of fund managers, or of accuracy for thousands of analysts, toward the group average in proportion to their noise. The raw extremes are mostly luck (empirical Bayes).
- **Forecasting with many parameters.** Vector autoregressions with a "Minnesota" prior that shrinks each variable toward a random walk forecast macroeconomic series far better than unrestricted ones (Litterman, 1986).
- **Covariance matrices.** The shrinkage estimators of Portfolio Construction are the same idea applied to Σ.

When the data are informative, Bayesian and frequentist answers converge. In regular models with large samples, the posterior is approximately normal and centred on the MLE, with the sandwich-free variance $\mathcal{J}^{-1}/n$. The answers diverge exactly where the data are weak. This happens in small samples, under weak identification, or with many parameters. There the prior determines the result, so report and defend it like any other assumption.

5.7 Choosing among them

The table compares the estimators of this section on the same attributes.

Estimator	Key assumption	Robust to	Typical use	Typical failure
OLS	Regressors uncorrelated with the error	Non-normality; heteroskedasticity and dependence, with the right meat	Linear conditional means	Endogeneity
IV / 2SLS	Valid, relevant instruments	Endogeneity of the regressor	Causal effects from natural experiments	Weak or invalid instruments (§7.3)
Efficient GMM	Valid moment conditions	The distributional form	Many theory-based moments	Finite-sample bias with many moments
Maximum likelihood	The full distribution is right	Nothing beyond the model	Volatility, discrete outcomes, latent states	Inconsistency under misspecification
Quasi-ML	Only the parts of the model that matter	A wrong distribution, in robust cases	GARCH; count-type outcomes	Invalid standard errors if the sandwich is skipped
Quantile regression	A linear conditional quantile	Outliers in Y	Tails, value-at-risk, heterogeneity	Misreading distributional shifts as individual effects

Estimator	Key assumption	Robust to	Typical use	Typical failure
Ridge, LASSO, Bayes	The prior or penalty is sensible	Overfitting	Prediction with many regressors	Biased coefficients; invalid naive inference (§10.3)

The ordering from §1.3 still holds. An analyst makes three choices: the orthogonality condition, the meat, and the estimator. [Practice] The estimator is usually the least consequential of the three. A well-identified parameter estimated by plain OLS with the right standard errors beats a badly identified one estimated by an elegant method. §12.2 collects the exact algebraic relationships among the estimators in this section.

§5 Key takeaways

1. OLS, IV, GMM, and maximum likelihood are one estimator. Each sets a sample average of a moment function to zero, and each gets its standard error from the same sandwich.
2. IV is a ratio: the effect of the instrument on the outcome divided by its effect on the regressor. It uses only the variation in the regressor that the instrument drives.
3. Never compute 2SLS standard errors by running the second stage by hand.
4. Over-identification gives a J-test. The test catches instruments that disagree. It cannot catch instruments that are wrong in the same way.
5. Efficient GMM performs poorly in finite samples with many moments. Pre-specified weights are often more reliable and more economically meaningful.
6. MLE is efficient because, under correct specification, the meat equals minus the bread. Under misspecification, use the sandwich, and know which quasi-likelihoods remain consistent.
7. A penalty is a prior. Shrinkage trades bias for variance, and when many related quantities are estimated at once, it provably beats the unbiased estimates.
8. The estimator is the least consequential of the three choices. Get the orthogonality condition and the meat right first.

Part III — Identification

6. Endogeneity: why a coefficient is not an effect

§3.2 stated the central problem of the field in one line. A regression estimates β_s plus a bias term. The bias term is zero only when the regressors are uncorrelated with the structural error. This section analyses that bias term. It explains what the term means in causal language, the five mechanisms that produce it, and randomisation, the one idea that removes it by construction. §7 then surveys the designs that try to approximate randomisation when nobody randomised.

6.1 Potential outcomes

The **potential outcomes** framework is the cleanest language for causal questions. Neyman proposed it, and Rubin (1974) developed it. For each unit i and a binary treatment D_i, imagine two outcomes. The first, $Y_i(1)$, is what would happen if the unit were treated. The second, $Y_i(0)$, is what would happen if it were not. The unit's causal effect is the difference, $\tau_i = Y_i(1) - Y_i(0)$. The observed outcome is the potential outcome that matches the unit's treatment status:

$$Y_i = D_i\, Y_i(1) + (1 - D_i)\, Y_i(0).$$

The fundamental problem of causal inference is that τ_i is never observed, because no unit is both treated and untreated at the same moment (Holland, 1986). Every causal estimate is therefore a statement about *averages*. The analyst must choose which average to estimate. The table lists the standard choices.

Estimand	Definition	Answers
Average treatment effect (ATE)	$\mathbb{E}[\tau_i]$	What if everyone were treated, versus no one?
Average effect on the treated (ATT)	$\mathbb{E}[\tau_i \mid D_i = 1]$	What did the treatment do for those who got it?
Average effect on the untreated (ATU)	$\mathbb{E}[\tau_i \mid D_i = 0]$	What would extending it to the rest do?
Conditional average effect (CATE)	$\mathbb{E}[\tau_i \mid X_i = x]$	How does the effect vary with characteristics?
Local average effect (LATE)	The effect for units whose treatment an instrument changes	What does IV actually estimate? (§7.3)

When effects are the same for everyone, these averages coincide. When effects differ, the averages differ too. Differing effects are the normal case. A method that is unbiased for one average is generally biased for another. State the estimand before choosing the design (§1.3).

Index inclusion is an example. For a stock added to the S&P 500, $Y_i(1)$ is its return over the following month. $Y_i(0)$ is the return it would have had if it had not been added. The ATT is the average inclusion effect on stocks that were added. It is the natural estimand. It differs from the effect that adding a randomly chosen stock would have.

SUTVA. The framework assumes the **stable unit treatment value assumption**. Under it, a unit's potential outcomes do not depend on other units' treatments, and the treatment has only one version. The first half fails frequently in finance. The return to a trading strategy depends on how many others trade it. Adding one stock to an index moves the prices of its close substitutes. A regulation that constrains some banks pushes activity toward the other banks. This contaminates the control group that the design uses to measure the regulation's effect. When interference occurs, a design estimates a contrast between treated and control units *in the presence of spillovers*. That contrast may be very different from the effect of treating everyone.

6.2 The selection bias decomposition

Consider what a simple comparison of treated and untreated units estimates. Adding and subtracting the treated units' untreated potential outcome gives

$$\underbrace{\mathbb{E}[Y_i \mid D_i = 1] - \mathbb{E}[Y_i \mid D_i = 0]}_{\text{observed difference}} = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i = 1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y_i(0) \mid D_i = 1] - \mathbb{E}[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

The selection-bias term compares the *untreated* outcomes of the two groups. It measures how much the units that got treated would have differed from the others even without treatment. It is the potential-outcomes version of the omitted-variable term of §3.5.

Consider share repurchases again. Suppose managers buy back stock when they believe it is undervalued. Then repurchasing firms would have earned higher returns than other firms *even had they not repurchased*, so $\mathbb{E}[Y_i(0) \mid D_i = 1] > \mathbb{E}[Y_i(0) \mid D_i = 0]$. The selection bias is positive, and the naive return difference overstates whatever the buyback itself contributes. Selection bias can also be negative. People who go to hospital are less healthy afterwards than people who do not, because the untreated health of those who seek treatment is worse. In a naive comparison, that difference can overwhelm any benefit of the treatment.

6.3 Why randomisation works

If treatment is assigned at random, it is independent of the potential outcomes: $D_i \perp (Y_i(1), Y_i(0))$. The untreated potential outcome then has the same distribution in both groups. The selection-bias term is therefore exactly zero. The simple difference in means estimates the ATT, which equals the ATE. In regression language, a randomly assigned regressor is uncorrelated with the structural error by construction, so the endogeneity term of §3.2 vanishes.

Randomisation balances *unobserved* characteristics as well as observed ones, in expectation. No other method does this. The randomised experiment is therefore the reference point for every design in §7. Each design argues that some feature of the world assigned treatment "as good as randomly" for some units. [Practice] **Ask of any observational study what experiment it is trying to approximate and where the approximation breaks.**

Finance has a few genuine experiments. The best known is the SEC's **Regulation SHO pilot**. The SEC sorted the stocks in the 2004 Russell 3000 by listing market and trading volume. It designated every third one a pilot stock, exempt from short-sale price tests from May 2005 to August 2007 (SEC, 2007). Dozens of papers have used the pilot to estimate the effects of short-selling constraints on outcomes from liquidity to earnings management (Fang, Huang & Karpoff, 2016). The pilot's heavy reuse taught a further lesson. Heath, Ringgenberg, Samadi & Werner (2023) show that many results that were significant in isolation do not survive a correction for multiple testing (§4.8). The correction applies because many outcomes are tested against the same experiment. A clean experiment does not protect against the garden of forking paths.

6.4 Five routes to a correlated error

Every failure of exogeneity comes from one of five mechanisms. Naming the mechanism indicates the likely direction of the bias and what could fix it.

1. Omitted variables (confounding). Something that affects the outcome also affects the regressor. The bias follows the OVB formula of §3.5. For example, a regression of fund returns on fund size that omits manager skill is biased if skill attracts inflows, which raise size, and also raises returns.

2. Measurement error. The regressor is measured with noise, $X_i = X_i^* + v_i$, where v_i is independent of everything else. The error in the regression on the observed X_i then includes $-\beta v_i$. This term is correlated with X_i, and

$$\operatorname{plim} \hat{\beta} = \beta\,\lambda, \qquad \lambda = \frac{\operatorname{Var}(X^*)}{\operatorname{Var}(X^*) + \operatorname{Var}(v)}.$$

The **reliability ratio** λ lies between 0 and 1, so classical measurement error **attenuates** the coefficient toward zero. A reliability of 0.5 halves it. Measurement error in the *outcome* adds noise without bias. With several regressors, measurement error in one regressor biases the others too, in directions that are hard to sign.

Analysts most often miss a consequence in panels. **Fixed effects and differencing remove signal and keep noise.** Suppose the true regressor is persistent, with a correlation of 0.9 between adjacent periods. Suppose the measurement error is independent over time, with a reliability in levels of 0.8. First-differencing cuts the variance of the true regressor to $2(1 - 0.9) = 0.2$ of its level. It doubles the noise variance to $2 \times 0.25 = 0.5$ in the same units. The reliability of the differenced regressor is $0.2/(0.2 + 0.5) = 0.29$, down from 0.8 (Griliches & Hausman, 1986). A "within-firm" estimate that is much smaller than the pooled one may therefore reflect attenuation, rather than confounding in the pooled regression.

Many regressors in finance are estimated with error. Examples are betas from a first-pass time-series regression (§9.4), expected returns proxied by characteristics, expectations proxied by analyst forecasts, and liquidity proxied by spreads.

3. Simultaneity. The outcome also affects the regressor, because the two are determined together. The founding example is supply and demand. Two equations set quantity and price jointly. A regression of quantity on price estimates a mixture of the two curves, weighted by which curve's shocks happen to be larger. If demand shocks dominate, the regression

recovers something close to the *supply* curve (Working, 1927). Economists invented instrumental variables to solve exactly this problem (§7.3; Stock & Trebbi, 2003). Finance equivalents include order flow and returns, fund flows and performance, and trading volume and volatility. In the first pair, informed traders buy because they expect returns, and returns attract momentum traders.

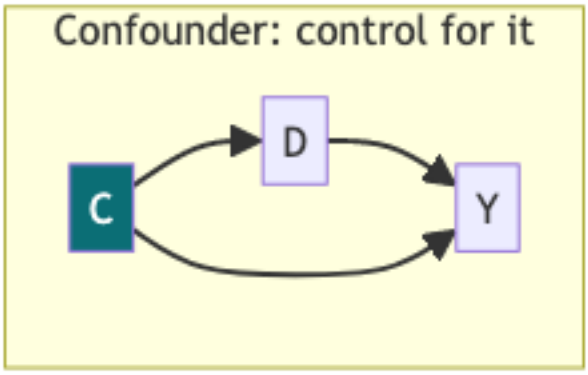
4. Sample selection. The outcome is observed only for a subsample chosen in a way related to the outcome. Wages are observed only for people who work, and fund returns only for funds that survive. Heckman (1979) showed that selection is an omitted variable. Within the selected sample, the expected error depends on the regressors through the probability of selection. Correcting for selection credibly requires a variable that affects selection but not the outcome. This requirement is an exclusion restriction.

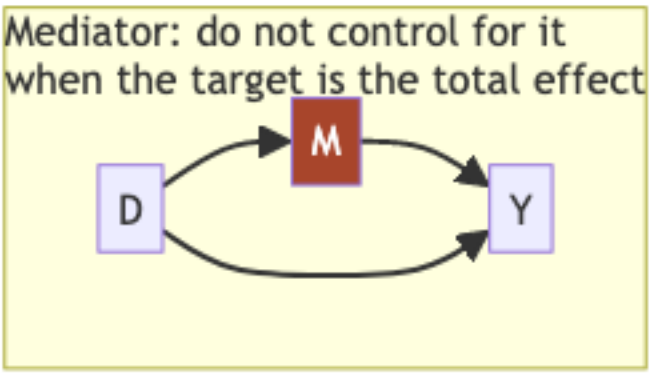
The finance case is **survivorship bias**. [Fact] Brown, Goetzmann, Ibbotson & Ross (1992) showed that poorly performing funds disappear from the data. A sample truncated by survival can therefore display apparent persistence in performance even when none exists. A related problem is look-ahead bias in sample construction. Backtesting on today's index constituents, for example, guarantees that every stock in the sample survived to today.

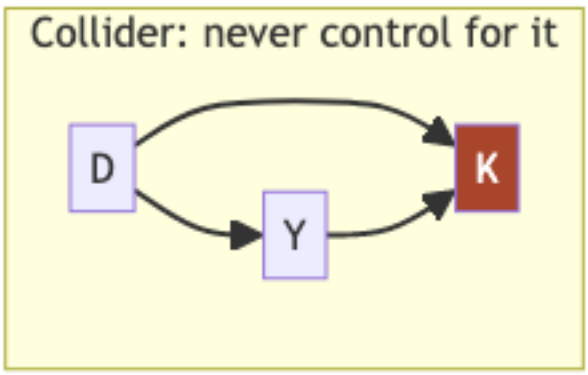
5. Bad controls. Conditioning on the wrong variable *creates* a correlation between the regressor and the error where none existed. This mechanism is the least intuitive of the five, because an action normally taken to reduce bias causes it. §6.5 treats it separately.

6.5 Causal graphs: confounders, mediators, colliders

A **directed acyclic graph** (DAG) draws variables as nodes and direct causal effects as arrows (Pearl, 2009). Its value for applied work is that it settles mechanically which variables to control for. Three elementary structures cover almost every case. The diagram shows them.



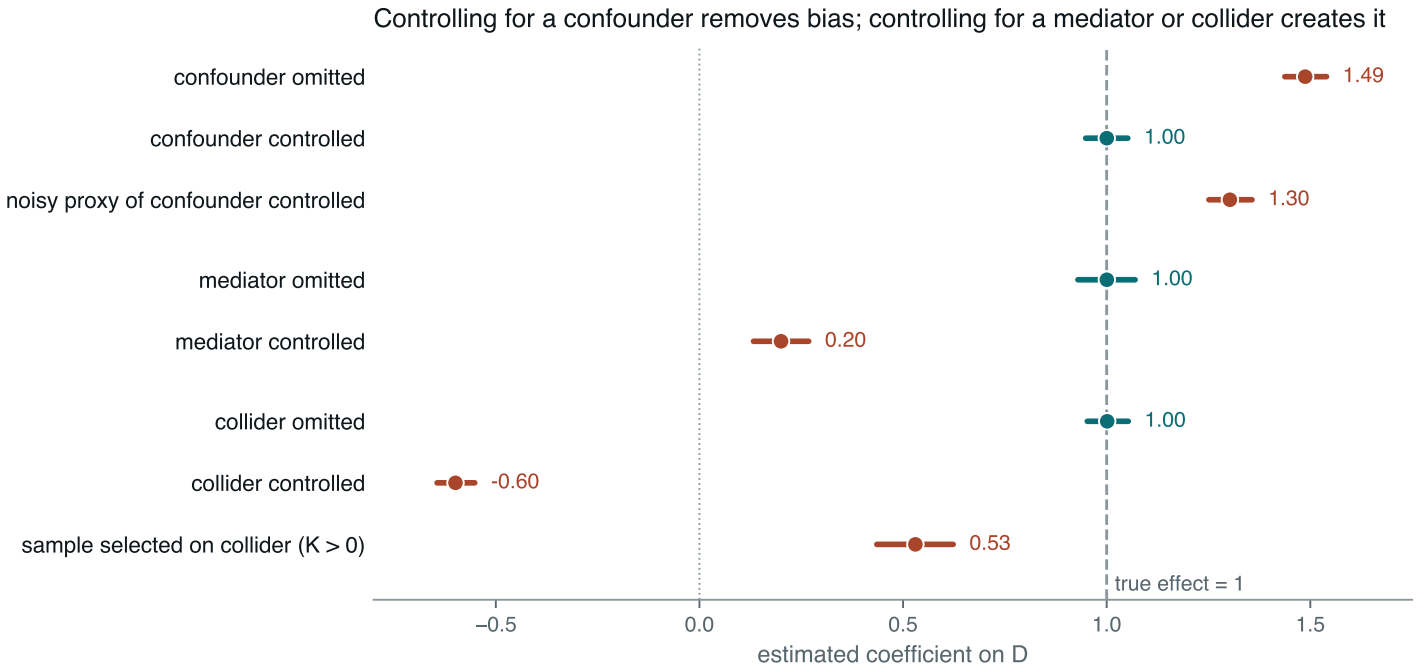




- **A confounder** ($C \to D$ and $C \to Y$) opens a non-causal "back-door" path from D to Y. Conditioning on C blocks it.
- **A mediator** ($D \to M \to Y$) carries part of the causal effect. Conditioning on M removes that part, leaving only the direct effect.
- **A collider** ($D \to K \leftarrow Y$) blocks the path between its causes *by default*. Conditioning on K opens the path. This creates an association between D and Y that is unrelated to the effect of D.

The formal rule is the **back-door criterion**. A set of controls identifies the effect of D on Y under two conditions. It blocks every path from D to Y that begins with an arrow *into* D. It also contains no variable caused by D.

The simulation below illustrates the three cases. In every scenario the true effect of D on Y is 1. [Simulated] Omitting a confounder gives 1.49, and controlling for it gives 1.00. Controlling for a noisy proxy of it gives 1.30, which removes less than 40% of the bias. The proxy has a reliability of 0.5. Omitting a mediator correctly gives the total effect of 1.00. Controlling for it gives only the direct part, 0.20. Omitting a collider correctly gives 1.00. Controlling for it gives -0.60, which reverses the sign. Even restricting the sample to units with a positive value of the collider gives 0.53.



A simple example shows the collider mechanism. Suppose skill and personal connections are independent among all managers. Suppose a firm hires anyone whose skill *plus* connections exceeds a threshold. Among hired managers, those with weak connections must have strong skill to have been hired. Skill and connections are therefore negatively correlated in the hired sample, although they are unrelated in the population. Selecting on being hired, which is the collider, created the correlation.

Finance has many colliders:

- **Survival and database inclusion.** A firm or fund stays in the data if its performance and its characteristics jointly allow it to survive. Any study restricted to survivors conditions on a collider.
- **Post-treatment controls.** If analyst coverage raises liquidity and liquidity affects returns, "controlling for liquidity" when estimating the effect of coverage on returns removes a mediated channel. If liquidity is also affected by returns, it is a collider as well.
- **Selecting on outcomes.** Studying "stocks that hit a 52-week high" or "funds that raised a second fund" conditions on a variable that depends on past returns.

Two further lessons follow. First, **"only control for pre-treatment variables" is necessary but not sufficient.** A pre-treatment variable can still be a collider on a path between unobserved causes of treatment and outcome, although such cases usually produce small biases (Cinelli, Forney & Pearl, 2024). Second, graphs and potential outcomes complement each other. Graphs are best at deciding what to control for. Potential outcomes are best at defining estimands, handling heterogeneous effects, and connecting to design-based inference (Imbens, 2020).

6.6 Selection on observables: regression, matching, and weighting

When no natural experiment is available, the fallback is to assume that the observed covariates capture everything that jointly drives treatment and outcomes. Formally, the assumption has two conditions:

- **Unconfoundedness** (conditional independence): $(Y_i(1), Y_i(0)) \perp D_i \mid X_i$. Among units with the same covariates, treatment is as good as random.
- **Overlap:** $0 < p(X_i) < 1$, where $p(x) = \Pr(D_i = 1 \mid X_i = x)$ is the **propensity score**. Every kind of unit has some chance of being in either group.

Under both conditions, comparing treated and untreated units with the same covariates and then averaging identifies the ATE: $\text{ATE} = \mathbb{E}\big[\mathbb{E}[Y_i \mid D_i = 1, X_i] - \mathbb{E}[Y_i \mid D_i = 0, X_i]\big]$. Rosenbaum & Rubin (1983) showed that if unconfoundedness holds given X, it holds given the scalar $p(X)$ alone. That result turns matching on many covariates into matching on one number.

The table lists the four families of estimator.

Estimator	What it models	Idea
Regression adjustment	The outcome, $\mathbb{E}[Y \mid D, X]$	Predict both potential outcomes for everyone and average the difference
Matching	Neither, directly	Compare each treated unit with untreated units with similar X or similar $p(X)$
Inverse probability weighting	The treatment, $p(X)$	Reweight so that both groups resemble the full population: $\mathbb{E}\left[\frac{D_i Y_i}{p(X_i)} - \frac{(1-D_i)Y_i}{1-p(X_i)}\right]$
Doubly robust (augmented IPW)	Both	Consistent if *either* the outcome model or the propensity model is right. It is the basis of §10.5

None of these estimators relaxes unconfoundedness. They differ only in how they use X. The choice among them matters less than whether X contains the confounders. [Practice] Matching is a way of doing covariate adjustment, not a research design. It inherits the adjustment's assumption.

Overlap is the one condition that can be checked. Plot the estimated propensity score separately for treated and untreated units. Where the distributions do not overlap, the estimate extrapolates the effect from functional form. Inverse probability weights also grow without bound as $p(X)$ approaches 0 or 1. Trimming the sample to the region of common support changes the estimand, and the change is explicit.

The LaLonde question, forty years on. Dehejia & Wahba (1999) revisited LaLonde's data. They found that propensity score methods came close to the experimental benchmark on a subsample with two years of pre-programme earnings. Smith & Todd (2005) showed that the result was sensitive to the sample and the specification. [Contested] The most recent reassessment, by Imbens & Xu (2025), finds that modern methods give robust estimates once covariate overlap is ensured. It also finds that robustness does not establish credibility. Validation exercises, such as placebo tests, have to judge whether the estimates can be read causally, not goodness of fit. The view here is that modern methods do not recover the benchmark merely because they are modern. The validation, not the estimator, separates good estimates from bad ones.

6.7 Sensitivity analysis: how much confounding would overturn a result

Unconfoundedness cannot be tested, because the confounders in question are unobserved. Instead, ask how strong an unobserved confounder would have to be to overturn the result. Then argue about whether a confounder that strong is plausible. Three approaches are in use.

- **Selection on observables as a guide.** Altonji, Elder & Taber (2005) proposed using the degree of selection on observed characteristics to bound selection on unobserved ones.
- **Coefficient stability, done properly.** Oster (2019) formalised the common practice of watching how a coefficient moves when controls are added. Movement is informative only relative to how much explanatory power the controls add. The method therefore uses the change in both the coefficient and the R^2. It reports δ. This is the ratio of selection on unobservables to selection on observables that would drive the effect to zero, given an assumed maximum attainable R^2. Oster's suggested defaults are a maximum R^2 of 1.3 times the R^2 with controls, and $|\delta| > 1$ as the robustness threshold. [Contested] Researchers debate both the default and the premise that observed controls are representative of unobserved ones.
- **Partial R^2 benchmarks.** Cinelli & Hazlett (2020) extend the OVB formula to report a **robustness value**. This is the minimum share of the residual variance of both the treatment and the outcome that a confounder would have to explain to eliminate the estimate. The number can be benchmarked against observed covariates, as in "a confounder would need to be three times as strongly related to treatment as firm size is."

[Practice] **Recommendation: report any estimate that rests on selection on observables with a sensitivity statistic.** An effect that a confounder half as strong as firm size would erase is a hypothesis. An effect that would need a confounder ten times as strong is evidence.

§6 Key takeaways

1. Potential outcomes define causal effects, and no unit's effect is ever observed. Choose the average (ATE, ATT, LATE, CATE) before choosing the method.

2. A naive comparison equals the effect on the treated plus selection bias. Selection bias is the difference in what the two groups would have experienced without treatment.
3. Randomisation removes selection bias by construction, including bias from unobserved characteristics. Every observational design is an argument that something mimicked it.
4. Endogeneity has five sources: omitted variables, measurement error, simultaneity, sample selection, and bad controls. Naming the source indicates the direction of the bias.
5. Measurement error attenuates, and fixed effects make it worse. Differencing a persistent regressor can cut its reliability from 0.8 to about 0.3.
6. Control for confounders. Do not control for mediators when the target is the total effect. Never control for colliders. Controlling for a collider can reverse the sign of an estimate.
7. Survivorship, database inclusion, and selection on outcomes are colliders, and they are common in financial data.
8. Regression, matching, weighting, and doubly robust estimators all rest on the same untestable assumption. Overlap can be checked, and a sensitivity statistic should be reported.

7. Research designs

7.1 How to read this section

A **research design** is an argument that some source of variation in the regressor is as good as random, together with an estimator that uses only that variation. In terms of the master form of §1.5, each design provides a particular orthogonality condition at a particular cost. The table summarises the condition and the cost for each design.

Design	The orthogonality condition it provides	The cost
Randomised experiment (§7.2)	Treatment is independent of potential outcomes, by construction	Cost, feasibility, and external validity
Instrumental variables (§7.3)	An instrument is uncorrelated with the structural error	An exclusion restriction, and a local estimand
Panel fixed effects (§7.4)	Within-unit changes in the regressor are uncorrelated with within-unit changes in the error	Strict exogeneity. Only time-invariant confounders are removed
Difference-in-differences (§7.5)	Treatment timing is unrelated to untreated trends	Parallel trends
Regression discontinuity (§7.6)	Crossing a cutoff is unrelated to the error near the cutoff	An estimand that exists only at the cutoff
Synthetic control (§7.7)	A weighted donor pool tracks the treated unit's untreated path	Long pre-periods, and good pre-treatment fit
Structural estimation (§7.8)	Whatever the economic model implies	The model has to be right

§6.6 covered selection on observables, the fallback design. Every design below is described with the same fields: **intuition**, **estimand**, **identifying assumption**, **estimation**, **inference**, **diagnostics**, **failure modes**, **in finance**, and **when preferred**. The shared fields allow direct comparison of failure modes across designs. The failure modes have the most practical value, so they get the most space.

7.2 Randomised experiments

Intuition. Assign treatment by coin flip, so that nothing about a unit can influence whether it is treated, and compare average outcomes.

Estimand. The ATE, which equals the ATT. If some units do not comply with their assignment, the comparison by *assignment* estimates the **intention-to-treat** effect, the effect of being offered treatment. The effect of treatment itself on those who comply is then an instrumental-variables problem, with assignment as the instrument (§7.3).

Identifying assumption. Random assignment, SUTVA (§6.1), and no attrition that differs by treatment arm.

Estimation. The difference in means. Adding pre-treatment covariates in a regression does not affect unbiasedness, but it can improve precision considerably. Interacting the treatment indicator with demeaned covariates guarantees that adjustment does not reduce precision in large samples (Lin, 2013). Never add covariates measured *after* treatment (§6.5).

Inference. Robust standard errors, clustered at the unit of randomisation. In small experiments, **randomisation inference** is exact. Recompute the statistic under every reassignment that could have happened. Then see where the actual statistic falls among them.

Diagnostics. A balance table of pre-treatment covariates across arms tests the randomisation *procedure*. With many covariates, a few will differ by chance. That is not evidence of a problem. Also check attrition by arm and compliance rates.

Failure modes.

- **Non-compliance** dilutes the intention-to-treat effect and changes the estimand.
- **Differential attrition** turns an experiment back into an observational study. If the treated units who drop out are those for whom treatment failed, the surviving comparison is biased.
- **Spillovers** contaminate the control group. Randomising at a higher level, such as markets rather than stocks or days rather than orders, can contain them. This costs many clusters.
- **Small samples** produce chance imbalances large enough to matter, which stratified or paired randomisation reduces.
- **Many outcomes** create multiple testing (§4.8), which is why pre-specifying the primary outcome is standard.
- **External validity.** An experiment identifies an effect for its population, its context, and its scale. The experiment cannot detect general-equilibrium effects that appear only when the treatment is scaled up.

In finance. Genuine experiments are rare but valuable. Examples are the Regulation SHO pilot (§6.3), the SEC's Tick Size Pilot, and field experiments in household finance. The most common experiment a quant runs is an **A/B test of an execution algorithm**. Randomise at the level of the order or the day. Cluster inference by day, because every order on a given day shares the same market conditions. Check for interference, because two algorithms trading the same names can move each other's prices.

When preferred. Whenever an experiment is feasible. Every other design in this section is a substitute for it.

7.3 Instrumental variables

Intuition. Find a variable that moves the regressor for reasons unrelated to the outcome's other causes. Use only the variation it creates (§5.2).

Estimand. With a constant effect, IV estimates the structural coefficient. With effects that differ across units, IV estimates a **local average treatment effect** (LATE). The LATE is the average effect among **compliers**, the units whose treatment status the instrument actually changes (Imbens & Angrist, 1994; Angrist, Imbens & Rubin, 1996). Units that receive treatment whatever the instrument says contribute nothing. Units that go untreated whatever the instrument says also contribute nothing. Two consequences follow. Different valid instruments can legitimately give different answers, because they shift different compliers. A failed over-identification test may also reflect heterogeneous effects rather than invalid instruments.

Identifying assumptions. IV rests on four assumptions, and the last two are untestable.

1. **Relevance.** The instrument moves the treatment. This one is testable.
2. **Independence.** The instrument is as good as randomly assigned.
3. **Exclusion.** The instrument affects the outcome *only* through the treatment.
4. **Monotonicity.** The instrument pushes everyone's treatment in the same direction, so there are no "defiers."

Estimation. 2SLS. With many instruments, limited-information maximum likelihood (LIML) is less biased.

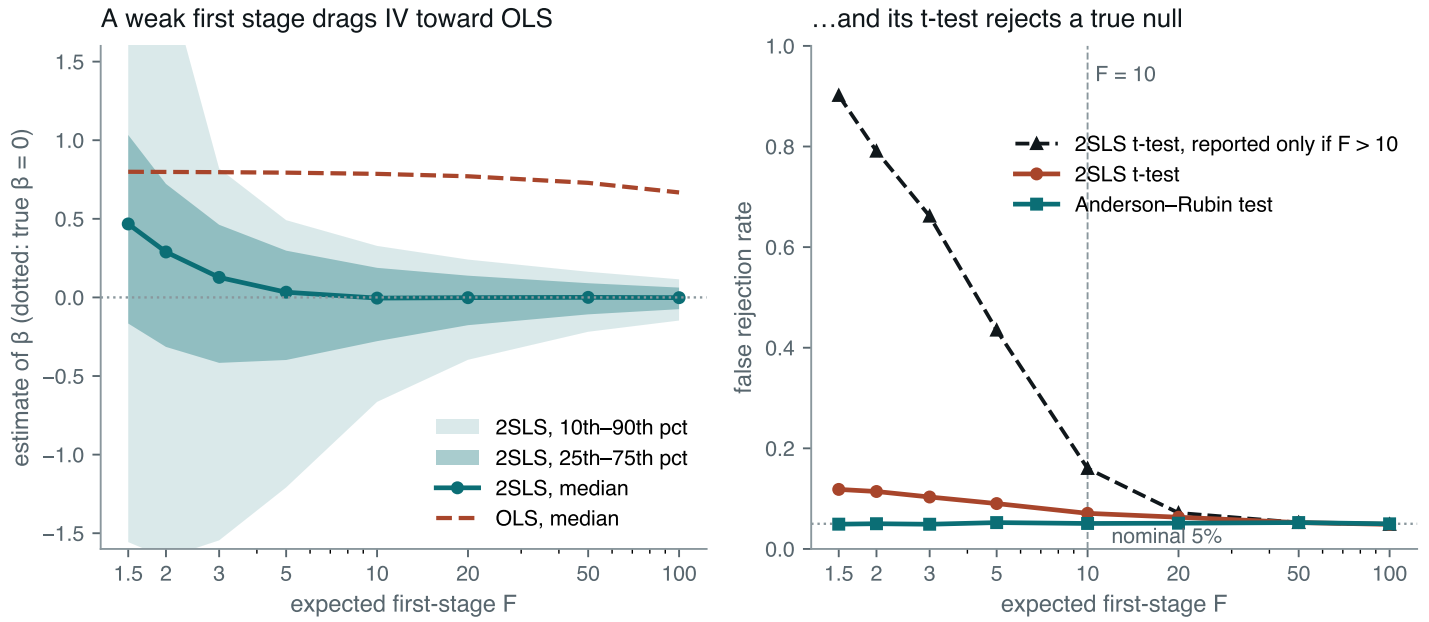
Inference. Robust or clustered standard errors when the instrument is strong. When it may be weak, use tests whose size does not depend on instrument strength. The **Anderson–Rubin test** (Anderson & Rubin, 1949) regresses $Y - X\beta_0$ on the instruments and tests whether they matter. It is valid at any instrument strength in the just-identified case. Its confidence set is the collection of values of β_0 that it does not reject. Moreira (2003) extends the idea to over-identified models.

Diagnostics.

- **The first stage.** The regression of the endogenous regressor on the instruments, $X_i = Z_i'\pi + v_i$, is the first stage. Report it, with a first-stage F-statistic for $\pi = 0$ that is robust to the dependence in the data. With heteroskedastic or clustered errors, that statistic is the "effective F" of Montiel Olea & Pflueger (2013), not the classical one.
- **The reduced form.** Regress the outcome directly on the instrument. The reduced form is an ordinary regression and is immune to weak-instrument problems. If it is zero, there is no effect to find.
- **Balance and placebos.** The instrument should not predict pre-determined characteristics, or outcomes that the treatment cannot affect.

Failure modes.

Weak instruments. When the first stage is weak, 2SLS is biased toward OLS, and its t-test rejects true nulls too often. The figure below simulates a just-identified model with strong endogeneity (a correlation of 0.8 between the first-stage and structural errors) and a true effect of zero. [Simulated] With an expected first-stage F of 2, the median 2SLS estimate is 0.29, which is over a third of the OLS estimate of 0.80. With an expected F of 10, the median bias is gone. The estimates still spread widely, and the t-test still rejects at 7%.



The dashed line in the right panel is the more important result. It shows how the t-test performs in only those samples whose first-stage F exceeds 10. A literature that screens on $F > 10$ reports exactly these samples. [Simulated] When the true expected F is 5, the samples that pass the screen reject the true null 44% of the time. The screen selects samples in which the first stage looks strong *by chance*. The first-stage and structural errors are correlated, so these samples are also the ones where the estimate is furthest from the truth.

The published thresholds agree. [Fact] For one instrument, keeping the size of a nominal 5% test below 10% requires a first-stage F above 16.38 (Stock & Yogo, 2005). Lee, McCrary, Moreira & Porter (2022) show that a true 5% test needs F above 104.7. Equivalently, when F is 10 the test needs a critical value of 3.43 instead of 1.96. They re-examine 57 IV papers in the *American Economic Review* and find that about half of the results presumed significant are not significant.

Invalid exclusion. The instrument affects the outcome through another channel. Weather is a popular instrument, and Mellon (2025) catalogues close to two hundred variables that published studies have linked to weather. Each one is a potential channel that violates the exclusion restriction of every other weather-IV study.

Weak instruments amplify small violations. The large-sample bias of IV is $\text{Cov}(Z, u)/\text{Cov}(Z, X)$. A weak first stage divides a small correlation with the error by a small correlation with the treatment. **An instrument that is both slightly invalid and weak can be worse than OLS.**

Many instruments. 2SLS with many instruments over-fits the first stage and drifts toward OLS. [Fact] Bound, Jaeger & Baker (1995) showed that *randomly generated* instruments could closely reproduce a famous estimate of the return to schooling. The original estimate used quarter of birth, interacted with many other variables, as instruments.

Fragile inference in practice. [Contested] Young (2022) re-examined 1,309 IV regressions from 30 papers in the American Economic Association's journals. Young found that non-independent errors and high leverage distort size and power. Results then depend on a few clusters or observations. First-stage F pretests were largely uninformative. IV estimates rarely rejected the OLS estimate, although they differed substantively from it.

The wrong estimand. LATE is the effect for compliers, who may be unrepresentative. [Contested] Researchers debate whether this is a feature or a flaw (Deaton, 2010; Imbens, 2010).

Shift-share instruments. "Bartik" instruments combine exposure shares with aggregate shocks. The identifying variation can come from the shares or from the shocks. The assumptions needed differ in each case (Goldsmith-Pinkham, Sorkin & Swift, 2020; Borusyak, Hull & Jaravel, 2022).

In finance. The mutual-fund flow instrument is a cautionary example. Large outflows force funds to sell their holdings. Researchers therefore used flow-induced selling as a source of price pressure unrelated to fundamentals (Coval & Stafford, 2007; Edmans, Goldstein & Jiang, 2012). A large literature then used it to study how mispricing affects corporate decisions. [Contested] Wardlaw (2020) showed that the standard construction of the measure is mechanically a function of the stock's own realised return in the outflow quarter. With that component removed, the price effect is small and does not reverse. Many downstream results then fail to hold. An instrument that contains the outcome fails the exclusion restriction. Two more constructive uses are demand-system asset pricing, which instruments for prices with features of investors' mandates (Koijen & Yogo, 2019), and monetary-policy surprises measured from futures prices (§8.5).

When preferred. Prefer IV when some feature of the world moves the treatment for reasons plausibly unrelated to the outcome. The exclusion restriction must also be defensible channel by channel.

7.4 Panel data and fixed effects

Intuition. Compare each unit with itself at other times, which removes everything about the unit that does not change.

Model and estimand.

$$Y_{it} = \alpha_i + \lambda_t + X'_{it}\beta + u_{it},$$

Here α_i absorbs every time-invariant characteristic of unit i, observed or not. The term λ_t absorbs every shock common to all units at date t. The coefficient β comes from **within-unit changes** in X, relative to the average change at that date.

Identifying assumption. Strict exogeneity: $\mathbb{E}[u_{it} \mid X_{i1}, \ldots, X_{iT}, \alpha_i] = 0$. The error at any date is unrelated to the regressors at *every* date. That rules out time-varying confounders. It also rules out feedback from past outcomes to current regressors.

Estimation. The **within estimator** subtracts each unit's time average from every variable and runs OLS. By Frisch–Waugh–Lovell (§3.4), this is identical to including a dummy variable for every unit. With two periods, first-differencing gives the same estimate. The two-way version also removes date means. High-dimensional variants absorb interactions such as industry-by-year effects.

Random effects assume that α_i is uncorrelated with the regressors. In exchange, they also use the between-unit variation. In economics that assumption is rarely credible, because the unobserved unit characteristic is usually the reason for concern. The **Hausman test** compares the two estimators (Hausman, 1978). Mundlak (1978) showed that adding the unit averages of the regressors to a random-effects model reproduces the fixed-effects coefficients exactly. This turns the choice into a regression specification.

Inference. Cluster by unit, which allows for serial correlation within units (§4.4). Cluster two-way if units also share shocks at a date.

Diagnostics. Measure how much of the regressor's variance is within units. Count how many units change their treatment, and check whether they are typical. Check whether the estimate changes sharply between pooled and fixed-effects specifications. If it does, check whether the change reflects confounding or attenuation (§6.4).

Failure modes.

- **Fixed effects remove only time-invariant confounders.** A confounder that changes over time, such as a firm's growth opportunities, stays in the error.
- **Feedback violates strict exogeneity.** If a bad year leads a firm to change its leverage, current leverage depends on past errors. With a lagged dependent variable and fixed effects, the within estimator is biased by roughly $-(1+\phi)/(T-1)$ in panels with T periods (Nickell, 1981). With ten periods and $\phi = 0.5$, the bias is about -0.17, a third of the true coefficient. GMM estimators that use lagged levels as instruments address the bias (Arellano & Bond, 1991). They become weak-instrument problems when ϕ is near one.
- **Fixed effects amplify measurement error**, as §6.4 showed, because demeaning removes signal and keeps noise.
- **Identification comes from a few switchers.** If the regressor rarely changes within units, the estimate comes from the small and possibly unusual set of units where it does change. State laws and a CEO's characteristics are examples.
- **Fixed effects can absorb the variation of interest.** With firm fixed effects, the estimated effect of a slow-moving characteristic such as governance quality comes from its noise.
- **Substitutes for fixed effects give inconsistent estimates.** [Fact] Gormley & Matsa (2014) show this for two common shortcuts in finance. The shortcuts are subtracting the industry mean from the dependent variable ("industry-adjusting") and adding the group mean of the dependent variable as a control. Use the fixed-effects estimator instead.

In finance. Empirical corporate finance uses firm fixed effects widely, in studies of capital structure, investment, and payout policy. Fund and manager fixed effects separate skill from style. Stock fixed effects in return-predictability regressions deserve caution, because demeaning a persistent predictor over a short window creates the finite-sample biases of §9.2.

When preferred. Prefer fixed effects when the main worry is a stable unobserved characteristic of each unit. The regressor must also change within units for reasons unrelated to the outcome's recent history.

7.5 Difference-in-differences

Intuition. Compare the change over time in a treated group with the change in an untreated group. The untreated group's change estimates what would have happened to the treated group.

Estimand. The ATT, for the treated group, in the post-treatment period.

The two-by-two case. With a treated and a control group, each observed before and after,

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{\text{treated, post}} - \bar{Y}_{\text{treated, pre}}) - (\bar{Y}_{\text{control, post}} - \bar{Y}_{\text{control, pre}}).$$

This equals the coefficient on the interaction in a regression of Y on a treated-group indicator, a post-period indicator, and their product. The first difference removes each group's fixed level. The second difference removes the common time shock.

Identifying assumption. Parallel trends: without treatment, the treated group's average outcome would have changed by the same amount as the control group's. The design also assumes no anticipation, meaning that outcomes do not respond before treatment begins, and SUTVA. Parallel trends is a statement about a particular *scale*. If the groups start at different levels, trends cannot be parallel in both levels and logs, so the choice of scale is part of the assumption.

Estimation. Run the **two-way fixed effects** (TWFE) regression $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + u_{it}$, where D_{it} switches on when unit i receives treatment. The **event-study** version replaces τD_{it} with separate coefficients for each period relative to treatment, including periods before it.

The staggered-adoption problem. With a single treatment date, TWFE equals the 2×2 estimator. When units receive treatment at *different* dates and effects vary over time, it does not. The example below works through the reason by hand. Take two groups over three periods. The early group receives treatment from period 2, and the late group from period 3. Untreated outcomes are flat, so parallel trends holds exactly. Write τ_{E2} and τ_{E3} for the early group's treatment effects in periods 2 and 3, and τ_{L3} for the late group's effect in period 3.

By Frisch–Waugh–Lovell (§3.4), the TWFE coefficient comes from regressing Y on the treatment indicator after removing unit and period means from the indicator. For a balanced panel, that residual is $\tilde{D}_{it} = D_{it} - \bar{D}_{i\cdot} - \bar{D}_{\cdot t} + \bar{D}$. The table gives its values.

	Period 1	Period 2	Period 3
Treatment D_{it}, early group	0	1	1
Treatment D_{it}, late group	0	0	1
Residual $\tilde{D}_{it}$, early group	$-1/6$	$+1/3$	$-1/6$
Residual $\tilde{D}_{it}$, late group	$+1/6$	$-1/3$	$+1/6$

The sum of squared residuals is $1/3$. The unit and period effects in the outcome are orthogonal to $\tilde{D}$, so only the treatment effects contribute to $\sum \tilde{D}_{it} Y_{it}$, and

$$\hat{\tau}_{\text{TWFE}} = 3\Big(\tfrac{1}{3}\,\tau_{E2} - \tfrac{1}{6}\,\tau_{E3} + \tfrac{1}{6}\,\tau_{L3}\Big) = \tau_{E2} - \tfrac{1}{2}\,\tau_{E3} + \tfrac{1}{2}\,\tau_{L3}.$$

The early group's second-period effect enters with a negative weight. In period 3 the early group, already treated, serves as the *control* for the late group. TWFE therefore subtracts growth in its treatment effect as if it were a trend. If every effect equals 1, TWFE correctly gives 1. If effects grow with exposure, $(\tau_{E2}, \tau_{E3}, \tau_{L3}) = (1, 2, 1)$, TWFE gives 0.5, while the average effect on the treated is $4/3$. If they grow faster, $(1, 4, 1)$, TWFE gives -0.5, **a negative estimate when every treatment effect is positive.** [Fact] Goodman-Bacon (2021) and de Chaisemartin & D'Haultfœuille (2020) document this mechanism.

The remedy is to compare treated units only with units that are not yet treated or never treated. Several estimators do this. One estimates group-by-period effects and then aggregates them (Callaway & Sant'Anna, 2021). One uses interaction-weighted event studies (Sun & Abraham, 2021). One imputes untreated outcomes from untreated observations (Borusyak, Jaravel & Spiess, 2024). Roth, Sant'Anna, Bilinski & Poe (2023) give a guide to choosing among them. Baker, Larcker & Wang (2022) show that the problem affects published work in finance and accounting, where staggered designs are common.

Inference. Cluster at the level at which treatment is assigned. Bertrand, Duflo & Mullainathan (2004) give the warning on this point (§4.4). The few-treated-clusters problem of §4.4 applies fully when a policy changes in only a few states.

Diagnostics.

- **The event-study plot.** Estimated effects in the periods before treatment should be near zero.
- **Placebo dates and placebo outcomes.** Assign treatment to an earlier date, or look at outcomes it could not have affected.
- **The scale.** Check whether conclusions survive switching between levels and logs.

Pre-trend tests have low power, so passing them is weak evidence. Reporting only designs that pass them also biases the resulting estimates (Roth, 2022). A more transparent approach bounds how far post-treatment trends could depart from parallel, in proportion to the departures seen before treatment. It then reports the range of effects consistent with that bound (Rambachan & Roth, 2023).

Failure modes.

- **Non-parallel trends from selection into timing.** Units adopt a treatment *because* of how their outcomes are evolving. [Fact] Workers who enter training programmes experience an earnings decline just before entering. The decline reverses whether or not the training works, so a naive DiD overstates the effect (Ashenfelter & Card, 1985). Firms adopt policies when they are in trouble. Mean reversion then looks like a treatment effect.
- **Anticipation.** In financial markets, prices respond at announcement, not at the effective date, so a DiD that uses the effective date misses most of the effect.
- **Staggered timing with heterogeneous effects**, described above.
- **Spillovers** to the control group, and changes in the composition of either group over time.
- **Policy endogeneity and institutional context.** [Contested] Karpoff & Wittry (2018) show that the effect of state antitakeover laws on a firm's takeover protection depends on other state laws, the firm's existing defences, and court decisions. Tests that ignore this context can draw misleading inferences from the laws used as natural experiments.

In finance. Staggered state antitakeover laws (Bertrand & Mullainathan, 2003), bank deregulation, and regulatory changes that affect some firms or markets before others.

When preferred. Prefer DiD when a policy or shock affects some units at some times. A comparison group must plausibly share the treated group's trends. Enough pre-treatment periods must exist to check this.

7.6 Regression discontinuity

Intuition. Suppose a rule assigns treatment according to whether a continuous **running variable** crosses a cutoff. Units just above and just below the cutoff are then nearly identical except for treatment. The jump in the average outcome at the cutoff is the treatment effect.

Estimand. The effect *at the cutoff*: $\tau(c) = \mathbb{E}[Y_i(1) - Y_i(0) \mid R_i = c]$, where R_i is the running variable and c is the cutoff. In a **sharp** design, treatment switches on exactly at the cutoff. In a **fuzzy** design, only the probability of treatment jumps. The ratio of the outcome jump to the treatment jump is then an IV estimate, a LATE for compliers at the cutoff (Hahn, Todd & van der Klaauw, 2001).

Identifying assumption. The average potential outcomes are continuous in the running variable at the cutoff. Nothing else jumps there, and units cannot precisely sort themselves to one side.

Estimation. Fit separate local linear regressions on each side of the cutoff, within a bandwidth, typically weighting nearer observations more heavily. Take the difference of the two fitted values at the cutoff. Choose the bandwidth by a data-driven mean-squared-error rule. Keep the fit local. High-order polynomials over the whole range put erratic weight on observations far from the cutoff and give misleading confidence intervals (Gelman & Imbens, 2019).

Inference. A bandwidth that minimises mean squared error accepts some bias. That bias invalidates conventional confidence intervals. Use **robust bias-corrected** intervals (Calonico, Cattaneo & Titiunik, 2014). A discrete running variable, such as an integer rank or a rounded score, needs special treatment (Kolesár & Rothe, 2018).

Diagnostics.

- **Manipulation.** A jump in the *density* of the running variable at the cutoff suggests units sorting themselves (McCrary, 2008; Cattaneo, Jansson & Ma, 2020).
- **Covariate balance.** Pre-determined characteristics should not jump at the cutoff.
- **Placebo cutoffs and bandwidth sensitivity.** Fake cutoffs should show no jumps, and the estimate should not depend on one bandwidth. A "donut" design that drops the observations closest to the cutoff checks for sorting right at the threshold.

Failure modes.

- **Manipulation of the running variable**, which recreates selection at exactly the point where the design needs none.
- **Compound treatments.** Other rules use the same cutoff, so the jump bundles several effects.

- **A coarse running variable**, which leaves no observations genuinely close to the cutoff.
- **Extrapolation.** The effect at the cutoff can differ from the effect anywhere else.
- **Specification search** over bandwidths, kernels, and polynomial orders.

In finance. The canonical application is the annual reconstitution of the Russell 1000 and Russell 2000 indices. Russell ranks stocks by market capitalisation in May. A stock just below the 1,000th rank enters the Russell 2000 at the top, with a large index weight. A stock just above that rank enters the Russell 1000 at the bottom, with a small weight. Researchers have used the design to estimate the effect of index weight on prices (Chang, Hong & Liskovich, 2015) and of passive ownership on governance (Appel, Gormley & Keim, 2016). The complications illustrate the failure modes above. Russell's capitalisation measure is proprietary, and researchers can only approximate it. Since 2007, a "banding" rule has made a stock's assignment depend on its previous index as well as its rank. Both features blur the discontinuity and change which comparisons are valid. [Contested] Researchers have argued at length about the right specification (Appel, Gormley & Keim, 2024). Other thresholds used in finance include debt covenant limits (Chava & Roberts, 2008), close shareholder votes, and credit-rating boundaries.

When preferred. Whenever a rule assigns treatment by a threshold on a variable that units cannot precisely control. For the effect it identifies, it is the most credible observational design.

7.7 Synthetic control

Intuition. Suppose a single aggregate unit receives treatment, such as a country, a state, or a market. Build a weighted average of untreated units that tracked the treated unit closely before treatment. The weighted average's path after treatment is the counterfactual.

Estimand. The effect on the treated unit, period by period after treatment.

Identifying assumption. The weighted donor combination continues to track the treated unit's untreated outcome after treatment. A sufficient condition is that outcomes follow a factor structure, and that the donors match the treated unit's factor loadings. The design also requires no spillovers onto the donors, and no other shock to the treated unit at the same time.

Estimation. Choose non-negative donor weights that sum to one. Pick the weights that minimise the distance between the treated unit and the weighted donors in pre-treatment outcomes and predictors (Abadie & Gardeazabal, 2003; Abadie, Diamond & Hainmueller, 2010). The constraints make the counterfactual an interpolation of real units and not an extrapolation.

Inference. Inference uses placebos. Apply the method to every donor as if it had been treated. Then find where the treated unit's post-treatment divergence, relative to its pre-treatment fit, ranks among the placebos. With 19 donors, being the most extreme of 20 units corresponds to a p-value of 0.05.

Diagnostics. Check the pre-treatment fit first. If the synthetic unit does not track the treated unit before treatment, do not use it. Also check that the weights are sparse and interpretable, and that results survive dropping each donor in turn.

Failure modes.

- **Poor pre-treatment fit**, often because the treated unit is extreme and lies outside what any combination of donors can reproduce.
- **Short pre-periods**, which let the weights fit noise rather than structure.
- **Specification search** over predictors and donor pools, which can manufacture results (Ferman, Pinto & Possebom, 2020).
- **Contaminated donors** that were themselves affected by the treatment.

Synthetic difference-in-differences combines synthetic-control unit weights with time weights and DiD's fixed effects. It is more robust than either synthetic control or DiD alone when the pre-period fit is imperfect (Arkhangelsky, Athey, Hirshberg, Imbens & Wager, 2021). Abadie (2021) sets out the data requirements.

In finance. A regulation or market-structure change in one country's market, or a shock to a single exchange, with other markets as donors.

When preferred. One or a few treated aggregate units, a long pre-treatment series, and a pool of comparable untreated units.

7.8 Structural estimation

Intuition. Write down a model of the decisions that generated the data, with parameters for preferences, technology, constraints, and beliefs. Estimate the parameters. Then use the model to answer counterfactual questions, including questions about interventions that have never happened.

Estimand. Deep parameters, and the counterfactuals they imply.

Identifying assumption. The assumption is the model itself: its functional forms, distributional assumptions, and equilibrium concept. It also includes whatever variation in the data determines each parameter. **The model is the identification.** Good structural work shows which features of the data identify which parameters. A reader can then judge whether identification comes from variation or from functional form.

Estimation. Maximum likelihood, GMM, or the **simulated method of moments**, which simulates the model and chooses parameters to match selected moments of the data.

Inference. The sandwich, or the bootstrap. Andrews, Gentzkow & Shapiro (2017) show how to report which moments determine which estimates. This makes the identification argument inspectable.

Diagnostics. Fit to moments that were *not* used in estimation, and the plausibility of counterfactuals.

Failure modes.

- **Identification by functional form**, where the estimate is determined by an assumption no data could check.
- **Misspecification that propagates to counterfactuals**, often invisibly, because the counterfactual is by definition outside the data.
- **Computational fragility**: local optima and sensitivity to starting values.
- **Many modelling choices**, which give the analyst many degrees of freedom.
- **Parameters that are not really invariant.** The Lucas critique (§1.2) motivates structural models, but it applies to them too if the "deep" parameters would change with policy.

In finance. Researchers estimate consumption-based asset pricing models by GMM (§5.3). They estimate dynamic corporate finance models by simulated moments (Hennessy & Whited, 2007). They estimate term-structure models by the Kalman filter. Demand-system asset pricing borrows the structural demand estimation of industrial organisation (Berry, Levinsohn & Pakes, 1995; Koijen & Yogo, 2019).

The debate. The longest-running argument in modern empirical economics concerns how much weight to give design-based versus structural evidence. The design-based side argues that credible identification of a well-defined effect is better than an ambitious model whose assumptions the data cannot check (Angrist & Pischke, 2010). The structural side argues that design-based estimates are local. They often answer questions nobody asked. They cannot say anything about interventions outside the data without a model anyway (Keane, 2010; Heckman, 2010; Deaton, 2010). [Contested] The view taken here is that the two approaches are complements. The best current work uses quasi-experimental variation to restrict a structural model's parameters. Credibly identified effects then constrain the model's extrapolations.

When preferred. Prefer structural estimation when the question is a counterfactual outside the range of the data, a welfare calculation, or a mechanism. Prefer it also when the relevant policy would change the environment that generated the data.

7.9 Comparison

The table compares the designs of this section on the attributes that discriminate between them.

Design	Estimand	Identifying assumption	What the data can check	Characteristic failure
Randomised experiment	ATE; intention-to-treat	Random assignment	Balance, attrition, compliance	Spillovers; external validity
Regression discontinuity	Effect at the cutoff	Continuity at the cutoff	Density, covariate balance, placebo cutoffs	Manipulation; extrapolation
Instrumental variables	LATE for compliers	Exclusion, independence, monotonicity	First stage, reduced form, placebos	Weak or invalid instruments
Difference-in-differences	ATT	Parallel trends	Pre-trends (weakly), placebos	Selection into timing; staggered bias
Synthetic control	Effect on one treated unit	Donors track the untreated path	Pre-treatment fit, placebos	Poor fit; specification search
Panel fixed effects	Within-unit effect	Strict exogeneity	Within variation; switchers	Time-varying confounders; feedback
Selection on observables	ATE or ATT	Unconfoundedness	Overlap only	Unobserved confounding
Structural	Deep parameters; counterfactuals	The model	Untargeted moments	Misspecification

[Practice] This chapter ranks the designs by credibility for the effect each design identifies, assuming competent execution. Randomised experiments rank first, then regression discontinuity. Next come instrumental variables with a genuinely random instrument and difference-in-differences with a well-understood policy. Synthetic control and panel fixed effects follow. Selection on observables ranks last. A structural model is as credible as its weakest assumption. The ranking concerns *internal* validity. An experiment on the wrong population can be less useful than a careful observational study of the right one. A regression discontinuity gives no information about units far from its cutoff.

§7 Key takeaways

1. Every research design obtains an orthogonality condition by making assumptions. Ask which condition the design delivers, and whether its assumptions are worth accepting, before asking about estimators.
2. IV estimates the effect for compliers, not for everyone. Weak instruments bias it toward OLS and make its t-test misleading. A true 5% test needs a first-stage F near 105, not 10, and screening on $F > 10$ makes reported results worse, not better.
3. Weak instruments turn small exclusion violations into large biases. Always report the reduced form.
4. Fixed effects remove only time-invariant confounders, fail under feedback from outcomes to regressors, and amplify measurement error. Industry-adjusting the dependent variable is not a substitute.
5. With staggered adoption and effects that change over time, two-way fixed effects can report a negative effect when every true effect is positive. Use estimators that compare treated units only with not-yet-treated or never-treated ones.
6. Pre-trend tests are weak evidence. Report the sensitivity of the result to plausible departures from parallel trends.
7. Regression discontinuity is the most credible observational design for the effect at its cutoff. It gives no information beyond the cutoff. Test for manipulation, and use robust bias-corrected intervals.
8. Synthetic control is for one treated aggregate unit with a long pre-period. Poor pre-treatment fit disqualifies it.
9. Structural models answer questions that designs cannot, at the cost of assumptions that designs do not need. The strongest evidence combines the two.

Part IV — Time series

8. Time-series econometrics

A cross-section is many units observed once. A time series is one unit observed many times, and that difference changes almost everything. A time series has only one history, so "repeated sampling" needs a new interpretation. Observations are dependent, so the meat of every standard error changes. Variables can move without limit, which breaks the asymptotic approximations of §2. This section covers the ideas a practitioner needs. Simple and Log Returns §6 applies several of them to returns specifically, and Market Regimes and Hidden Markov Models takes regime-switching and change-point models much further.

8.1 What changes when observations are ordered

Stationarity replaces random sampling. A process is **covariance stationary** if its mean and variance are constant over time, and if the covariance between two observations depends only on how far apart they are. **Ergodicity** requires dependence to decline fast enough for time averages to converge to population averages. Together, the two conditions let a single long history substitute for many independent draws. Without them, the average of one path need not estimate anything.

The autocorrelation function $\rho_j = \mathrm{Corr}(y_t, y_{t-j})$ summarises linear dependence. The **partial** autocorrelation at lag j is the correlation between y_t and y_{t-j} after removing the effect of the intermediate lags. It equals the coefficient on y_{t-j} in a regression on j lags.

Three strengths of "unpredictable" correspond to the three strengths of "unrelated" in §2.2. **White noise** is uncorrelated with its past. A **martingale difference sequence** has a conditional mean of zero given its past. An **i.i.d.** sequence is independent of its past. Returns come close to the middle condition and are far from the last, because their variance is predictable.

Testing for autocorrelation shows why the distinction matters. The Ljung–Box statistic (Ljung & Box, 1978) sums squared sample autocorrelations and compares the total with a chi-squared distribution. That comparison is valid when the series is i.i.d. On returns with volatility clustering it rejects too often, because the variance of the squared autocorrelations that it assumes is too small. The **variance ratio** compares the variance of q-period returns with q times the variance of one-period returns,

$$\mathrm{VR}(q) = \frac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)} = 1 + 2\sum_{j=1}^{q-1}\Big(1 - \frac{j}{q}\Big)\rho_j,$$

which equals 1 for uncorrelated returns and exceeds 1 under positive autocorrelation. Lo & MacKinlay (1988) built a version whose standard errors are robust to heteroskedasticity. They rejected the random walk for weekly US stock index returns. Section 4.5.1 of Momentum in Financial Markets develops the variance ratio as a trading statistic.

8.2 Autoregressions and the Wold decomposition

The AR(1) model, $y_t = c + \phi y_{t-1} + \varepsilon_t$ with $|\phi| < 1$, shows most features of time-series dynamics with one parameter. The table lists its main properties.

Property	Formula	With $\phi = 0.9$
Mean	$\mu = c/(1-\phi)$	
Variance	$\sigma^2/(1-\phi^2)$	5.3 times the innovation variance
Autocorrelation at lag j	ϕ^j	0.59 at lag 5
Response at horizon h to a unit shock	ϕ^h	0.35 after 10 periods
Half-life of a shock	$\ln(0.5)/\ln\phi$	6.6 periods
Forecast	$\mathbb{E}[y_{t+h} \mid \mathcal{F}_t] = \mu + \phi^h(y_t - \mu)$	Reverts to the mean
Effective sample size for the mean	$T(1-\phi)/(1+\phi)$	5.3% of T (§2.5)

Higher-order **autoregressions** add lags. **Moving-average** models, $y_t = \varepsilon_t + \theta_1\varepsilon_{t-1} + \ldots$, model a shock whose effect lasts a fixed number of periods and then stops. **ARMA** models combine the two.

The Wold decomposition explains why these simple models are so general. Every covariance-stationary process equals a deterministic component plus an infinite moving average of its own one-step-ahead linear forecast errors. **Linear ARMA models can reproduce the autocovariances of any stationary process, and they capture nothing beyond the autocovariances.** Volatility clustering, asymmetric responses, and any non-linear predictability do not appear in the autocovariances, so these models cannot detect them (§8.6).

The Box–Jenkins workflow, from Box and Jenkins's 1970 book, is still the right order of operations. Identify a candidate model from the sample autocorrelations and partial autocorrelations, estimate it, and check that its residuals look like white noise. Only then forecast. Choosing lag lengths by information criteria involves a trade-off. The **AIC** (Akaike, 1974) targets forecast accuracy and tends to choose too many lags. The **BIC** (Schwarz, 1978) penalises parameters more heavily. It picks the true order with probability approaching one when the true model is among the candidates. [Practice] **Recommendation: use AIC or cross-validation for forecasting, and BIC for describing the dynamics.** Historically, simple ARIMA models forecast US macroeconomic aggregates about as well as the large structural models of the 1960s (§13).

8.3 Unit roots and spurious regression

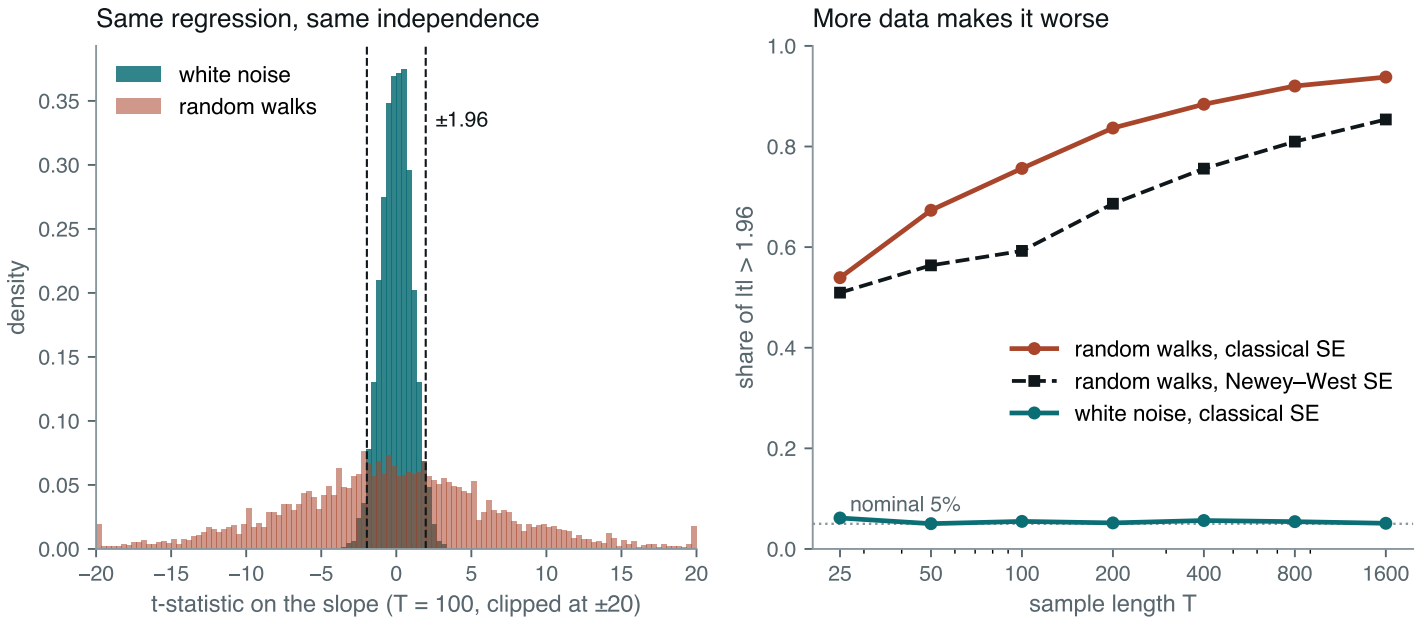
The random walk $y_t = y_{t-1} + \varepsilon_t$ is the AR(1) with $\phi = 1$, and at that value every property in the table above fails. The variance grows linearly with time, so the process has no fixed variance to estimate. Shocks never decay, so the half-life is infinite. The forecast has no mean to revert to. A series like this is **integrated of order one**, written I(1). Its first difference is stationary, and its level is not.

Testing for a unit root. The **Dickey–Fuller** test regresses the change on the lagged level, $\Delta y_t = \alpha + \gamma y_{t-1} + \sum_j \delta_j \Delta y_{t-j} + \varepsilon_t$, and tests $\gamma = 0$ (Dickey & Fuller, 1979). Under the null, the regressor is itself a random walk, so the t-statistic does not have a t distribution. Its 5% critical value with a constant is about -2.86, far below the -1.65 of a standard one-sided test. The test also has low power. With samples of typical length, it cannot reliably tell $\phi = 1$ from $\phi = 0.97$. The second value gives a very different process, with a half-life of 23 periods. The **KPSS** test takes stationarity as the null (Kwiatkowski, Phillips, Schmidt & Shin, 1992). Running both tests is common.

[Contested] Researchers debated at length whether macroeconomic series have unit roots. Nelson & Plosser (1982) could not reject a unit root in most US macroeconomic series. Perron (1989) showed that allowing for a few structural breaks reversed many of those conclusions. The practical lesson is that unit-root tests are rarely decisive. [Practice] Base the decision to difference a series on the economics and on the question as much as on a test.

Spurious regression. A regression of one random walk on another, independent one will usually report a strong, significant relationship. Yule (1926) described the phenomenon, Granger & Newbold (1974) quantified it, and Phillips (1986) explained it. Under the null, the estimated slope converges to a random variable and not to zero, and the t-statistic grows without bound at rate $\sqrt{T}$.

The figure shows how severe the problem is. [Simulated] With 100 observations of two independent random walks, the conventional t-test rejects "no relationship" 76% of the time at a nominal 5%. With 1,600 observations it rejects 94% of the time, so **more data make it worse.** Newey–West standard errors lower the rates to 59% and 85% but do not fix the problem. The median R^2 is about 0.17 at every sample length, and a third of the regressions report an R^2 above 0.3.



The problem has three remedies. Differencing both series answers a different question, about how *changes* relate. Including lagged levels of both variables restores standard inference for some coefficients. If the levels truly move together, one can model the relationship as cointegration. In finance, price levels, dividend–price ratios, interest rates, and volatility are all at or near unit roots. A regression of one price level on another is spurious until shown otherwise.

8.4 Cointegration and error correction

Two I(1) series are **cointegrated** if some linear combination of them, $y_t - \beta x_t$, is stationary. The series share a common stochastic trend and move together. The gap between them keeps reverting. The combination is the **cointegrating relationship**.

Estimation. The Engle & Granger (1987) procedure has two steps. First, regress y_t on x_t in levels. When the series are cointegrated, this regression is not spurious. The estimate of β is **superconsistent**, converging at rate T rather than $\sqrt{T}$ (Stock, 1987). Superconsistency arises because any wrong β leaves a non-stationary residual that makes the sum of squared residuals large. Second, test the residual for a unit root, using Engle–Granger critical values. These are more negative than Dickey–Fuller ones, because the least-squares step chooses the residual to look stationary. The Johansen (1991) procedure estimates the whole system by maximum likelihood and allows for several cointegrating relationships.

The error-correction model. Cointegration is equivalent to a model in which changes respond to the previous period's deviation from equilibrium:

$$\Delta y_t = \alpha \left(y_{t-1} - \beta x_{t-1} \right) + \text{lagged changes} + \varepsilon_t,$$

Here $\alpha < 0$ measures the speed at which a gap closes. This equivalence is the Granger representation theorem.

In finance. Pairs trading is cointegration in practice. Traders trade the spread between two related prices because they expect it to revert (Gatev, Goetzmann & Rouwenhorst, 2006). Spot and futures prices, yields of different maturities, and log prices and log dividends are other examples. Whether to model a pair as a stationary spread or a stationary ratio is a separate decision (Simple and Log Returns §6.4).

Failure modes.

- **Multiple testing.** Screening 1,000 candidate pairs with a 5% cointegration test finds about 50 "cointegrated" pairs even if none is.
- **Low power and breaks.** Cointegration tests cannot reliably separate slow mean reversion from none. A relationship that held in the sample can break precisely when a position is open.
- **Inference on β.** Superconsistency does not make the levels regression's t-statistics standard, and inference needs estimators designed for it, such as dynamic OLS with leads and lags.

8.5 Vector autoregressions, impulse responses, and identification

The reduced-form VAR. Stack several series in a vector $\mathbf{y}_t$ and regress each on lags of all of them:

$$\mathbf{y}_t = \mathbf{c} + A_1\mathbf{y}_{t-1} + \cdots + A_p\mathbf{y}_{t-p} + \mathbf{e}_t, \qquad \text{Var}(\mathbf{e}_t) = \Sigma_e.$$

OLS estimates each equation. Sims (1980) proposed VARs as an alternative to large structural models, whose identifying restrictions Sims called "incredible." With k variables and p lags, the model has k^2p coefficients, so shrinkage priors are standard for forecasting (§5.6).

Granger causality asks whether the past of one series improves forecasts of another, given the second series's own past (Granger, 1969). It is a test of predictive content and nothing more. [Fact] Stock prices Granger-cause GDP because prices anticipate economic news, not because the stock market drives output. Forecasts of rain Granger-cause rain.

Structural VARs and the identification problem. The reduced-form errors $\mathbf{e}_t$ are combinations of economically meaningful **structural shocks** ε_t, such as a monetary policy shock or an oil supply shock. The relation is $\mathbf{e}_t = B\varepsilon_t$. With structural shocks normalised to unit variance, $\Sigma_e = BB'$. That gives $k(k+1)/2$ equations for the k^2 unknowns in B, so $k(k-1)/2$ restrictions are needed. This is the simultaneity problem of §6.4 in dynamic form. The table lists the standard solutions.

Approach	The restriction	Main weakness
Recursive (Cholesky) ordering	Variables are ordered so that each responds to earlier ones within the period, and not to later ones	Results depend on an ordering that is often arbitrary
Long-run restrictions	Some shocks have no permanent effect on some variables (Blanchard & Quah, 1989)	Sensitive to how low-frequency dynamics are modelled
Sign restrictions	Shocks move variables in theoretically signed directions	Identify a set of models, not a single one
External instruments	An outside measure of the shock serves as an instrument (Mertens & Ravn, 2013)	The instrument's relevance and exclusion

High-frequency identification is the external-instrument approach with the most influence on finance. The change in federal funds futures prices in a narrow window around a monetary policy announcement measures the *surprise* component of the decision (Kuttner, 2001; Gürkaynak, Sack & Swanson, 2005). Within thirty minutes nothing else systematically moves interest rates, so the surprise is plausibly exogenous. The approach is an event study (§9.3) used as an instrument. [Contested] A surprise can also reveal the central bank's private information about the economy, which complicates the approach. A rate rise can then signal strength as well as tightening (Nakamura & Steinsson, 2018).

Impulse responses and local projections. An **impulse response** traces the effect of a one-time structural shock on each variable over subsequent horizons. A VAR computes it by iterating the estimated dynamics forward. **Local projections** estimate it directly, with one regression per horizon of y_{t+h} on the shock and controls (Jordà, 2005). Local projections are more robust to misspecified dynamics and noisier. [Fact] In population, with unrestricted lag structures, the two methods estimate the same impulse responses. The two methods therefore differ in finite-sample bias and variance, not in what they estimate (Plagborg-Møller & Wolf, 2021). The dependent variable at horizon h overlaps with its neighbours, so local-projection inference faces the overlap problem of §4.5. Augmenting the regression with lags of the outcome makes that inference much simpler (Montiel Olea & Plagborg-Møller, 2021).

8.6 Volatility: ARCH, GARCH, and realised variance

The stylised fact. [Fact] Asset returns are nearly uncorrelated, but their squares and absolute values are strongly and persistently autocorrelated. Calm periods and turbulent periods cluster (Cont, 2001). Volatility is predictable even when returns are not.

ARCH and GARCH. Engle (1982) modelled the conditional variance as a function of past squared shocks. The GARCH(1,1) model (Bollerslev, 1986) adds the past conditional variance:

$$\sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2,$$

where $\varepsilon_t = \sigma_t z_t$ and z_t is an i.i.d. shock with unit variance. Its properties follow directly.

- **The long-run variance** is $\omega/(1-a-b)$, provided $a+b<1$.
- **Persistence** is $a+b$. A variance shock has a half-life of $\ln 0.5/\ln(a+b)$ periods. The half-life is 34 days at 0.98 and 69 days at 0.99. [Fact] Estimates for daily equity index returns typically give $a+b$ in the high 0.9s.
- **GARCH(1,1) is an ARMA(1,1) in squared shocks.** Writing $v_t = \varepsilon_t^2 - \sigma_t^2$, which is a martingale difference, gives $\varepsilon_t^2 = \omega + (a+b)\varepsilon_{t-1}^2 + v_t - b\,v_{t-1}$. The results of §8.2 apply directly.

Researchers estimate GARCH models by maximum likelihood. Gaussian quasi-maximum likelihood with sandwich standard errors is consistent even though returns are not normal (§5.4). **Asymmetric** versions let volatility respond more to negative returns than to positive ones, which fits equity data better (Nelson, 1991; Glosten, Jagannathan & Runkle, 1993).

Realised variance. With intraday data, one can *measure* variance instead of modelling it. Summing squared high-frequency returns over a day estimates that day's variance. The error shrinks as the sampling interval shrinks, until market microstructure noise dominates (Andersen, Bollerslev, Diebold & Labys, 2003). This applies Merton's asymmetry (§2.5). Finer sampling leaves the precision of a mean return unchanged and greatly improves the precision of a variance. The **HAR** model regresses tomorrow's realised variance on its daily, weekly, and monthly averages. It is simple, and few models outperform it (Corsi, 2009).

Evaluating volatility forecasts raises a further problem. The target is never observed, so analysts compare forecasts with a noisy proxy such as a squared return. Patton (2011) showed that only certain loss functions, notably mean squared error and the "QLIKE" loss, rank forecasts consistently when the proxy is noisy but unbiased. Other losses can prefer the wrong forecast. [Fact] In the most thorough comparison, Hansen & Lunde (2005) compared 330 ARCH-type models. For exchange rates, they found no evidence that any model outperforms a GARCH(1,1). For IBM stock returns, models with a leverage effect were clearly better.

8.7 Structural breaks and parameter instability

A relationship estimated over a long sample assumes that the parameters were constant throughout. Tests for **structural breaks** check that assumption.

- **A known break date.** The Chow (1960) test compares coefficients estimated before and after the date.
- **An unknown break date.** Computing the Chow statistic at every candidate date and taking the largest is a multiple-testing problem. The maximum therefore has its own, larger critical values (Andrews, 1993). A Chow test at a date chosen *after looking at a plot* is this sup-test with the wrong critical values.
- **Several breaks.** Bai & Perron (1998) estimate the number and location of multiple breaks.

Breaks and forecasting. When parameters change, a rolling estimation window adapts to the change, and an expanding window is more precise. The choice is therefore a bias–variance trade-off. Pesaran & Timmermann (2007) show that including some pre-break data can improve forecasts even after a break. It trades a little bias for a large reduction in variance. [Practice] Averaging forecasts across several window lengths is a common robust alternative to choosing one. Regime-switching and change-point models treat instability as a process rather than an event. Market Regimes and Hidden Markov Models §5 covers them.

8.8 Forecasting and forecast evaluation

Choose the loss function first. The conditional mean minimises the mean squared error. The conditional median minimises the absolute error. A conditional quantile minimises the asymmetric "tick" loss of quantile regression (§5.5). A forecast is good or bad *for a given loss*. Evaluating a forecast with a different loss from the one it targets gives misleading rankings.

Evaluate out of sample using only information available at the time. A **pseudo-out-of-sample** exercise re-estimates the model at each forecast date using only data available at that date, with either a rolling or an expanding window. "Only data available at that date" is stricter than it sounds. It covers the choice of model, the choice of predictors, the tuning of hyperparameters, and the data themselves. Agencies revise macroeconomic data after release. A forecast evaluated with final revised data uses information nobody had (Croushore & Stark, 2001). Accounting data become available only months after the period they describe. The out-of-sample R^2 of §3.8 is the standard summary.

Comparing two forecasts. The **Diebold–Mariano** test takes the difference between two forecasts' losses at each date and tests whether its mean is zero, with a HAC standard error (Diebold & Mariano, 1995). Diebold and Mariano designed it to compare *forecasts*. The test is invalid for comparing a model with a larger model that nests it when the researcher estimates both. Under the null that the extra predictors are useless, the larger model still incurs estimation error from estimating them. That error inflates its loss and makes the test too conservative, so the test misses real predictive content. Clark & West (2007) adjust for exactly that estimation noise. Giacomini & White (2006) reframe the question as a comparison of forecasting *methods*, estimation windows included.

The equity premium debate is the best-known application, and researchers have not resolved it. [Contested] The main positions follow.

- Goyal & Welch (2008) examined the predictors proposed in the literature, such as dividend yields, earnings yields, interest rates, and others. They found that the predictors predicted poorly both in and out of sample over the preceding 30 years, were unstable, and would not have helped an investor.
- Campbell & Thompson (2008) showed that imposing weak economic restrictions, such as a non-negative forecast of the premium, lets many predictors beat the historical mean out of sample. The R^2 values are small but economically meaningful (§3.8).
- Cochrane (2008) argued that the right test is joint. If dividend growth is unpredictable, as it appears to be, then the variation in dividend yields has to come from predictable returns.
- Rapach, Strauss & Zhou (2010) found that simple averages of the individual forecasts beat the historical mean out of sample, even though most individual predictors did not. Goyal, Welch & Zafirov (2024) updated the original evaluation with newer predictors and found the record still weak.

The view taken here is that equity premium predictability is real, small, and unstable. The econometric problems of §9.2 make its in-sample strength look larger than it is.

Combine forecasts. Averaging forecasts from different models is one of the most reliable improvements in forecasting (Bates & Granger, 1969). [Fact] Simple equal weights usually beat weights estimated to be optimal. The error in estimating the weights explains this "forecast combination puzzle". That error outweighs the gain from optimising the weights (Smith & Wallis, 2009). The logic is the same as for the shrinkage of §5.6.

§8 Key takeaways

1. Stationarity and ergodicity let one history stand in for many samples. Without them, time averages need not estimate anything.
2. Returns are close to a martingale difference sequence and far from i.i.d. Tests that assume independence are mis-sized on returns, so use heteroskedasticity-robust versions.
3. Linear time-series models describe autocovariances completely, and they cannot detect predictable volatility or non-linear dependence.
4. Unit-root tests have non-standard critical values and low power. Decide whether to difference on economics as much as on tests.
5. A regression of one random walk on an independent one rejects "no relationship" most of the time, and more data make the problem worse. Newey–West does not fix it.
6. Cointegration makes levels regressions meaningful, and screening many pairs for it is a multiple-testing exercise.
7. Granger causality is predictive precedence, not causation. Structural VARs need identifying restrictions, and high-frequency event windows are the most credible source of them for monetary shocks.
8. Volatility is highly persistent and predictable. GARCH(1,1) is an ARMA(1,1) in squared shocks, and realised variance turns high-frequency data into precise measurements of variance.
9. Choose the loss first. Evaluate out of sample with only the information available at the time, use Clark–West instead of Diebold–Mariano for nested models, and average forecasts.

Part V — Practice

9. Financial econometrics

Everything in Parts I–IV applies to financial data, and this section covers where the problems are most severe. Finance combines the problems of the earlier sections in an unusually unfavourable way. The signals are small. The noise is large and fat-tailed. Observations depend on each other across both time and assets, and the predictors are persistent. Thousands of researchers have searched the data for decades, and publishing a finding changes the process that generated it.

9.1 Why financial data are statistically hostile

The table lists the properties of financial data that cause the most trouble, with their consequences.

Property	Consequence	Where
Low signal-to-noise. Monthly return predictability rarely exceeds an R^2 of a few percent	The t-statistic of a strategy's mean is roughly its Sharpe ratio times the square root of the number of years, so a Sharpe ratio of 0.5 needs about 16 years to reach $t = 2$ (18 once the estimation-uncertainty correction of §2.6 is included)	§2.5, §4.7
Fat tails and volatility clustering	Classical and i.i.d. standard errors are wrong, and estimates of higher moments are imprecise	§2.4, §8.6
Common factors across assets	Thousands of stocks at one date are far fewer independent observations; inference clusters by date	§4.4, §9.5
Persistent predictors. Valuation ratios and interest rates are near unit roots	Biased slopes and oversized tests in predictive regressions	§8.3, §9.2
Overlapping horizons	A 40-year sample of annual returns has about 40 observations, whatever the sampling frequency	§4.5
Non-stationarity and adaptation	Relationships break, and published anomalies decay	§8.7, §9.6
Survivorship and look-ahead	Samples that condition on the future, and datasets revised after the fact	§6.4, §9.7
Collective data mining	Decades of research on the same databases turn every new result into the best of an unknown number of tries	§4.8, §9.6
Microcaps	They are about 60% of listed US stocks and about 3% of market value (Fama & French, 2008), so equal-weighted results describe them	§9.6

The effect of publication can be measured. [Fact] McLean & Pontiff (2016) studied 97 return predictors from the academic literature. They found portfolio returns 26% lower in the period between the end of the original sample and publication, and 58% lower after publication. The 26% figure is an upper bound on the share of the original result that was statistical overfitting. The difference between the two numbers is about a third of the original return. It is consistent with investors trading on published results.

9.2 Predictive regressions

The canonical return-forecasting regression pairs a return equation with an equation for a persistent predictor such as the dividend yield:

$$r_{t+1} = \alpha + \beta\, x_t + u_{t+1}, \qquad x_{t+1} = \theta + \phi\, x_t + v_{t+1}.$$

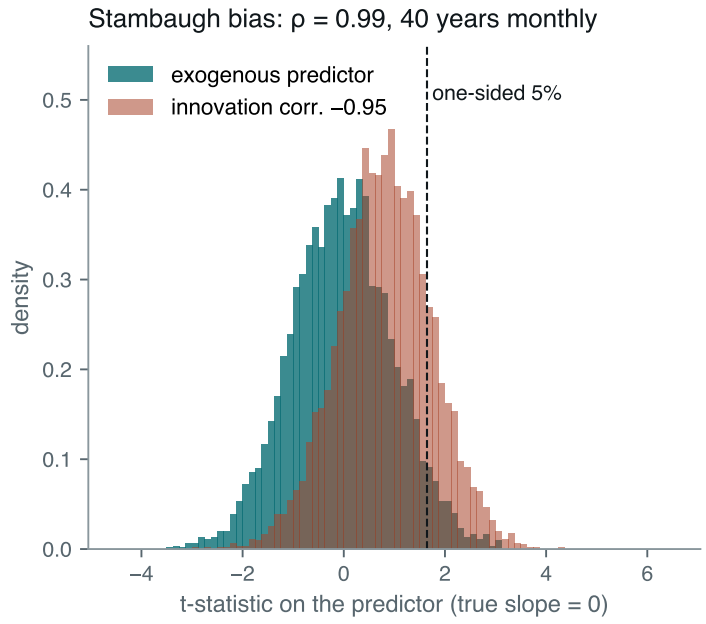
For valuation ratios, the two innovations are strongly negatively correlated. A positive return shock raises the price, and the higher price mechanically lowers the dividend yield in the same period.

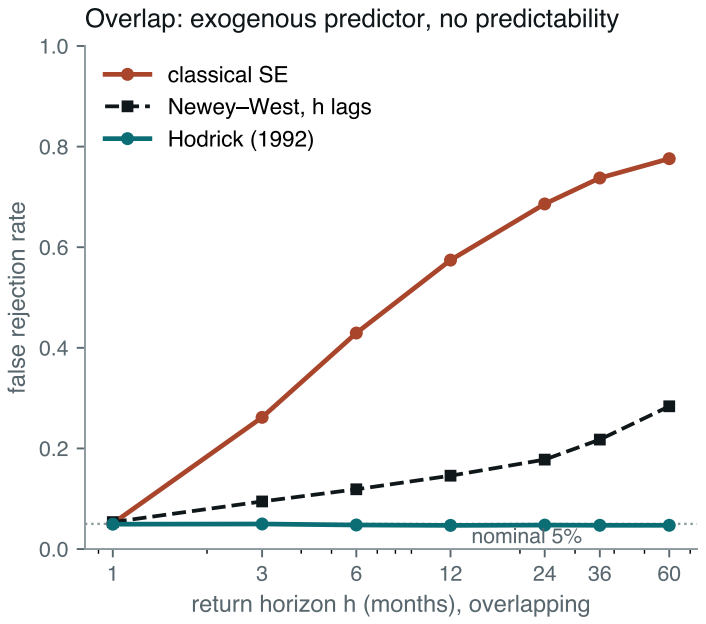
Stambaugh bias. Because x_t is a lagged dependent variable in its own equation, $\hat{\phi}$ is biased downward, by roughly $-(1+3\phi)/T$ (§2.3). The correlation of the innovations transmits that bias to the return equation's slope (Stambaugh, 1999):

$$\mathbb{E}[\hat{\beta} - \beta] \approx \frac{\sigma_{uv}}{\sigma_v^2}\, \mathbb{E}[\hat{\phi} - \phi],$$

where σ_{uv} is the covariance of the innovations and σ_v^2 is the variance of the predictor's innovation. With $\sigma_{uv} < 0$ and $\hat{\phi}$ biased down, $\hat{\beta}$ is biased *up*, toward finding that a high dividend yield predicts high returns.

The left panel of the figure simulates 40 years of monthly data with no predictability at all, a predictor with $\phi = 0.99$, and an innovation correlation of -0.95. [Simulated] The estimate of the predictor's autoregressive coefficient is 0.009 too low on average, close to the approximation's 0.008. The average estimated slope is about 1.4 standard errors above zero (the approximation gives 1.2). The average t-statistic is 0.82 instead of zero. A one-sided 5% test for positive predictability rejects 18% of the time.





Long-horizon regressions. Researchers once thought that regressing returns over h months on today's predictor revealed predictability that short horizons missed, because R^2 values grow with the horizon. Much of that growth is mechanical. The right panel makes the predictor exogenous, which removes the Stambaugh problem, and keeps only the overlap of §4.5. [Simulated] With no predictability, the median R^2 at a 60-month horizon is 5.5%, and one simulated sample in ten reports an R^2 above 27%. Classical standard errors reject a true null 57% of the time at 12 months and 78% of the time at 60. Newey–West with h lags rejects 15% and 28% of the time. Hodrick (1992) standard errors, which impose the null of no predictability when estimating the meat, stay at 5%. Valkanov (2003) derives the non-standard behaviour of long-horizon regressions formally. Boudoukh, Richardson & Whitelaw (2008) show that estimates at different horizons are highly correlated. The long-horizon evidence therefore adds little to the short-horizon evidence it comes from.

What to do.

- **Estimate at the shortest sensible horizon**, and treat long-horizon results as a presentation of short-horizon evidence, not as additional evidence.
- **Use inference designed for persistent predictors.** Bias-corrected slopes, Hodrick standard errors, or the IVX approach, which remains valid whether the predictor is stationary, near a unit root, or exactly at one (Kostakis, Magdalinos & Stamatogiannis, 2015).
- **Evaluate out of sample** (§8.8), with the caveat that out-of-sample tests have low power and can miss real but small predictability.
- **Test the joint implications.** If a valuation ratio does not predict cash flows, it must predict returns (§8.8).

9.3 Event studies

The design. An event study measures how security prices respond to news, such as an earnings announcement, a merger, an index inclusion, or a regulatory decision. It compares returns around the event with what they would normally have been (Fama, Fisher, Jensen & Roll, 1969; MacKinlay, 1997). The procedure has three steps.

1. Fix the event date and an **event window**, such as the day before to the day after.
2. In an earlier **estimation window**, fit a model of normal returns, typically the market model $r_{it} = a_i + b_i r_{mt} + e_{it}$.
3. Compute each day's **abnormal return**, $AR_{it} = r_{it} - (\hat{a}_i + \hat{b}_i r_{mt})$, sum it over the event window to get a **cumulative abnormal return** (CAR), and average across events.

The identifying assumption. Prices incorporate the news within the window, nothing else moves the stock during the window, and the normal-return model is adequate. Over short windows the last condition barely matters. Expected daily returns are tiny compared with event-day volatility, so any reasonable model gives nearly the same answer (Brown & Warner, 1985).

The joint-hypothesis problem. Any test of whether prices react "correctly" is jointly a test of a model of expected returns (Fama, 1970). Over a three-day window the model is nearly irrelevant. Over three years it dominates the result.

Failure modes.

- **Event clustering.** When many firms share an event date, as with an industry-wide regulation, their abnormal returns are cross-sectionally correlated. Conventional t-statistics are then too large. Form a portfolio of the affected firms and treat it as one observation, or adjust for the correlation (Kolari & Pynnönen, 2010).
- **Event-induced variance.** Volatility rises on event days, which invalidates tests that standardise by the estimation-window variance.
- **Leakage and anticipation.** If news leaks before the window, the measured response is too small.
- **Confounding news** inside the window.
- **Long horizons.** Over months or years, small errors in the expected-return model compound. Long-horizon returns are skewed, and overlapping events are correlated. [Contested] Barber & Lyon (1997) show that common long-horizon test statistics are misspecified. Fama (1998) argues that long-term return anomalies tend to disappear under reasonable changes in method. Mitchell & Stafford (2000) show that correcting for the correlation between overlapping events removes much of the evidence for long-run abnormal performance after corporate events. Behavioural-finance researchers dispute how general that conclusion is.

The difference-in-differences literature uses "event study" for a related but different method. It is a DiD with separate coefficients for each period around treatment (§7.5, §12.3).

9.4 Testing asset pricing models

A **factor model** says that a few common sources of risk, the **factors**, explain the differences in expected returns across assets. The vector f_t holds the factor returns. An asset's **loadings** (its betas) b_i measure how strongly its return moves with each factor. The model predicts that expected excess returns follow the loadings. Assets with more exposure to a priced factor earn more, and no other exposure earns a premium. Testing a factor model means testing that prediction.

Time-series tests. When the factors are themselves traded portfolio returns, regress each test asset's excess return on the factor returns, $r^e_{it} = \alpha_i + b_i' f_t + e_{it}$. The model predicts that every α_i is zero. The Gibbons, Ross & Shanken (1989) test checks them jointly. It has a direct economic interpretation. It asks whether adding the test assets to the factors would significantly raise the highest attainable Sharpe ratio.

Cross-sectional tests: Fama–MacBeth. The two-pass procedure of Fama & MacBeth (1973) first estimates each asset's factor loadings from time-series regressions. Then, at every date t, it regresses the cross-section of returns on the estimated loadings, or on firm characteristics. This gives a slope $\hat{\lambda}_t$, the period's reward per unit of exposure. The estimate is the time-series average of $\hat{\lambda}_t$. Its standard error is the standard deviation of $\hat{\lambda}_t$ divided by $\sqrt{T}$.

The logic of the standard error defines what the procedure fixes and what it does not. **Each date contributes one observation, so the procedure handles correlation across assets within a date automatically. It does not handle correlation of $\hat{\lambda}_t$ over time.** With persistent characteristics the slopes are serially correlated, and the time series of $\hat{\lambda}_t$ needs a Newey–West standard error. It also does not correct for firm-level persistence in the residuals (§9.5).

Errors in variables. The loadings used in the second pass are estimates. The second-pass slopes therefore suffer from the attenuation of §6.4, and their standard errors are too small. Shanken (1992) derived the correction to the standard errors. Grouping stocks into portfolios reduces the noise in the loadings, but it also discards cross-sectional variation. Ang, Liu & Schwarz (2020) therefore argue for individual stocks with an appropriate correction. Jegadeesh, Noh, Pukthuanthong, Roll & Wang (2019) use loadings estimated in separate subsamples as instruments.

Two problems specific to these tests.

- **Useless factors look priced.** [Fact] Kan & Zhang (1999) show that a factor unrelated to returns can appear significantly priced in two-pass regressions, and that the problem gets *worse* with longer samples. This is the weak-instrument problem of §7.3 in asset-pricing form. A factor with near-zero loadings makes the second pass divide by a number near zero. Tests robust to this problem exist (Kleibergen, 2009).
- **The usual test assets are easy to fit.** Portfolios sorted on size and book-to-market have a strong factor structure. Almost any candidate factor correlated with the size and value factors therefore fits their average returns with a high cross-sectional R^2. Lewellen, Nagel & Shanken (2010) recommend adding test portfolios unrelated to the sorting characteristics, such as industries. They also recommend reporting a GLS R^2, which is tied to the efficiency of the model's factor-mimicking portfolio and imposes a stricter test.

The SDF approach. One can write any linear factor model as a stochastic discount factor that is linear in the factors, and estimate it by GMM from $\mathbb{E}[M_{t+1}R^e_{t+1}] = 0$ (§5.3). The Hansen & Jagannathan (1997) distance then measures how far a misspecified model is from pricing the assets correctly, in units an investor can interpret.

9.5 Panel standard errors in finance

Finance panels have two dimensions of dependence. A **firm effect** is persistence within a firm over time, for example in leverage or in a firm's residual returns. A **time effect** is correlation across firms at a date, as with market-wide shocks. The two need different corrections. The literature long used the wrong one for its setting. [Fact] Petersen (2009) surveyed 207 panel-data papers in the three leading finance journals and found that 42% did not adjust standard errors for dependence in the residuals. Petersen also found that the common practice in asset pricing, Fama–MacBeth, corrects for time effects and not for firm effects, while White standard errors correct for neither. The table matches the correction to the dependence.

Dependence in the residuals	Use
Time effect only (cross-sectional correlation at each date)	Fama–MacBeth, or cluster by date, given many dates
Firm effect only (persistence within firms)	Cluster by firm
Both	Two-way clustering, or date fixed effects plus firm clustering if the time effect is common to all firms
Time effects that persist across dates	Driscoll–Kraay standard errors (Driscoll & Kraay, 1998), or Newey–West on the Fama–MacBeth slopes

The limits of §4.4 still apply. Two-way clustering with ten years of annual data has only ten time clusters, which is not enough.

9.6 Multiple testing, the factor zoo, and replication

Finance has run the multiple-testing experiment of §4.8 on a very large scale. Researchers have published several hundred variables as predictors of the cross-section of stock returns, and most researchers tested them on the same US databases. [Contested] The field disputes how many of them are real. The main positions follow.

- **Raise the threshold.** Harvey, Liu & Zhu (2016) argue that, given the search, a new factor should clear $t > 3.0$.
- **Most anomalies fail.** Hou, Xue & Zhang (2020) re-examine 452 anomalies with NYSE breakpoints and value-weighted returns, which limit the influence of microcaps. 65% fail to reach $|t| \geq 1.96$, and 82% fail a multiple-testing hurdle of 2.78.
- **Most anomalies replicate.** Jensen, Kelly & Pedersen (2023) build 153 factors in 93 countries. Using a Bayesian model that accounts for multiple testing, they find that the large majority replicate, against the 35% implied by Hou, Xue & Zhang's criteria. The factors cluster into 13 themes and work out of sample and across countries. Chen & Zimmermann (2022) reproduce nearly all of several hundred published predictors using the original papers' own methods, and publish the code.
- **The case for higher hurdles is hard to identify.** Chen (2025) argues that the data cannot determine how much higher the hurdle should be, because results that failed the old hurdle were never published. False-discovery rates for published findings can be estimated well, and they are modest.

The view taken here is that these positions conflict less than their titles suggest. Hou, Xue & Zhang ask whether an anomaly survives an implementation that downweights microcaps, which is closer to asking whether the anomaly is economically important. Jensen, Kelly & Pedersen, and Chen & Zimmermann, ask whether the original statistical finding reproduces. Both camps find that many anomalies are concentrated in small, illiquid stocks and have weakened since publication. The practical implication is the same from either side. **Recommendation: before believing a new predictor, test it value-weighted, net of costs, out of sample, and against the existing factor themes.**

Data snooping in trading rules is the same problem applied to whole trading rules instead of single predictors. Sullivan, Timmermann & White (1999) applied White's reality check (§4.8) to several thousand technical trading rules on a century of daily Dow Jones data. They adjusted for the full search that produced the best rule. In the ten years that followed their original sample, the best rules were not profitable. Earlier, Lo & MacKinlay (1990) had shown that forming test portfolios on characteristics chosen *because* they are known to be related to returns biases asset-pricing tests toward rejection.

9.7 Backtests are econometrics

A backtest estimates the mean return of a strategy, and the mean is the hardest parameter in finance to estimate (§2.5). The estimate comes from dependent data (§4.5). It follows a search over configurations (§4.8). It frequently uses information that was not available at the time (§6.4). It deducts costs that the analyst assumed rather than measured. Every failure in this chapter can appear in a backtest, and the output flags none of them. The table lists the common failures with their fixes.

Failure	Mechanism	Fix
Look-ahead in the data	Restated financials, revised macroeconomic data, today's index constituents	Point-in-time data with realistic reporting lags
Look-ahead in the method	Full-sample normalisation, z-scores, or hyperparameters tuned on the test period	Strict walk-forward estimation of *everything*
Survivorship	Delisted securities, or their final returns, missing from the data	Include delistings and delisting returns
Best of many configurations	The reported strategy is the maximum of a noisy search	Holdout data; deflated Sharpe ratio; reality check (§4.8)
Overlapping positions	Holding periods longer than the rebalancing interval	Hodrick or HAC standard errors; the T/h rule (§4.5)
Microcap dominance	Equal weights on stocks that cannot be traded in size	Value weights, NYSE breakpoints, liquidity filters
Assumed costs	Spreads and impact omitted or understated	Cost models calibrated to trade size and capacity
Too short a sample	A Sharpe ratio of 0.5 needs about 16 years to reach $t = 2$	Compute the power first (§4.7)

Two numbers from earlier sections belong next to every backtest. Suppose a strategy's true Sharpe ratio is 0.3 and an analyst tests it for ten years. A backtest that clears $t = 2$ then reports a Sharpe ratio of about 0.8 on average (§4.7). The average published anomaly also lost more than half its return after publication (§9.1). The expected out-of-sample performance of a strategy selected from a search is materially below its backtest, even if the analyst did nothing wrong. Portfolio Construction, Appendix A.45–A.46, gives the standard error of a Sharpe ratio and the deflated Sharpe ratio. Momentum in Financial Markets and Market Regimes and Hidden Markov Models cover walk-forward and purged validation in detail.

§9 Key takeaways

1. Financial data combine small signals, fat tails, cross-sectional and serial dependence, persistent predictors, and decades of collective data mining. Assume every problem in Parts I–IV is present until shown otherwise.
2. Published return predictors lose about a quarter of their return out of sample and more than half after publication.
3. Persistent predictors whose innovations are correlated with returns bias predictive slopes upward. Long-horizon regressions add little independent evidence and inflate R^2 mechanically.
4. Short-window event studies are robust to the choice of expected-return model, and long-window ones depend mainly on it. Correct for the clustering of events in time.
5. Fama–MacBeth standard errors handle correlation across assets within a date, not persistence over time or within firms. First-pass betas are noisy regressors.
6. Useless factors can look priced in two-pass tests, and size- and value-sorted test portfolios are easy to fit. Add unrelated test assets.
7. Identify whether the residuals have firm effects, time effects, or both before choosing a standard error. Many published papers did not.

8. The replication debate depends partly on definitions. Either way, test a new predictor value-weighted, net of costs, out of sample, and against the existing factor themes.
9. A backtest is an econometric estimate of the hardest parameter in finance. Expect its out-of-sample performance to be materially lower, even when the analyst did nothing wrong.

10. Prediction, model selection, and machine learning

Machine learning has changed empirical finance more than any other development of the past fifteen years. Many people misunderstand its relationship to econometrics in both directions. Econometricians sometimes treat machine learning as curve-fitting without inference. Machine-learning practitioners sometimes treat econometrics as a collection of outdated linear models. In fact the two fields answer different questions, share most of their mathematics, and are most powerful in combination.

10.1 Two questions: $\hat{Y}$ and $\hat{\beta}$

Breiman (2001) described "two cultures" of statistical modelling. One assumes a data-generating model and estimates its parameters. The other treats the mechanism as unknown and optimises predictive accuracy. Shmueli (2010) put the same point as *explaining* versus *predicting*. Mullainathan & Spiess (2017) give the sharpest form for applied work. Some problems need $\hat{Y}$, a good prediction of the outcome. Others need $\hat{\beta}$, a good estimate of a parameter.

Machine learning methods target $\hat{Y}$ problems and handle them very well. Their parameters serve prediction, not interpretation. When predictors are correlated, a LASSO fitted on two random halves of the same data can select quite different variables while producing nearly identical predictions. The predictions are stable, but the coefficients behind them are unstable. **Feature importances from a predictive model are statements about the model, not about the world.**

Some decisions need only a prediction. Kleinberg, Ludwig, Mullainathan & Obermeyer (2015) call them "prediction policy problems." Deciding whether to carry an umbrella requires a forecast of rain, not the causal effect of umbrellas on rain. In finance, return forecasting, risk models, and default prediction are $\hat{Y}$ problems. Two examples of $\hat{\beta}$ problems are evaluating whether a hedging programme reduced a firm's cost of capital and evaluating whether a regulation reduced liquidity.

The two errors are symmetric. Reading a predictive model's coefficients causally is the mistake of §1.4. Using a causal estimate to predict outcomes in a different population or period assumes external validity, which the design never established.

10.2 Overfitting, measured

Consider linear regression with k estimated coefficients, n observations, and error variance σ^2. The expected mean squared error in the estimation sample is $\sigma^2(1 - k/n)$. The expected error on new observations with the same regressors is $\sigma^2(1 + k/n)$. [Fact] The gap, about $2\sigma^2 k/n$, is the **optimism** of in-sample fit. Mallows's C_p and the AIC rest on it. With 20 coefficients and 100 observations, in-sample error understates the noise variance by 20%. Out-of-sample error exceeds it by 20%.

This arithmetic is decisive for return prediction. A short derivation shows why. Suppose the true predictors explain a share ρ^2 of the variance of returns, and OLS estimates them with k coefficients. In units of the return variance, the out-of-sample mean squared error of the forecast is roughly $(1-\rho^2)(1+k/n)$. The historical mean's is roughly $1+1/n$. Out-of-sample R^2 compares the two, $R^2_{\text{OS}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{null}}$ (§3.8). It is approximately

$$R^2_{\text{OS}} \approx \rho^2 - \frac{k}{n}\left(1-\rho^2\right) \approx \rho^2 - \frac{k}{n},$$

which is positive only if $n > k/\rho^2$. With a true monthly R^2 of 1% and ten predictors, that threshold is 1,000 months, or 83 years. **An unrestricted regression of monthly returns on ten individually sensible predictors is expected to lose to the historical mean out of sample, even if every predictor is truly informative.** The calculation is this chapter's own, not a published result. It explains why shrinkage, sign restrictions, and forecast combination (§8.8) matter so much in this setting. Each one reduces the effective k.

A current debate concerns this same logic. Kelly, Malamud & Zhou (2024) argue that heavily over-parameterised models can predict returns *better* than parsimonious ones, a "virtue of complexity." Their models have far more parameters than observations and strong ridge shrinkage. [Contested] Nagel (2025) shows that the method, trained in short rolling windows, reduces to a volatility-timed momentum strategy. That strategy happened to perform well historically. On artificial data in which returns reverse, the method constructs the same strategy. There it performs poorly.

10.3 Model selection: AIC, BIC, and cross-validation

Choosing among models trades fit against complexity. The three standard tools target different things, as the table shows.

Criterion	Formula or procedure	What it targets	Tendency
AIC	$-2\log L + 2k$	Expected out-of-sample log loss	Picks larger models. Good for prediction
BIC	$-2\log L + k\log n$	The true model, if it is among the candidates	Picks smaller models. Consistent for the true order
Cross-validation	Fit on part of the data, score on the held-out part, rotate	Out-of-sample loss, directly	Accurate only if its splits prevent leakage (§10.6)

L is the maximised likelihood, and k is the number of estimated parameters.

Inference after selection is invalid. Researchers often choose a model with the data and then report that model's standard errors as if the model had been fixed in advance. The resulting confidence intervals are too narrow. After data-driven selection, the distribution of an estimator is not normal. The distribution cannot be estimated uniformly well (Leeb & Pötscher, 2005). [Fact] The most common instance in applied work is choosing control variables by their significance. The selection drops a control that is insignificant in the outcome equation but strongly correlated with the treatment. The omitted-variable bias of §3.5 then returns. §10.5 gives the fix.

10.4 Regularisation

Shrinkage (§5.6) is the main defence against overfitting when the number of candidate predictors is large. Four forms are common:

- **Ridge** shrinks all coefficients proportionally. It suits many small, correlated effects. Return predictability usually consists of such effects.
- **LASSO** shrinks coefficients and sets some exactly to zero (Tibshirani, 1996). Its selections are unstable when predictors are correlated, and the retained coefficients are biased toward zero.
- **Elastic net** mixes the two penalties.
- **Trees, boosting, and neural networks** allow nonlinearity and interactions. Depth, learning rate, early stopping, and dropout regularise them, instead of an explicit coefficient penalty.

Every penalty strength or architecture choice is a hyperparameter, and cross-validation tunes it. Gu, Kelly & Xiu (2020) give the benchmark evidence for return prediction. With more than 900 candidate predictors, ordinary least squares on all of them had a negative out-of-sample R^2. Penalised and dimension-reduction linear models recovered small positive values. Tree ensembles and neural networks did best. The best neural network reached a monthly out-of-sample R^2 of about 0.4% for individual stocks. By the arithmetic of §3.8, those numbers are economically meaningful. By the arithmetic of §10.2, they are about what one should expect careful regularisation to deliver.

10.5 Machine learning for causal parameters

Machine learning is now used *inside* causal estimators. This is the most important recent development at the boundary between prediction and causal estimation. The problem it solves is common. The target is the effect θ of a treatment D on Y, and X contains many potential confounders. The confounders may outnumber what a regression can handle without overfitting, or they may enter in unknown non-linear ways.

Why naive approaches fail. A LASSO of Y on D and X drops controls that are weakly related to Y, even when they are strongly related to D. The regularisation itself then creates omitted-variable bias.

Post-double selection. Belloni, Chernozhukov & Hansen (2014) first select the controls that predict Y. They then select, separately, the controls that predict D. Finally, they regress Y on D and the *union* of the two sets. This keeps a confounder that is strongly related to either side. Inference on θ is valid under approximate sparsity.

Double/debiased machine learning (DML). Chernozhukov et al. (2018) generalise the idea. Take the partially linear model $Y = \theta D + g(X) + u$ and $D = \pi(X) + v$. Here $\pi(X)$ is a flexible first stage, the non-linear counterpart of the coefficient π of §7.3. Write $\ell(X) \equiv \mathbb{E}[Y \mid X] = \theta\,\pi(X) + g(X)$ for the best prediction of Y from X alone. This function differs from g, because X predicts Y partly through D. The estimator has two steps.

1. Predict Y from X with any machine-learning method to get $\widehat{\ell}(X)$, and predict D from X the same way to get $\widehat{\pi}(X)$.
2. Regress the residual $Y - \widehat{\ell}(X)$ on the residual $D - \widehat{\pi}(X)$. The slope is $\widehat{\theta}$.

This is Frisch–Waugh–Lovell (§3.4) with flexible first stages. It rests on two ingredients:

- **Neyman orthogonality.** The moment condition $\mathbb{E}\big[(Y - \ell(X) - \theta(D - \pi(X)))(D - \pi(X))\big] = 0$ is insensitive, to first order, to small errors in the two prediction functions. The regularisation bias of the machine learning step therefore does not affect $\hat{\theta}$.
- **Cross-fitting.** The method estimates the prediction functions on folds of the data that it does not use to compute the residuals. Their overfitting therefore does not affect the second step.

The result is a $\sqrt{n}$-consistent, approximately normal $\hat{\theta}$ with an ordinary sandwich standard error. It is the general moment-condition form of §1.5 with a carefully chosen ψ. For effects that vary with characteristics, **causal forests** estimate the conditional average treatment effect with valid confidence intervals (Wager & Athey, 2018). Athey & Imbens (2019) survey these methods.

Machine learning does not create identification. DML still requires unconfoundedness given X (§6.6). It makes adjusting for many controls feasible without bias. It does not make the controls sufficient. Machine learning is an estimator, not a research design.

10.6 Validation with dependent data

Cross-validation estimates out-of-sample loss accurately only if the held-out data are truly unseen. Random K-fold cross-validation assumes that the observations are exchangeable. With time series, overlapping labels, or assets that share shocks, that assumption fails. Information then leaks from the training folds into the test folds. Leaks take three common forms:

- **Serial dependence.** A test observation's neighbours in time belong to the training set and carry information about it.
- **Overlapping labels.** A 20-day forward return on Monday and one on Tuesday share 19 days. With one in training and one in testing, the test partly scores the model on data it learned from.
- **Cross-sectional dependence.** Stocks on the same date share market and industry shocks, so splitting assets rather than dates leaks the date's shock.

The remedies follow from the leaks:

- **Walk-forward (forward-chaining) validation.** Train only on the past, and test on the future.
- **Purging and embargo.** Remove training observations whose label windows overlap the test period. Also remove a buffer after it (López de Prado, 2018; Market Regimes and Hidden Markov Models §11.3 and Appendix A.39).
- **Group by date** when splitting panels.
- **Nested validation.** Tune hyperparameters inside the training folds only. Keep a final holdout and use it once.

Hyperparameter search is multiple testing. This is one more problem beyond the leaks above. The cross-validated loss of the best of 200 configurations is an optimistic estimate of that configuration's out-of-sample loss, for exactly the reason given in §4.8. The holdout measures how optimistic the estimate is.

§10 Key takeaways

1. Machine learning answers $\widehat{Y}$ questions, and econometric designs answer $\widehat{\beta}$ questions. Feature importances describe a model, not the world.
2. In-sample fit overstates out-of-sample fit by about $2\sigma^2 k/n$. For return prediction, an unrestricted regression needs roughly k/ρ^2 observations just to beat the historical mean.
3. AIC targets prediction, BIC targets the true model, and cross-validation targets out-of-sample loss directly. Standard errors after data-driven model selection are invalid.
4. Shrinkage is the main defence against overfitting. Realistic out-of-sample monthly R^2 values for individual stocks are fractions of a percent, and they are economically meaningful.
5. Selecting controls by significance recreates omitted-variable bias. Use double selection or double machine learning, whose orthogonal moments and cross-fitting deliver valid inference on a causal parameter.
6. Machine learning is an estimator, not a research design. DML still needs unconfoundedness.
7. Random K-fold cross-validation leaks on time series, overlapping labels, and panels. Walk forward, purge, embargo, group by date, and treat hyperparameter search as multiple testing.

11. The practice of applied econometrics

The previous sections describe methods. This one describes the habits that decide whether methods produce knowledge. Almost none of the material is mathematically difficult. Researchers routinely skip almost all of it.

11.1 Where the effort should go

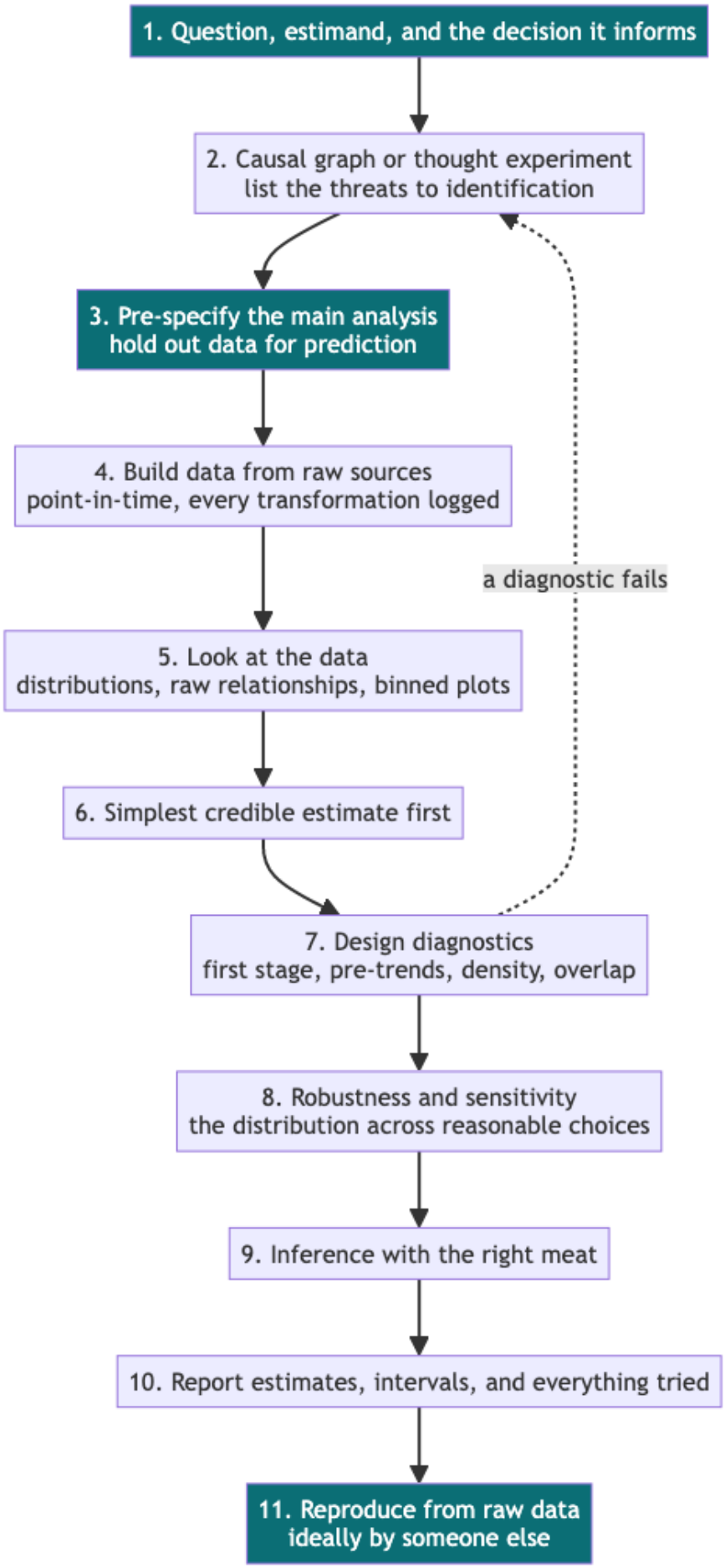
[Practice] The list ranks the parts of an analysis by how much each typically affects whether a result is right:

1. **The question and the estimand.** This part fixes the decision that the answer will inform and the exact population quantity that would inform it. A wrong estimand makes everything downstream irrelevant.
2. **The data.** This part covers how each variable was measured, when it was known, and who is missing. Data errors are the most common cause of wrong results, and the least discussed.
3. **The identification strategy**, or for a prediction problem, the validation design. Here the orthogonality condition holds or fails.
4. **The specification**: functional form, controls, sample definition.
5. **Inference**: the meat.
6. **The estimator.**
7. **Software and computation.**

Researchers usually allocate effort in close to the reverse order, because the interesting tools are at the bottom of the list and the tedious work is at the top. The allocation that works follows the list. The early stages are most of the analysis, not preliminaries to it.

11.2 A workflow

The diagram sets out the workflow, including the loop back to identification when a diagnostic fails.


1. Question, estimand, and the decision it informs
2. Causal graph or thought experiment
list the threats to identification
3. Pre-specify the main analysis
hold out data for prediction
4. Build data from raw sources
point-in-time, every transformation logged
5. Look at the data
distributions, raw relationships, binned plots
6. Simplest credible estimate first
7. Design diagnostics
first stage, pre-trends, density, overlap
a diagnostic fails
8. Robustness and sensitivity
the distribution across reasonable choices
9. Inference with the right meat
10. Report estimates, intervals, and everything tried
11. Reproduce from raw data
ideally by someone else

Researchers most often skip two steps, so they deserve emphasis. **Step 3** separates choices from the data that evaluate them (§4.8). A note dated before the analyst examines the outcome data is enough for most internal research. **Step 5** catches problems that no estimator can catch. Anscombe (1973) constructed four datasets with identical means, variances, correlations, and regression lines. The four datasets look completely different when plotted. Binned scatter plots are the standard way to look at a conditional mean with many observations. They have their own pitfalls. One is residualising the controls in a way that distorts the plot (Cattaneo, Crump, Farrell & Feng, 2024).

The dotted arrow matters as much as the solid ones. When a design diagnostic fails, return to the identification argument. Do not search for a specification in which the diagnostic passes.

11.3 Data work

[Practice] Data errors, not econometric errors, cause most wrong results in empirical finance. Six data errors recur:

- **Joins.** Merges that silently duplicate or drop rows. Count rows before and after every join, and check key uniqueness on both sides.
- **Units and scales.** Returns can be in percent in one table and decimals in another. Other mismatches are thousands versus millions, currencies, and share counts before and after splits.
- **Timing.** Check when each value was *knowable*. Firms release accounting data months after the fiscal period they describe. For this reason, the standard academic convention matches fiscal-year accounting data with returns only from the middle of the following calendar year. Macroeconomic series are revised after release (§8.8). Timestamps need time zones. Closing prices from different exchanges are not simultaneous.
- **Corporate actions and delistings.** Splits, spin-offs, ticker changes, and the final returns of delisted securities.
- **Outliers.** Winsorising at the 1st and 99th percentiles is a choice of estimand as well as a cleaning step. A winsorised mean is a different quantity. In finance the extreme observations are often the economically important ones. Justify every outlier rule by the question, and report it.
- **Missing data.** Dropping incomplete observations is harmless only if missingness is unrelated to the outcome. In finance it rarely is. Firms stop reporting as they approach distress. Funds stop reporting after bad performance.

11.4 Specification and robustness

Functional form. A linear specification estimates the best linear approximation to the conditional mean (§3.2). That approximation is fine when it is close and misleading when it is not. Binned plots, splines, and tests for omitted non-linearity, such as the RESET test (Ramsey, 1969), show which case applies.

Diagnostics are also specification choices. Running a diagnostic, respecifying when it fails, and reporting the specification that passes is a specification search. Decide in advance which diagnostic results would change the specification. Report them whatever they show.

Robustness tables versus specification curves. A conventional robustness table shows a handful of alternatives that the author chose. It therefore shows the tail of a distribution that the author selected. A **specification curve** estimates the model under every combination of reasonable analytical choices, sorts the estimates, and displays which choices produced which estimates (Simonsohn, Simmons & Nelson, 2020). The idea goes back to Leamer's "extreme bounds" (Leamer, 1983), which asked whether a coefficient keeps its sign across all specifications. The idea also goes back to Sala-i-Martin (1997), who argued for looking at the whole distribution of estimates across millions of control sets instead of at the extremes. The claim that a specification curve supports is this: "across the choices a reasonable analyst might make, the estimate lies in this range."

Sensitivity to what was not observed belongs in the same section of a report as sensitivity to what was observed (§6.7).

11.5 Reporting

- **Report magnitudes with intervals**, in units a decision-maker understands (§4.10).
- **Show the design, not just the estimate.** For IV, the first stage and the reduced form. For difference-in-differences, the event-study plot. For regression discontinuity, the binned outcome around the cutoff and the density test. For synthetic control, the pre-treatment fit and the placebo distribution. For selection on observables, the overlap of propensity scores.
- **Report every analysis attempted.** State the number of specifications estimated, the choices made after seeing the data, and which results were confirmatory and which exploratory.
- **Make it reproducible.** Leading economics journals now require data and code sufficient to reproduce the results, and internal research should meet the same standard.

The reason for these norms is measurable. [Fact] Brodeur, Lé, Sangnier & Zylberberg (2016) found a shortage of published test statistics just below conventional significance thresholds and an excess just above them in top economics journals. Brodeur, Cook & Heyes (2020) examined thousands of hypothesis tests in 25 leading journals. They found that the problem varies by method. It is substantial in IV and difference-in-differences studies. It is much smaller in randomised trials and regression discontinuity designs. Designs that leave more discretion to the researcher show more signs of it.

[Practice] In a quantitative research group, the equivalent of a pre-registration is a **research log** that records every backtest run, not just the ones that the group kept. The log replaces a guess at the number of things tried with a count. The deflated Sharpe ratio and every multiple-testing correction need that number as input (§4.8).

11.6 A catalogue of failure modes

The table lists the failure modes of the whole chapter and points to the section that explains each.

Failure	Mechanism	How it shows up, and how to check	Fix
Confounding	A common cause of regressor and outcome	Coefficient moves when controls are added; check stability relative to R^2	Research design; sensitivity analysis (§3.5, §6.7)
Bad control	Conditioning on a mediator or collider	Estimate changes sharply, even in sign, when a control is added; check timing	Drop post-treatment and collider controls (§6.5)
Survivorship and selection	The sample depends on the outcome	Results differ between the survivor sample and the full universe	Point-in-time universe with delistings (§6.4)
Measurement error	Noise in the regressor	Attenuation; within estimates far below pooled	A second measure as an instrument (§6.4)
Simultaneity	The outcome moves the regressor	Implausible signs; check timing and economics	IV; high-frequency identification (§6.4, §8.5)
Weak instruments	A tiny first stage	Small robust first-stage F; IV close to OLS	Anderson–Rubin inference; better instruments (§7.3)
Invalid exclusion	The instrument affects the outcome directly	Placebo outcomes respond	Map every channel; bound the violation (§7.3)
Staggered two-way fixed effects	Already-treated units used as controls	Negative weights; odd event-study patterns	Clean-comparison DiD estimators (§7.5)
Non-parallel trends	Selection into treatment timing	Pre-trends; dips before treatment	Sensitivity bounds; better comparison groups (§7.5)
Ignored clustering	Correlation within groups	Classical standard errors far below clustered	Cluster at the assignment level (§4.4)
Few treated clusters	The meat comes from a handful of sums	Huge t-statistics from one or two treated groups	Wild cluster bootstrap; randomisation inference (§4.4)
Overlapping observations	Shared returns across rows	t-statistics and R^2 grow with horizon	Hodrick standard errors; count T/h (§4.5, §9.2)
Persistent predictors	Stambaugh bias	High persistence and correlated innovations	Bias correction; IVX (§9.2)
Spurious regression	Independent trending series	High R^2; non-stationary residuals	Difference, or test for cointegration (§8.3)
Structural breaks	Parameters change	Drifting rolling estimates; out-of-sample collapse	Break tests; window averaging (§8.7)
Overfitting	Too many parameters for the signal	In-sample far above out-of-sample	Shrinkage; fewer parameters (§10.2)
Validation leakage	Dependence across folds	Cross-validated score far above live performance	Walk-forward, purging, grouping by date (§10.6)
Specification search	Data-dependent choices among many possible analyses (the “garden of forking paths”)	Test statistics bunched just above thresholds	Pre-specification; holdout; specification curves (§4.8, §11.4)
Winner’s curse	Low power plus a significance filter	Effects shrink when replicated	Compute power first; shrink estimates (§4.7)
Look-ahead	Information unavailable at the time	Backtest far above live	Point-in-time data (§9.7)
Manual two-stage estimation	Second-stage residuals use fitted values	Wrong standard errors	Estimate 2SLS as one step (§5.2)
Generated regressors	A regressor estimated in a first step	Standard errors too small	Bootstrap both steps together (Pagan, 1984)
Mechanical correlation	Regressor and outcome share a component	Significance with no economic mechanism	Rebuild the variables (§7.3)
Regression to the mean	Selecting units for extreme values	Extremes “revert” with or without treatment	A comparison group selected the same way
Aggregation reversal	Pooling groups of different composition	Pooled and within-group slopes differ in sign	Within-group estimates; fixed effects (§3.4)

Failure	Mechanism	How it shows up, and how to check	Fix
Extrapolation	Estimates where the data are sparse	Poor overlap; estimates driven by functional form	Trim; state the local estimand (§6.6, §7.6)

§11 Key takeaways

1. Effort should go to the question, the data, and identification before specification, inference, and estimators. Researchers usually spend effort in close to the reverse order.
2. Most wrong results in empirical finance are data errors: joins, units, timing, corporate actions, and delistings.
3. Plot the data before estimating anything. Summary statistics can be identical for data that look nothing alike.
4. Diagnostics that trigger respecification are specification choices. Decide in advance what they would change.
5. Report a specification curve instead of a curated robustness table. The claim is the range of estimates across reasonable choices.
6. Show the design (first stage, event-study plot, density test, pre-period fit, overlap), not just the estimate.
7. Published test statistics bunch just above significance thresholds, most of all in the designs with the most researcher discretion. Keep a log of every analysis attempted.

Part VI — Synthesis

12. Taxonomy and equivalences

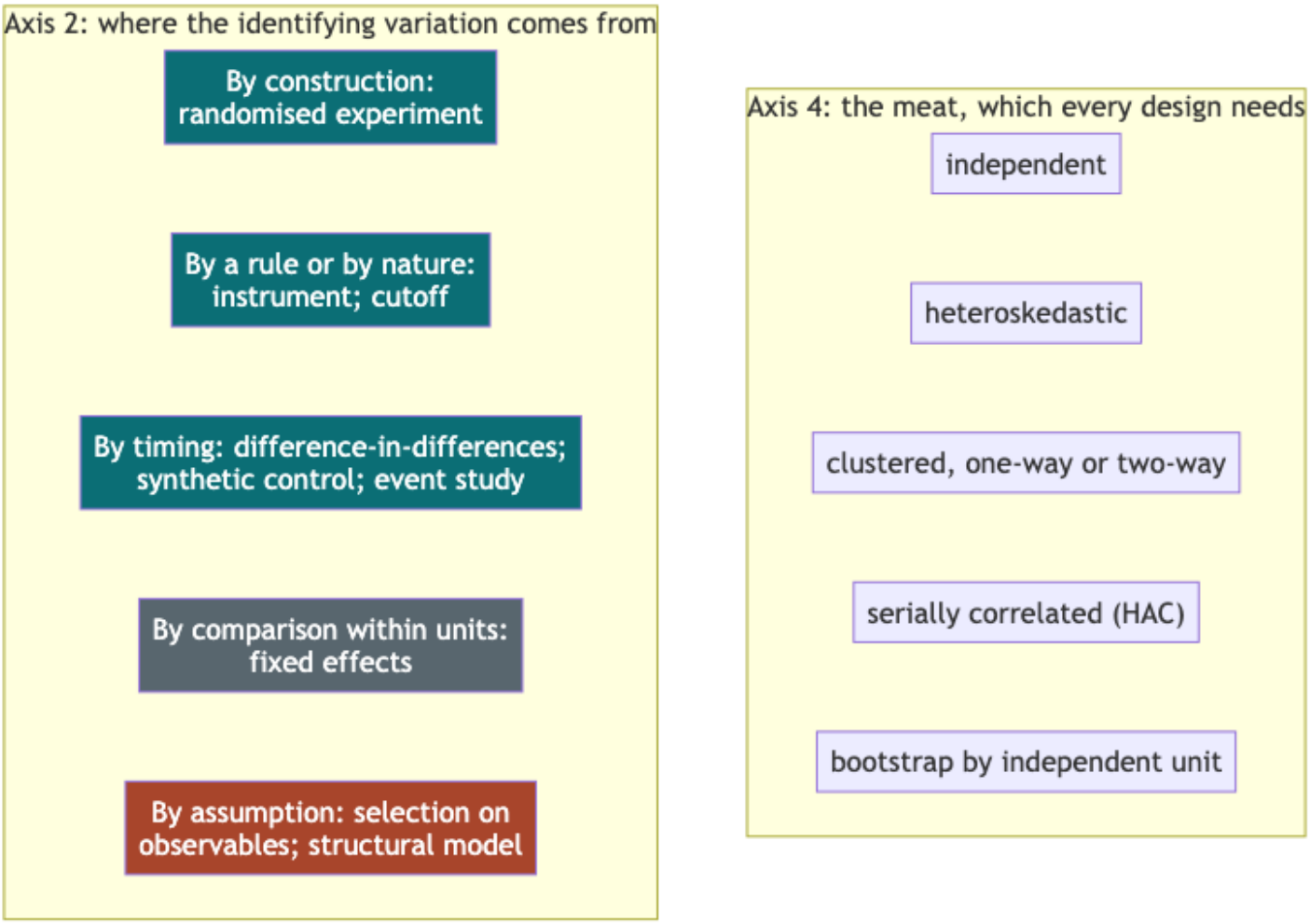
12.1 The design space as a product

The methods in a flat list ("OLS, IV, GMM, fixed effects, DiD, clustered errors, Newey–West, LASSO…") answer different questions, and the list does not show that. Every empirical study is a *coordinate* in a product space of four independent choices. Most methods in the literature name a choice on one axis only. The table defines the four axes.

Axis	What it decides	Choices
1. Estimand	Which question is answered	A projection (description); a forecast under a loss; ATE, ATT, LATE, or CATE; a structural parameter
2. Identifying variation	What is assumed uncorrelated with the error (the orthogonality condition)	Randomisation; controls; an instrument; within-unit changes; treatment timing; a cutoff; donor weights; a model
3. Estimator	How the sample moment condition is formed and solved	OLS; IV and GMM; maximum and quasi-maximum likelihood; quantile and robust M-estimators; shrinkage; machine-learning-assisted (DML)
4. Meat	What is independent of what	Independent; heteroskedastic; clustered one-way or two-way; HAC; Driscoll–Kraay; Fama–MacBeth; bootstrap by the independent unit

A prediction problem replaces axis 2 with a **validation design**: in-sample fit, random cross-validation, walk-forward validation, or purged and grouped splits (§10.6).

In this view, "difference-in-differences with clustered standard errors" is a coordinate: ATT, timing, OLS with fixed effects, and clustering by state. "A Fama–MacBeth regression" is also a coordinate: a projection or risk premium, characteristics as regressors, period-by-period OLS, and a meat that treats each date as one observation. The empty cells are informative too. For instance, a regression-discontinuity design can use a wild cluster bootstrap when the running variable is assigned at the cluster level. The diagram shows the two axes that are most often confused.



The shading runs from variation that is random by construction to variation that is random only by assumption. The diagram draws the meat separately because the meat is not a design. Every design needs one. Choosing it is a question about the data, not about identification.

12.2 Equivalences

The table lists relationships between methods that look different but are provably the same. The status column says whether each identity is exact, holds only asymptotically, or holds under a stated condition.

Relationship	Status	Where
Estimators		
OLS is the method of moments with $\psi_i = X_i(Y_i - X_i'\beta)$	Exact	§3.3
OLS is the Gaussian maximum likelihood estimator of β	Exact	§5.4
Just-identified IV, 2SLS, and GMM with any weighting matrix coincide	Exact	§5.2–§5.3
2SLS is GMM with $W = (\mathbf{Z}'\mathbf{Z}/n)^{-1}$, and efficient GMM under homoskedasticity	Exact; efficiency conditional	§5.3
Simple IV is the reduced-form slope divided by the first-stage slope	Exact	§5.2
A multiple-regression coefficient is the slope on the residualised regressor (Frisch–Waugh–Lovell)	Exact	§3.4
Unit dummies and within-unit demeaning give the same coefficients	Exact	§7.4
First differences equal the within estimator when $T = 2$	Exact	§7.4
Random effects plus unit means of the regressors reproduce fixed effects (Mundlak)	Exact	§7.4
The 2×2 difference-in-differences equals the interaction coefficient in two-way fixed effects	Exact	§7.5
Staggered two-way fixed effects is a weighted average of 2×2 comparisons, with weights that can be negative	Exact decomposition	§7.5
Fuzzy regression discontinuity is IV with the cutoff indicator as the instrument	Exact, locally	§7.6
Ridge is a posterior mean under a normal prior; LASSO a posterior mode under a Laplace prior	Exact	§5.6
Double machine learning in the partially linear model is Frisch–Waugh–Lovell with flexible first stages	Exact in form	§10.5
Interpretation		
Omitted variable bias is a special case of the endogeneity term $\mathbb{E}[X_i X_i']^{-1}\mathbb{E}[X_i u_i]$	Exact	§3.2, §3.5
Selection bias in potential outcomes is omitted variable bias in regression language	Conceptually identical	§6.2
Heckman's selection correction adds the omitted inverse-Mills-ratio term, $\varphi(\cdot)/\Phi(\cdot)$	Exact under normality	§6.4
OLS with a binary treatment and saturated controls is a variance-weighted average of cell effects	Exact	§3.4
IV with heterogeneous effects estimates the effect for compliers (LATE)	Exact under monotonicity	§7.3
Inference		

Relationship	Status	Where
Classical standard errors are the sandwich with a homoskedastic, independent meat	Exact	§4.2
Cluster-robust with one observation per cluster is heteroskedasticity-robust	Exact up to a finite-sample factor	§4.4
Two-way clustered variance is firm-clustered plus time-clustered minus heteroskedasticity-robust	Exact	§4.4
Newey–West is a Bartlett-kernel estimate of the long-run variance (the spectral density at frequency zero)	Exact	§4.5
HC3 is close to the jackknife variance	Approximate	§4.3
Wald, likelihood ratio, and Lagrange multiplier tests agree	Asymptotic, under correct specification	§4.6
The F-statistic for one restriction is the squared t-statistic	Exact	§4.6
R^2 is the squared correlation between Y and $\hat{Y}$ (with an intercept)	Exact	§3.8
Time series and asset pricing		
A stationary AR(1) is an MA(∞) with coefficients ϕ^j	Exact	§8.2
Every covariance-stationary process is deterministic plus MA(∞) (Wold)	Exact	§8.2
GARCH(1,1) is ARMA(1,1) in squared shocks	Exact	§8.6
Cointegration is equivalent to an error-correction representation	Exact (Granger representation)	§8.4
Local projections and VARs estimate the same impulse responses	Exact in population	§8.5
The variance ratio is one plus a weighted sum of autocorrelations	Exact	§8.1
The GRS test asks whether test assets raise the factors' maximum Sharpe ratio	Exact	§9.4
A linear factor model is a stochastic discount factor linear in the factors	Exact	§9.4

12.3 Same name, different thing

The table lists terms that carry more than one meaning, so that a reader can tell which object a given use refers to.

Term	One meaning	Another meaning
Fixed effects	Unit-specific intercepts, allowed to correlate with the regressors (econometrics)	Non-random effects, as opposed to random effects in a mixed model (statistics)
Random effects	Intercepts assumed uncorrelated with the regressors (panel econometrics)	Hierarchical coefficients with a distribution, estimated with shrinkage (mixed models)
Robust	Standard errors valid under heteroskedasticity or dependence	Estimators insensitive to outliers; or a robustness check across specifications
Event study	Abnormal returns around news (finance, §9.3)	A difference-in-differences with coefficients for each period around treatment (§7.5)
Causality	The effect of an intervention (§6.1)	Granger causality: the past of one series predicts another (§8.5)

Term	One meaning	Another meaning
Structural	A model of economic behaviour (§7.8)	A VAR with identified shocks (§8.5); a structural *break* is a parameter change (§8.7)
Exogenous	Uncorrelated with the error (§3.2)	Strictly exogenous: uncorrelated at every date (§7.4); determined outside the economic model
Bias	Finite-sample bias, $\mathbb{E}[\hat{\theta}] - \theta_0$	Inconsistency, a probability limit away from the truth; the two are routinely conflated
Efficient	Minimum-variance among a class of estimators	Informationally efficient markets
Significance	Statistical: signal relative to noise	Economic: size relative to what matters (§4.10)
R^2	In-sample fit (§3.8)	Out-of-sample R^2 against a benchmark (§8.8); GLS R^2 in asset-pricing tests (§9.4)
Heterogeneity	Treatment effects that differ across units (§6.1)	Unobserved unit characteristics, as absorbed by fixed effects (§7.4)
Identification	The estimand is a function of the observable distribution (§1.3)	Uniqueness of parameters in a statistical model; recovering structural shocks in a VAR
Factor	A priced risk factor	A statistical factor from principal components; a categorical variable in software
Instrument	A variable satisfying relevance and exclusion (§7.3)	A tradable security

12.4 Which choice matters most

The four axes of the product space do not matter equally.

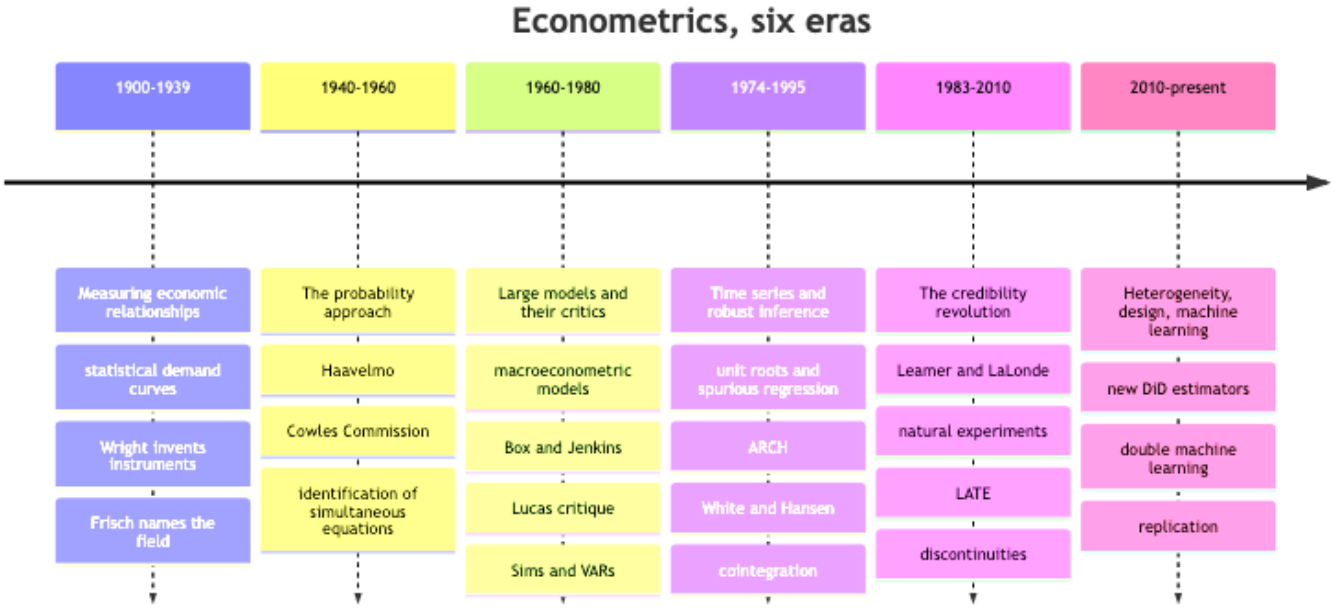
- **For causal questions, the identifying variation dominates.** LaLonde's comparison (§1.1) and its successors show estimates on the same data varying by more than the effect itself when the source of variation changes. [Practice] Under a *fixed* identifying assumption with good overlap, different estimators tend to agree with each other (Imbens & Xu, 2025). Their agreement with the truth depends on the assumption, not on the estimator.
- **For inference, the meat dominates.** Choosing the wrong dependence structure changes standard errors by factors of two to seven in realistic settings (§4.4, §4.5, §9.2). No estimator choice has an effect of that size on precision.
- **For prediction, the effective number of parameters relative to the signal dominates.** With a low signal-to-noise ratio, the k/n term of §10.2 decides whether a model beats a naive benchmark. That is why shrinkage and combination beat more complex models.
- **The estimator is rarely the binding choice.** The exceptions are the weak-identification cases (weak instruments, many moments, near-unit roots), where the default estimator's approximations fail.

§12 Key takeaways

1. Every empirical study is a coordinate on four axes: estimand, identifying variation, estimator, and meat. Most method names describe one axis only.
2. The meat is not a design. Every design needs one, and choosing it is a question about the dependence in the data.
3. Many apparently different methods are algebraically identical. IV is a ratio of regressions. Fixed effects are demeaning. DML is Frisch–Waugh–Lovell. Fuzzy RD is IV. Ridge is a Bayesian posterior.
4. Many shared names refer to different objects. "Event study," "robust," "fixed effects," "structural," and "causality" each mean at least two things.
5. Identifying variation dominates causal conclusions, the meat dominates inference, and the parameter count relative to the signal dominates prediction. The estimator is rarely the binding choice.

13. How the field evolved

The history explains why the field's concerns are what they are, so it merits a few thousand words. It explains why an econometrician's first concern is "identification". It also explains why researchers cluster standard errors as a matter of habit, and why a finance seminar asks about data mining before it asks about the model. This section summarises each era in four parts: its contribution, what changed as a result, its limitations, and what survives. The timeline shows the six eras.



13.1 Era I — Measuring economic relationships (1900–1939)

Thesis: economists combined statistics with economic theory, and the identification problem appeared at once.

Contribution. Economists began estimating demand curves from market data. Working (1927) showed that a scatter of prices and quantities reveals a demand curve only if supply shifts while demand stays fixed. This statement of the identification problem came two decades before the problem was named. The following year, Philip Wright's book on tariffs on animal and vegetable oils included an appendix. The appendix derived instrumental variables as a solution (Stock & Trebbi, 2003). Yule (1926) documented nonsense correlations between time series. Frisch & Waugh (1933) proved the partialling-out theorem. Ragnar Frisch coined the word "econometrics" and helped found the Econometric Society and its journal. The first volume of *Econometrica* carried Cowles (1933), which examined the recommendations of 45 professional forecasting agencies. It found no evidence that they did better than chance. This was the first test of market efficiency and the first finance paper in the field's leading journal.

What changed. Economists began to estimate economic relationships instead of asserting them.

Limitations. The era had no probability model for the errors, and economists handled identification case by case. The most ambitious project was Tinbergen's statistical business-cycle models. Keynes (1939) criticised it, and the era could not answer the critique. Keynes argued that the method needs a complete list of causes, relationships that stay stable, and a way to tell causation from correlation.

Lasting influence. Instrumental variables, partialling out, spurious correlation, and the identification problem all remain central. This era has the highest ratio of lasting ideas to papers.

13.2 Era II — The probability approach (1940–1960)

Thesis: economic data are draws from a probability model, and identification is a property of that model.

Contribution. Haavelmo (1943) showed that least squares applied to one equation of a simultaneous system is inconsistent. Haavelmo (1944) argued that researchers should treat economic data as realisations from a joint distribution generated by a system of behavioural equations. This is the framing of §1.2. The Cowles Commission developed this idea into a research

programme. Koopmans (1949) formalised identification, with the rank and order conditions for simultaneous equations. Anderson and Rubin developed estimation and inference for them (Anderson & Rubin, 1949).

What changed. Econometrics became a probabilistic discipline with its own theory, and "identification" acquired a precise meaning.

Limitations. The identifying restrictions came from economic theory, and researchers accepted them without testing them. Computing power and data were too limited to test much. Koopmans (1947) attacked the NBER's empirical business-cycle measurement as "measurement without theory." The attack opened an argument between theory-led and data-led empirical work that continues (§7.8).

Lasting influence. The structural tradition and the language of identification survive from this era. Frisch and Tinbergen shared the first economics Nobel prize in 1969. Haavelmo received the prize in 1989.

13.3 Era III — Large models and their critics (1960–1980)

Thesis: large structural models promised policy evaluation, but simple time-series methods forecast better than they did, and rational expectations undermined their theory.

Contribution. Economists built macroeconometric models with hundreds of equations for forecasting and policy analysis. Box and Jenkins's 1970 book systematised ARIMA modelling. Nelson (1972) found that simple ARIMA forecasts contained information that the forecasts of the large FRB–MIT–Penn model missed. He also found that the ARIMA forecasts were competitive with the model's. Granger (1969) gave a testable definition of predictive causality. Lucas (1976) argued that the models' estimated relationships would shift whenever policy changed. The relationships summarised decisions made under the old policy. Sims (1980) called the models' identifying restrictions "incredible" and proposed vector autoregressions instead.

What changed. Confidence in large structural models fell sharply. Forecasting and policy analysis became separate activities, and time-series methods gained ground.

Limitations. VARs gained credibility by giving up economic content. The Lucas critique applies with equal force to reduced-form relationships.

Lasting influence. Structural VARs remain the main tool of empirical macroeconomics (§8.5). The Lucas critique is the standard warning about relationships that break when deployed. Sargent and Sims shared the 2011 Nobel prize.

13.4 Era IV — Time series and robust inference (1974–1995)

Thesis: the classical assumptions were systematically wrong for economic data, and the field built tools that did not need them.

Contribution. The era produced work in four areas. On non-stationarity: Granger & Newbold (1974) on spurious regression, Dickey & Fuller (1979) on unit-root testing, and Engle & Granger (1987) on cointegration. On volatility: Engle (1982) on ARCH. On robust inference: White (1980) on heteroskedasticity, White (1982) on misspecified likelihoods, Hansen (1982) on GMM, and Newey & West (1987) on autocorrelation. On microeconometrics: Heckman (1979) on selection, discrete-choice models, and panel methods. Finance built its own econometrics in parallel. Examples are event studies (Fama, Fisher, Jensen & Roll, 1969), two-pass tests (Fama & MacBeth, 1973), Euler-equation GMM (Hansen & Singleton, 1982), and portfolio efficiency tests (Gibbons, Ross & Shanken, 1989). Campbell, Lo and MacKinlay's 1997 textbook collected this work.

What changed. Standard errors robust to heteroskedasticity and dependence became the default. Researchers modelled non-stationarity and time-varying volatility instead of ignoring them.

Limitations. Robust inference did not address identification. The specification searches that Leamer was about to attack continued unchanged.

Lasting influence. Nearly all of the standard-error practice of §4 survives from this era, as do GARCH, GMM in asset pricing, and cointegration. Heckman and McFadden shared the 2000 Nobel prize. Engle and Granger shared the 2003 prize. Hansen shared the 2013 prize with Fama and Shiller.

13.5 Era V — The credibility revolution (1983–2010)

Thesis: credible causal claims come from research design, not from modelling.

Contribution. Leamer (1983) argued that specification searches made most econometric conclusions fragile. LaLonde (1986) showed non-experimental estimators failing against an experimental benchmark (§1.1). The response was to look for variation that was random for reasons outside the model. Examples are a draft lottery (Angrist, 1990), quarter of birth (Angrist & Krueger, 1991), and a minimum-wage increase in one state but not its neighbour (Card & Krueger, 1994). The theory followed. It produced LATE (Imbens & Angrist, 1994) and weak instruments (Bound, Jaeger & Baker, 1995; Staiger & Stock, 1997). It also produced regression discontinuity (Hahn, Todd & van der Klaauw, 2001), synthetic control (Abadie & Gardeazabal, 2003), and clustering in difference-in-differences (Bertrand, Duflo & Mullainathan, 2004). Angrist and Pischke's *Mostly Harmless Econometrics* (2009) and their 2010 essay named the movement.

What changed. "What is your identification strategy?" became the first question in applied microeconomics and in much of empirical corporate finance. Randomised field experiments spread through development economics. Banerjee, Duflo, and Kremer shared the 2019 Nobel prize for experiments. Card, Angrist, and Imbens shared the 2021 prize for natural experiments.

Limitations. The designs produce local estimands and have limited external validity, as the structural side argued forcefully (Keane, 2010; Deaton, 2010; Heckman, 2010). The later evidence on p-hacking shows that designs constrain researcher discretion without removing it (§11.5).

Lasting influence. This approach is the dominant paradigm of applied microeconomics and of causal work in corporate finance.

13.6 Era VI — Heterogeneity, design, and machine learning (2010–present)

Thesis: researchers re-examined the credibility revolution's tools under heterogeneous effects and adopted machine learning as an estimator.

Contribution. Researchers showed that difference-in-differences with staggered timing fails under heterogeneous effects. New estimators followed within a few years (§7.5). Researchers reassessed weak-instrument practice (Andrews, Stock & Sun, 2019; Lee, McCrary, Moreira & Porter, 2022). They also reworked the foundations of clustering (Abadie, Athey, Imbens & Wooldridge, 2023). Machine learning entered causal inference with valid inference (Belloni, Chernozhukov & Hansen, 2014; Chernozhukov et al., 2018; Wager & Athey, 2018). Meta-research documented low power and selective reporting (Ioannidis, Stanley & Doucouliagos, 2017; Brodeur, Cook & Heyes, 2020). Journals began requiring replication packages. In finance, three developments changed empirical research. They were the large number of published factors and the debate over their replication (§9.6), machine-learning asset pricing (Gu, Kelly & Xiu, 2020), and demand-system asset pricing (Koijen & Yogo, 2019).

What changed. Two-way fixed effects stopped being the default for staggered designs. Weak-instrument-robust inference became expected. Machine-learning estimators gained valid confidence intervals. Replication became a condition of publication.

Limitations. Methods now change faster than practitioners can adopt them. Each new estimator carries its own assumptions. None of them substitutes for a credible source of variation.

Lasting influence. It is too early to judge. [Hypothesis] This section expects that heterogeneity-robust difference-in-differences and double machine learning will be routine within a decade. It also expects that the argument over whether complexity is a virtue in return prediction will still be open.

§13 Key takeaways

1. Working stated the identification problem in 1927, and Philip Wright invented instrumental variables in 1928 to estimate supply and demand. The field's central concern is as old as the field.
2. Haavelmo's probability approach made identification a property of a model, and the structural tradition followed from it.
3. Simple time-series models forecast better than large structural models, and the Lucas critique undermined their theory. VARs replaced them in macroeconomics.
4. Work in the 1980s made inference robust to heteroskedasticity, autocorrelation, and misspecification. It did not address identification.
5. The credibility revolution moved the source of credibility from the model to the research design. The cost was local estimands.
6. The current era is re-examining those designs under heterogeneous effects, adopting machine learning as an estimator, and addressing evidence of low power and selective reporting.

14. Synthesis

14.1 The framework on one page

The chapter reduces to three questions, one equation, and two ways to fail.

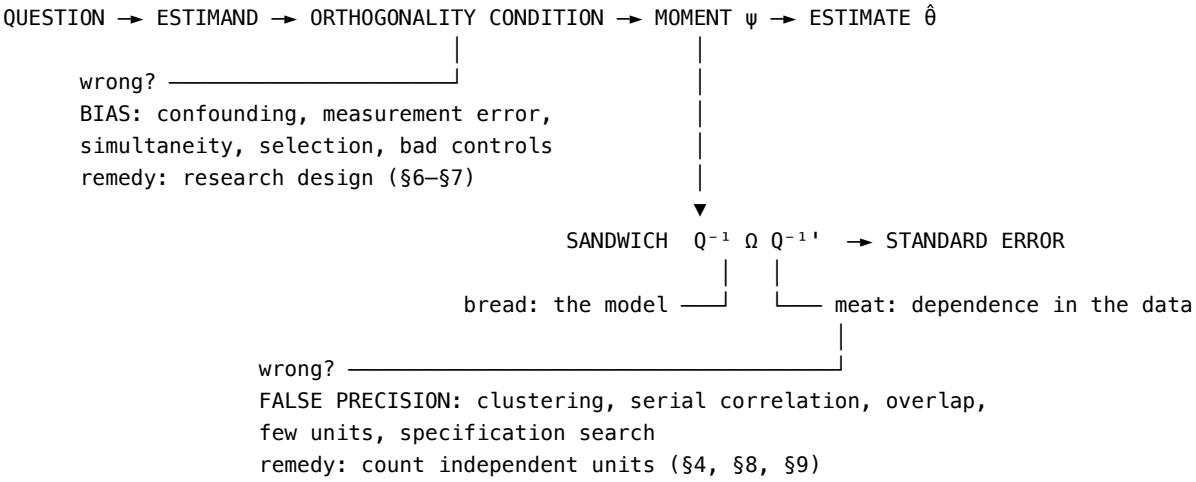
The three questions (§1.3) concern the estimand, its identification, and its precision. Analysts answer them in that order, and an error in an earlier answer affects every later one. Prediction problems replace the identification question with a sound validation design.

The one equation (§1.5–§1.6). Every estimator solves a sample moment condition. Every standard error is the sandwich around it:

$$\frac{1}{n}\sum_{i=1}^{n}\psi_i(\hat{\theta}) = 0, \qquad \text{Avar}\left(\sqrt{n}(\hat{\theta}-\theta_0)\right) = Q^{-1}\,\Omega\,Q^{-1\prime}.$$

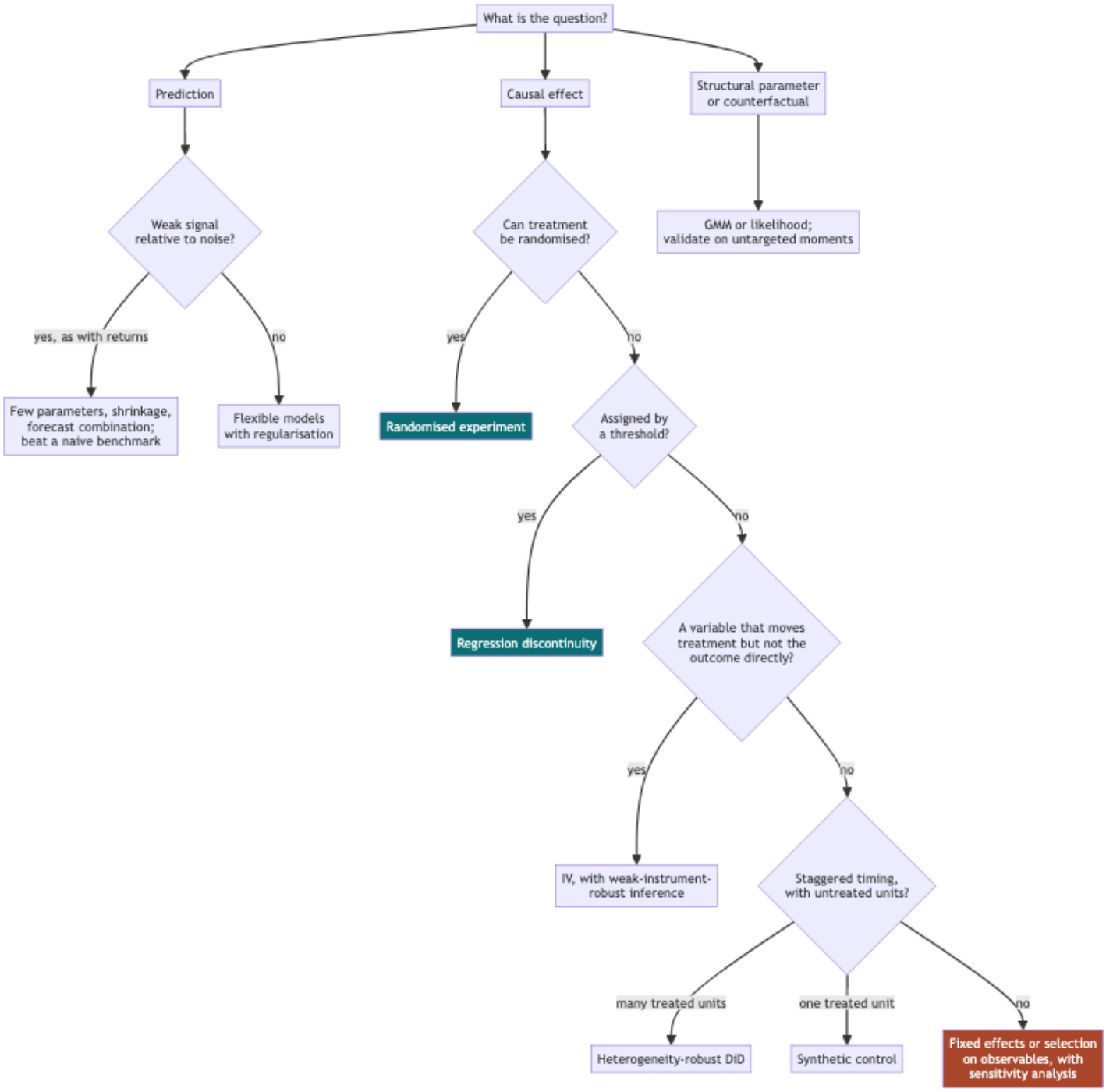
The choice of ψ states the orthogonality condition, which is the identifying assumption. The bread Q comes from the model. The meat Ω comes from the dependence in the data.

The two ways to fail. A false orthogonality condition causes bias: confounding, measurement error, simultaneity, selection, or a bad control (§6). A wrong meat causes false precision: ignored clustering, serial correlation, overlap, too few independent units, or a specification search that the standard error does not account for (§4). The research designs of §7 make the first failure less likely. The standard-error methods of §4, §8, and §9 prevent the second. When the goal is prediction, shrinkage and validation (§5.6, §10) trade a little bias for a large reduction in variance. The diagram traces both failures through the pipeline.



14.2 A decision tree

The tree branches on the questions that change the analysis and ends in named methods. The step listed after it applies on every branch.



On every branch: choose the meat by the dependence in the data. Count the independent units. Check when every variable was known. Report intervals together with every analysis attempted.

The shading on the selection-on-observables leaf is a deliberate warning. Analyses end up there when no better source of variation exists. In that case a sensitivity analysis (§6.7) is required.

14.3 A roadmap for building an empirical research capability

The roadmap is for a team or an individual building the capability to produce empirical results that hold up on new data. It proceeds in stages. Each stage has a gate, and the team must pass the gate before moving on. The first two stages are infrastructure rather than the interesting part, and skipping them is the usual reason later stages fail.

Stage	Build	Gate before moving on
0. Data	Point-in-time data with reporting lags; delistings and corporate actions; tests on every join	Reproduce a published series, such as a standard factor's returns, to within a small tolerance
1. Inference core	Regression with heteroskedasticity-robust, clustered, two-way, HAC, Driscoll–Kraay, and Fama–MacBeth errors; a simulation harness	Recover known answers in simulations with known dependence, including this chapter's clustering and overlap simulations
2. Designs	IV with Anderson–Rubin inference; heterogeneity-robust DiD; RD with robust bias-corrected intervals	Replicate one published design end to end
3. Time series and forecasting	Unit-root and cointegration tools; GARCH and HAR; walk-forward evaluation; Diebold–Mariano and Clark–West	Reproduce a known *negative* out-of-sample result
4. Prediction	Shrinkage and tree models; purged, grouped validation; a research log with trial counts	Every reported result has a computable number of trials behind it
5. Causal machine learning and structure	Double machine learning; causal forests; a small structural model estimated by simulated moments	DML matches OLS in simple simulated designs and removes regularisation bias in hard ones

A reading order follows the same sequence. Start with *Mastering 'Metrics* for intuition. For designs, read *Mostly Harmless Econometrics*, *The Mixtape*, or *The Effect*. Read Wooldridge for panels or Hansen for theory, and Hamilton for time series. For finance, read Campbell, Lo and MacKinlay, and Cochrane. For machine learning, read the *Applied Causal Inference* book. §15.1 has the details.

14.4 Advice for someone starting today

These recommendations condense the chapter for a practitioner starting empirical work.

1. **Write down the estimand before analysing the data.** If no population quantity can be named that would answer the question, no estimate will answer it.
2. **Ask which experiment the study approximates**, and where the approximation fails.
3. **Count the independent units** before trusting a standard error: clusters, treated clusters, and non-overlapping periods. The rows do not matter.
4. **Plot the raw data and the reduced form** before estimating anything more elaborate.
5. **Controls can introduce bias.** Draw the causal graph, and never control for anything the treatment could have caused.
6. **Treat any t-statistic below about 3 that a search produced as a hypothesis**, not a finding.
7. **Compute the minimum detectable effect first.** If the plausible effect is smaller, a significant result is more likely an exaggeration than a discovery.
8. **A backtest estimates a mean from dependent data after a search.** Expect it to shrink out of sample, even when the analyst made no error.
9. **Use the simplest estimator the design allows**, and spend the effort on data and diagnostics.
10. **Simulate the method on data with a known answer**, including data where the answer is zero. Simulations produced every figure in this chapter, and each one found a failure that the textbook formula misses.
11. **Keep a log of every analysis run.** A log is the only way to count the analyses attempted.
12. **For prediction, the only valid test sample is one the model never saw. For causation, the nearest equivalent is a placebo test.**

14.5 What is and is not known

Settled. The mathematics of estimation and inference is settled. It covers moment conditions, the sandwich, and the asymptotic theory behind both. The mechanisms of failure are also settled: omitted variables, measurement error, simultaneity, selection, colliders, spurious regression, and specification search. So is the need for inference that accounts for the dependence in the data. For causal questions, the source of variation matters far more than the choice of estimator, and this is settled as well.

Contested. Four questions remain open. The first is how much weight local, design-based estimates should get relative to structural models (§7.8). The second is how many published return predictors reflect real, exploitable effects (§9.6). The third is whether heavily over-parameterised models really improve return prediction (§10.2). The fourth is how far pre-analysis plans and multiple-testing hurdles should go, given their cost in missed discoveries (§4.8, §9.6).

Open. Three problems are both important and unsolved. The first is **inference with few independent episodes**. Finance has one history, a handful of crises, and a few regimes. No standard-error formula creates independence that the data lack. The second is **external validity**, which asks when an effect estimated in one market, period, or population transfers to another. Most of the field's tools do not address it. The third is **interference and reflexivity**. In markets, one participant's treatment changes the prices that the others face. This violates the no-interference assumption, a central assumption of causal inference. Publishing a finding also changes the process that produced it.

Econometrics cannot make data answer a question whose answer the data do not contain. It can say precisely which questions a given dataset can answer, under which assumptions, and with how much confidence. This chapter aims to make that capability usable.

15. References

This section groups the references by kind, because readers read the kinds differently. Each entry says why it matters. Where a free copy exists, the link points to it. That copy is usually a working-paper or author version, which can differ in detail from the published version. Publishers may paywall entries linked to a publisher page.

15.1 Textbooks and guides

- **Angrist & Pischke (2015).** *Mastering 'Metrics: The Path from Cause to Effect.* Princeton University Press. — The best first book. It teaches five research designs through examples, with almost no matrix algebra.
- **Angrist & Pischke (2009).** *Mostly Harmless Econometrics: An Empiricist's Companion.* Princeton University Press. — The design-based methods at graduate level. It is the source of the regression-as-projection presentation in §3.
- **Huntington-Klein (2021).** *The Effect: An Introduction to Research Design and Causality.* CRC Press; free online. — The clearest book on *thinking* about identification.
- **Cunningham (2021).** *Causal Inference: The Mixtape.* Yale University Press; free online. — Designs with code in several languages. Strong on causal graphs and difference-in-differences.
- **Hernán & Robins (2020).** *Causal Inference: What If.* Chapman & Hall/CRC; free online. — The epidemiology tradition's treatment of potential outcomes and graphs. Excellent on time-varying treatments.
- **Imbens & Rubin (2015).** *Causal Inference for Statistics, Social, and Biomedical Sciences.* Cambridge University Press. — The potential-outcomes framework in full, especially experiments and selection on observables.
- **Pearl (2009).** *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press. — The graphical approach to causation, including the back-door criterion of §6.5.
- **Wooldridge (2010).** *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. MIT Press. — The reference for microeconometrics and panels.
- **Hansen (2022).** *Econometrics.* Princeton University Press. — A modern, rigorous graduate text built on projection and the sandwich. It is the textbook closest to this chapter's organisation.
- **Hayashi (2000).** *Econometrics.* Princeton University Press. — GMM as the common framework of the whole field, with finance applications.
- **Hamilton (1994).** *Time Series Analysis.* Princeton University Press. — Still the standard graduate time-series reference.
- **Kilian & Lütkepohl (2017).** *Structural Vector Autoregressive Analysis.* Cambridge University Press. — A comprehensive treatment of identification in VARs.
- **Elliott & Timmermann (2016).** *Economic Forecasting.* Princeton University Press. — A rigorous treatment of loss functions, evaluation, combination, and breaks.
- **Hyndman & Athanasopoulos (2021).** *Forecasting: Principles and Practice*, 3rd ed. OTexts; free online. — Practical forecasting with code.
- **Campbell, Lo & MacKinlay (1997).** *The Econometrics of Financial Markets.* Princeton University Press. — The classic financial econometrics text: event studies, predictability, asset-pricing tests.
- **Cochrane (2005).** *Asset Pricing*, revised ed. Princeton University Press. — Asset pricing as GMM on the stochastic discount factor.
- **Cattaneo, Idrobo & Titiunik (2020).** *A Practical Introduction to Regression Discontinuity Designs: Foundations.* Cambridge University Press. — The implementation guide to regression discontinuity, by the authors of its standard tools.
- **Chernozhukov, Hansen, Kallus, Spindler & Syrgkanis (2024).** *Applied Causal Inference Powered by ML and AI.* Free online. — Double machine learning and related methods, from the authors who developed them.
- **Hastie, Tibshirani & Friedman (2009).** *The Elements of Statistical Learning*, 2nd ed. Springer; free online. — The machine-learning reference an econometrician can read.
- **Gelman, Hill & Vehtari (2020).** *Regression and Other Stories.* Cambridge University Press. — Applied regression with an emphasis on simulation and model checking.
- **López de Prado (2018).** *Advances in Financial Machine Learning.* Wiley. — Purged and embargoed validation and backtest overfitting. The book is opinionated, and its validation chapters alone are useful.

15.2 Classic books

This list links no stable free copies.

- **Wright, P. G. (1928).** *The Tariff on Animal and Vegetable Oils.* Macmillan. — Its Appendix B derives instrumental variables. See Stock & Trebbi (2003) for the history.
- **Tinbergen, J. (1939).** *Statistical Testing of Business-Cycle Theories.* League of Nations. — The first macroeconometric models. Keynes (1939) criticised them.

- **Box, G. E. P. & Jenkins, G. M. (1970).** *Time Series Analysis: Forecasting and Control.* Holden-Day. — The ARIMA methodology of §8.2.
- **Goldberger, A. S. (1991).** *A Course in Econometrics.* Harvard University Press. — Econometrics built from the conditional expectation function, and the origin of "micronumerosity."

15.3 Foundations and history

- **Yule (1926).** "Why Do We Sometimes Get Nonsense-Correlations between Time-Series?" *Journal of the Royal Statistical Society* 89(1), 1–63. — Spurious correlation between trending series, sixty years before the theory.
- **Working (1927).** "What Do Statistical 'Demand Curves' Show?" *Quarterly Journal of Economics* 41(2), 212–235. — The identification problem, stated before it had a name.
- **Frisch & Waugh (1933).** "Partial Time Regressions as Compared with Individual Trends." *Econometrica* 1(4), 387–401. — The partialling-out theorem of §3.4.
- **Cowles (1933).** "Can Stock Market Forecasters Forecast?" *Econometrica* 1(3), 309–324. — 45 forecasting services, with no evidence of skill. It was the first test of market efficiency.
- **Keynes (1939).** "Professor Tinbergen's Method." *Economic Journal* 49(195), 558–577. — The first great critique of econometric modelling. Most of its objections still apply.
- **Haavelmo (1943).** "The Statistical Implications of a System of Simultaneous Equations." *Econometrica* 11(1), 1–12. — Least squares is inconsistent in a simultaneous system.
- **Haavelmo (1944).** "The Probability Approach in Econometrics." *Econometrica* 12 (Supplement). — The founding statement of econometrics as inference about a probability model (§1.2).
- **Koopmans (1947).** "Measurement Without Theory." *Review of Economics and Statistics* 29(3), 161–172. — The theory-versus-data argument, in its first form.
- **Koopmans (1949).** "Identification Problems in Economic Model Construction." *Econometrica* 17(2), 125–144. — A formal treatment of identification.
- **Anderson & Rubin (1949).** "Estimation of the Parameters of a Single Equation in a Complete System of Stochastic Equations." *Annals of Mathematical Statistics* 20(1), 46–63. — LIML, and the test that remains valid with weak instruments (§7.3).
- **Nelson (1972).** "The Prediction Performance of the FRB-MIT-PENN Model of the U.S. Economy." *American Economic Review* 62(5), 902–917. — Simple ARIMA forecasts carried information a large structural model missed.
- **Lucas (1976).** "Econometric Policy Evaluation: A Critique." *Carnegie-Rochester Conference Series on Public Policy* 1, 19–46. — Estimated relationships shift when policy changes.
- **Sims (1980).** "Macroeconomics and Reality." *Econometrica* 48(1), 1–48. — "Incredible" identifying restrictions, and the VAR alternative.
- **Stock & Trebbi (2003).** "Retrospectives: Who Invented Instrumental Variable Regression?" *Journal of Economic Perspectives* 17(3), 177–194. — A stylometric investigation of the origin of IV, and a clear introduction to it.
- **Angrist & Pischke (2010).** "The Credibility Revolution in Empirical Economics: How Better Research Design Is Taking the Con out of Econometrics." *Journal of Economic Perspectives* 24(2), 3–30. — The manifesto of design-based empirical work.
- **Imbens & Xu (2025).** "Comparing Experimental and Nonexperimental Methods: What Lessons Have We Learned Four Decades after LaLonde (1986)?" *Journal of Economic Perspectives* 39(4), 173–201. — What modern methods fix and what they do not fix, on LaLonde's own data.

15.4 Regression, inference, and standard errors

- **Kendall (1954).** "Note on Bias in the Estimation of Autocorrelation." *Biometrika* 41(3/4), 403–404. — The finite-sample bias of the autoregressive coefficient behind Stambaugh bias.
- **Anscombe (1973).** "Graphs in Statistical Analysis." *The American Statistician* 27(1), 17–21. — Four datasets share one regression line. Plot the data.
- **Ramsey (1969).** "Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis." *Journal of the Royal Statistical Society, Series B* 31(2), 350–371. — The RESET test for omitted nonlinearity.
- **White (1980).** "A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity." *Econometrica* 48(4), 817–838. — Robust standard errors. They removed the need for the homoskedasticity assumption.
- **Pagan (1984).** "Econometric Issues in the Analysis of Regressions with Generated Regressors." *International Economic Review* 25(1), 221–247. — Why regressors estimated in a first step need corrected standard errors.
- **MacKinnon & White (1985).** "Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties." *Journal of Econometrics* 29(3), 305–325. — HC2 and HC3.
- **Kiefer & Vogelsang (2005).** "A New Asymptotic Theory for Heteroskedasticity-Autocorrelation Robust Tests." *Econometric Theory* 21(6). [[paywalled]] — Fixed-b asymptotics for HAC tests.

- **Newey & West (1987).** "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55(3), 703–708. — The HAC estimator of §4.5.
- **Moulton (1990).** "An Illustration of a Pitfall in Estimating the Effects of Aggregate Variables on Micro Units." *Review of Economics and Statistics* 72(2), 334–338. — The Moulton factor.
- **Angrist (1998).** "Estimating the Labor Market Impact of Voluntary Military Service Using Social Security Data on Military Applicants." *Econometrica* 66(2), 249–288. — Regression with a binary treatment as a variance-weighted average of cell effects.
- **Efron (1979).** "Bootstrap Methods: Another Look at the Jackknife." *Annals of Statistics* 7(1), 1–26. — The bootstrap.
- **Künsch (1989).** "The Jackknife and the Bootstrap for General Stationary Observations." *Annals of Statistics* 17(3), 1217–1241. — The moving block bootstrap.
- **Politis & Romano (1994).** "The Stationary Bootstrap." *Journal of the American Statistical Association* 89(428), 1303–1313. — Blocks of random length.
- **Benjamini & Hochberg (1995).** "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society, Series B* 57(1), 289–300. — The false discovery rate.
- **Driscoll & Kraay (1998).** "Consistent Covariance Matrix Estimation with Spatially Dependent Panel Data." *Review of Economics and Statistics* 80(4), 549–560. — Standard errors robust to cross-sectional and serial dependence.
- **Merton (1980).** "On Estimating the Expected Return on the Market: An Exploratory Investigation." *Journal of Financial Economics* 8(4), 323–361. — Means need calendar span. Variances need sampling frequency (§2.5).
- **Cont (2001).** "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues." *Quantitative Finance* 1(2), 223–236. — Heavy tails, volatility clustering, and the other stylised facts.
- **Lo (2002).** "The Statistics of Sharpe Ratios." *Financial Analysts Journal* 58(4), 36–52. — The standard error of a Sharpe ratio, with serial correlation.
- **Bertrand, Duflo & Mullainathan (2004).** "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1), 249–275. — Placebo laws significant in up to 45% of simulations.
- **Cameron, Gelbach & Miller (2008).** "Bootstrap-Based Improvements for Inference with Clustered Errors." *Review of Economics and Statistics* 90(3), 414–427. — The wild cluster bootstrap.
- **Conley & Taber (2011).** "Inference with 'Difference in Differences' with a Small Number of Policy Changes." *Review of Economics and Statistics* 93(1), 113–125. — Inference with very few treated groups.
- **Cameron, Gelbach & Miller (2011).** "Robust Inference with Multiway Clustering." *Journal of Business and Economic Statistics* 29(2), 238–249. — Two-way clustering.
- **Thompson (2011).** "Simple Formulas for Standard Errors That Cluster by Both Firm and Time." *Journal of Financial Economics* 99(1), 1–10. — Two-way clustering for finance panels.
- **Cameron & Miller (2015).** "A Practitioner's Guide to Cluster-Robust Inference." *Journal of Human Resources* 50(2), 317–372. — The practical reference on clustering.
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15.10 Prediction and machine learning

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- **Gelman & Carlin (2014).** "Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors." *Perspectives on Psychological Science* 9(6), 641–651. — The winner's curse, quantified.
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- **Mellon (2025).** “Rain, Rain, Go Away: 194 Potential Exclusion-Restriction Violations for Studies Using Weather as an Instrumental Variable.” *American Journal of Political Science* 69(3), 881–898. — One instrument, hundreds of channels.

15.12 If you only read ten things

In this order:

1. **Angrist & Pischke (2015)**, *Mastering ’Metrics*, for the intuition behind research designs.
2. **Huntington-Klein (2021)**, *The Effect*, for thinking about identification before estimation.
3. **Leamer (1983)** and **Angrist & Pischke (2010)**, read together, for why research design became the centre of the field.
4. **Cameron & Miller (2015)**, for what standard errors assume and how to cluster.
5. **Cinelli, Forney & Pearl (2024)**, for which controls help and which hurt.
6. **Roth, Sant’Anna, Bilinski & Poe (2023)**, for difference-in-differences as it is now done.
7. **Andrews, Stock & Sun (2019)**, for weak instruments and what to do about them.
8. **Stambaugh (1999)** and **Goyal & Welch (2008)**, for what goes wrong in return-prediction regressions.
9. **Harvey, Liu & Zhu (2016)** and **Jensen, Kelly & Pedersen (2023)**, read together, for the replication debate in finance.
10. **Chernozhukov et al. (2018)**, or the *Applied Causal Inference* book, for machine learning applied to causal questions.

Appendix A: Concepts and prerequisites

This appendix explains the concepts that the main text uses without explaining them. The intended reader is comfortable with mathematics but does not do econometrics for a living. This reader meets "fixed-b critical values" or "Neyman orthogonality" in the middle of an argument. The reader wants the idea, not a textbook chapter, and prefers not to leave the chapter to find it.

The entries follow **dependency** order, not alphabetical order. Seven parts group them, and the parts also follow dependency order, so later entries build on earlier ones. Each entry gives the idea in words first. It then gives the formal definition, the reason the concept appears in this chapter, and a pointer to deeper reading. Notation follows the notation table of the main text. Where a standard formula from another literature uses conflicting letters, the entry says so and keeps those letters local to the entry.

The *argument* of the chapter does not depend on this material. Implementing the methods depends on much of it.

Index. The table shows where each concept is first used.

Concept	First used	Concept	First used
Information sets and martingale differences	§1.4	Local polynomial regression	§7.6
Stationarity and ergodicity	§2.4	Approximate sparsity	§10.5
Convergence, Slutsky, continuous mapping	§1.6	Neyman orthogonality, cross-fitting	§10.5
The chi-squared distribution	§5.3	The Kalman filter	§7.8
Long-run variance, spectral density	§4.5	Granger representation theorem	§8.4
Fixed-b asymptotics	§4.5	Market microstructure noise	§8.6
Kullback–Leibler, pseudo-true values	§5.4	MSE and QLIKE for volatility	§8.6
Fisher information, Cramér–Rao	§5.4	IVX	§9.2
Priors as penalties	§5.6	Excess returns, Sharpe ratio	§2.4
Stein's paradox, empirical Bayes	§5.6	Risk aversion and the SDF	§1.5
Truncated normal, inverse Mills ratio	§6.4	Portfolio sorts and breakpoints	§1.4
Jackknife and block bootstraps	§4.3	Tangency portfolio, GLS R^2	§9.4
Wild cluster bootstrap	§4.4	The Shanken correction	§9.4
Randomisation inference	§4.4	Driscoll–Kraay standard errors	§1.6
Family-wise error and FDR	§4.8	The deflated Sharpe ratio	§4.8
Reality check, SPA, stepdown	§4.8	Trees, forests, boosting	§10.4
The Hausman test	§7.4	Mallows's C_p	§10.2
d-separation, back-door criterion	§6.5		
Oster's δ, robustness value	§6.7		
Weak-instrument tools	§7.3		
Dynamic panel GMM	§7.4		

Part I — Probability and asymptotics. This part covers the approximations behind every standard error in the chapter and the time-series conditions under which they hold.

A.1 Information sets, filtrations, and martingale differences

The idea. A forecast can use only what the forecaster knows when making it. An **information set** is everything known at time t. It includes past prices, past data releases, and the forecaster's own past forecasts. Information accumulates, so each

period's set contains the previous one. A sequence is a **martingale difference** if nothing in the information set helps predict its next *value on average*. The information may still help predict the size of the next value. Asset returns behave this way: their levels are unpredictable but their magnitudes are predictable.

Formally. A **filtration** is an increasing sequence of information sets (σ-algebras) $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots$; a process x_t is **adapted** if x_t is known at t. A sequence ε_t with $\mathbb{E}|\varepsilon_t| < \infty$ is a martingale difference sequence if

$$\mathbb{E}[\varepsilon_{t+1} \mid \mathcal{F}_t] = 0.$$

If second moments exist, this implies $\mathbb{E}[\varepsilon_{t+1}\, h] = 0$ for any h known at t. So the sequence is uncorrelated with every function of the past. It does not imply independence. The quantity $\mathbb{E}[\varepsilon_{t+1}^2 \mid \mathcal{F}_t]$ can vary, which is exactly what GARCH models (§8.6).

Why it appears here. The prediction estimand in §1.4 is a conditional expectation given $\mathcal{F}_t$. §2.2 and §8.1 use this definition to place returns between "uncorrelated" and "independent". The Hodrick standard errors of §4.5 and §9.2 impose the null that returns are a martingale difference sequence. The Euler equations of §5.3 are martingale-difference conditions on pricing errors.

Deeper. Hamilton (1994), chapter 7. Section 8 of Stochastic Processes builds filtrations from σ-algebras.

A.2 Stationarity and ergodicity

The idea. With one history, the only way to learn about a process is to average over time. That works only if two conditions hold. First, the process must be **stationary**. Its statistical behaviour does not change with the calendar, so early and late observations follow the same rules. Second, the process must be **ergodic**. A single long path eventually covers the full range of the process's behaviour, so the time average converges to the average over all possible paths. A process can be stationary without being ergodic. Flip a coin once, then report the outcome every day forever. The series is stationary, but its time average is the one flip, not one-half.

Formally. A process is **strictly stationary** if the joint distribution of $(y_{t_1+h}, \ldots, y_{t_k+h})$ does not depend on h. It is **covariance stationary** if $\mathbb{E}[y_t] = \mu$ and $\text{Cov}(y_t, y_{t-j}) = \gamma_j$ do not depend on t. A covariance-stationary process is **ergodic for the mean** if $\bar{y}_T \xrightarrow{p} \mu$; a sufficient condition is absolutely summable autocovariances, $\sum_j |\gamma_j| < \infty$, which also makes the long-run variance of A.5 finite.

Why it appears here. §2.4 states that the law of large numbers needs "independence or dependence that fades with distance". Ergodicity is the precise version of that condition. §8.1 makes stationarity the time-series substitute for random sampling. §8.3 covers what happens when stationarity fails.

Deeper. Hamilton (1994), chapter 3; Hansen (2022), the time-series chapters.

A.3 Convergence in probability and in distribution; Slutsky and continuous mapping

The idea. Asymptotic theory uses two senses of convergence. An estimate **converges in probability** to a number when, in large samples, it is almost certainly close to that number. That is consistency. A statistic **converges in distribution** when the *shape* of its sampling distribution approaches a limit. The statistic itself keeps varying from sample to sample. Convergence in distribution is what justifies normal critical values. Two standard results let the two kinds of limit combine as if they were ordinary numbers.

Formally. $X_n \xrightarrow{p} c$, written $\text{plim}\, X_n = c$, if $\Pr(|X_n - c| > \epsilon) \to 0$ for every $\epsilon > 0$. $X_n \xrightarrow{d} X$ if the distribution function of X_n converges to that of X at every point where the latter is continuous. The **continuous mapping theorem**: if g is continuous, $X_n \xrightarrow{p} c$ implies $g(X_n) \xrightarrow{p} g(c)$ and $X_n \xrightarrow{d} X$ implies $g(X_n) \xrightarrow{d} g(X)$. **Slutsky's theorem**: if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, and $X_n / Y_n \xrightarrow{d} X/c$ for $c \neq 0$.

Why it appears here. The sandwich derivation of §1.6 and §4.2 uses both kinds of limit. The bread converges in probability, and the scaled moment sum converges in distribution. Slutsky's theorem combines them. It also makes a t-statistic with an *estimated* standard error asymptotically normal (§2.6). The delta method of §2.6 is the continuous mapping theorem with a first-order expansion.

Deeper. Hansen (2022), the chapters on asymptotic theory; Hayashi (2000), chapter 2.

A.4 The chi-squared distribution and degrees of freedom

The idea. A test of several restrictions at once measures how far the estimates are from the null in several directions. Squared distances add up. If each direction contributes an independent standard normal discrepancy, the sum of the squares has a **chi-squared** distribution. Its **degrees of freedom** count the independent directions in which the null can fail.

Formally. If $Z_1, \dots, Z_q$ are independent $N(0,1)$, then $\sum_{j=1}^{q} Z_j^2 \sim \chi^2_q$, with mean q and variance $2q$. The 5% critical values are 3.84 for $q = 1$ (the square of 1.96), 5.99 for $q = 2$, and 18.31 for $q = 10$. A **Wald statistic** for q linear restrictions $R\theta = r$ is

$$(R\hat{\theta} - r)' \left[R \hat{V} R'\right]^{-1} (R\hat{\theta} - r) \xrightarrow{d} \chi^2_q$$

under the null, where $\hat{V}$ is the estimated variance of $\hat{\theta}$. It is a squared t-statistic generalised to several dimensions. In the J-test the degrees of freedom are $\ell - k$. Estimating k parameters lets the estimator set k combinations of the ℓ sample moments to zero. Only $\ell - k$ combinations remain that can reveal a violation.

Why it appears here. The chi-squared distribution underlies the J-test of §5.3, the Ljung–Box test of §8.1, the Hausman test (A.17), the Anderson–Rubin test (A.20), and every joint test in §4.6.

Deeper. Hansen (2022), the chapters on hypothesis testing and GMM.

A.5 The long-run variance as a spectral density at frequency zero

The idea. Any stationary series decomposes into cycles of different frequencies, from fast to slow. The **spectral density** measures how much of the variance lies at each frequency. A long average removes the fast cycles and keeps the slow cycles. The spectral density at the slowest frequency, zero, therefore determines the variance of a long average. So the "meat" of a time-series standard error and the spectral density at zero are the same quantity.

Formally. For a covariance-stationary vector series ψ_t with autocovariances Γ_j, the spectral density is $f(\omega) = \frac{1}{2\pi} \sum_{j=-\infty}^{\infty} \Gamma_j e^{-i\omega j}$. At $\omega = 0$,

$$2\pi f(0) = \sum_{j=-\infty}^{\infty} \Gamma_j = \Omega,$$

the long-run variance of §4.5. Kernel estimators such as Newey–West estimate $f(0)$ by a weighted sum of sample autocovariances. The **bandwidth** L sets how many autocovariances enter the sum. Equivalently, it sets how wide a band of low frequencies the estimator averages.

Why it appears here. §4.5 defines the HAC meat as $\sum_j \Gamma_j$. §12.2 lists Newey–West as a Bartlett-kernel estimate of the spectral density at zero.

Deeper. Hamilton (1994), chapters 6 and 10.

A.6 Fixed-b asymptotics

The idea. Conventional theory for HAC standard errors assumes that the bandwidth is a vanishing fraction of the sample. Under that assumption, analysts can treat the estimated long-run variance as exact. In real samples the bandwidth is a noticeable fraction of T, and the variance estimate is itself noisy. A t-statistic that divides by it therefore has fatter tails than a normal. **Fixed-b** theory holds the ratio $b = L/T$ fixed as the sample grows. It derives the distribution that the t-statistic actually has. It gives critical values that account for the noise in the denominator.

Formally. With b fixed, the HAC t-statistic converges to a nonstandard distribution that depends on the kernel and on b. Its critical values rise with b. For the equal-weighted cosine estimator with ν terms, the limit is exactly Student-t with ν degrees of freedom, which makes the method easy to use. Larger bandwidths reduce the bias of the variance estimate. They also add noise, and fixed-b critical values account for that noise. Lazarus, Lewis, Stock & Watson (2018) optimise this trade-off.

Why it appears here. §4.5 recommends a Newey–West bandwidth of $1.3\,T^{1/2}$ with fixed-b critical values, or the cosine estimator with $\nu = 0.4\,T^{2/3}$ and t_ν critical values.

Deeper. Kiefer & Vogelsang (2005); Lazarus, Lewis, Stock & Watson (2018).

Part II — Likelihood, information, and priors. This part covers what maximum likelihood estimates when the model is wrong, why it is efficient when the model is right, and how priors become penalties.

A.7 Kullback–Leibler divergence and pseudo-true parameters

The idea. A misspecified model cannot match the true distribution of the data, but some parameter values bring it closer than others. **Kullback–Leibler divergence** measures "closer" as the average shortfall in log-likelihood from using the model instead of the truth. Maximum likelihood on a wrong model converges to the parameter that minimises this divergence. This value is the **pseudo-true** value. Whether it is the parameter of interest depends on which parts of the model are wrong.

Formally. For a true density p and a model density q_θ,

$$\mathrm{KL}(p \,\|\, q_\theta) = \mathbb{E}_p\big[\log p(Y) - \log q_\theta(Y)\big] \geq 0,$$

with equality only when the two coincide. Maximising the expected log-likelihood $\mathbb{E}_p[\log q_\theta(Y)]$ is the same as minimising the divergence. So the MLE satisfies $\widehat{\theta} \xrightarrow{p} \theta^* = \arg\min_\theta \mathrm{KL}(p \,\|\, q_\theta)$. For Gaussian quasi-likelihood in a regression or GARCH model, θ^* equals the true conditional-mean and conditional-variance parameters whenever the model specifies those two equations correctly, even if the shocks are not normal.

Why it appears here. §5.4 uses this result for quasi-maximum likelihood. It shows which misspecified likelihoods still estimate the right parameter and why their standard errors need the sandwich.

Deeper. White (1982).

A.8 Fisher information, the information matrix equality, and the Cramér–Rao bound

The idea. If the log-likelihood falls steeply on either side of its peak, the data determine the parameter precisely. If it is flat, many values fit almost equally well. **Fisher information** measures that curvature at the true value. The **Cramér–Rao bound** converts the curvature into a limit on precision. No unbiased estimator can have a variance smaller than the inverse of the information. Maximum likelihood attains the bound in large samples when the model is correct.

Formally. With per-observation score $s_i(\theta) = \partial \log f(Y_i \mid X_i; \theta)/\partial\theta$, which has mean zero at θ_0, the information is

$$\mathcal{J}(\theta_0) = \mathbb{E}\big[s_i(\theta_0) s_i(\theta_0)'\big] = -\mathbb{E}\Big[\frac{\partial^2 \log f(Y_i \mid X_i; \theta_0)}{\partial\theta\, \partial\theta'}\Big].$$

The second equality, the **information matrix equality**, requires correct specification. The Cramér–Rao bound says $\mathrm{Var}(\tilde{\theta}) \geq \mathcal{J}(\theta_0)^{-1}/n$ for any unbiased $\tilde{\theta}$. For a normal mean with known variance σ^2, $\mathcal{J} = 1/\sigma^2$ and the bound is σ^2/n, which the sample mean attains. Under misspecification the two expectations differ. The MLE's variance is then the sandwich $H^{-1} J H^{-1}/n$, where H is minus the expected Hessian and J is the variance of the score.

Why it appears here. §5.4 explains the efficiency of MLE as the meat equalling minus the bread. That statement is this equality in sandwich terms.

Deeper. Hansen (2022), the chapters on maximum likelihood.

A.9 Priors as penalties

The idea. A Bayesian estimate combines two inputs. The likelihood summarises what the data say. The prior summarises what the analyst assumes before seeing the data. A prior that coefficients are probably small shrinks estimates toward zero. The strength of the shrinkage depends on how confident the prior is relative to how noisy the data are. The most probable coefficients under such a prior are exactly the solution of a penalised regression. Every penalty in machine learning therefore corresponds to a prior, and every prior corresponds to a penalty.

Formally. With $\mathbf{Y} \mid \beta \sim N(\mathbf{X}\beta, \sigma^2 I)$ and prior $\beta \sim N(0, \tau^2 I)$, the posterior mean and mode are

$$\big(\mathbf{X}'\mathbf{X} + \kappa I\big)^{-1} \mathbf{X}'\mathbf{Y}, \qquad \kappa = \sigma^2/\tau^2,$$

the ridge estimator of §5.6. A Laplace prior with density proportional to $e^{-|\beta_j|/c}$ gives a posterior mode that minimises $\|\mathbf{Y}-\mathbf{X}\beta\|^2+(2\sigma^2/c)\sum_j|\beta_j|$. This is the LASSO. The **Minnesota prior** for vector autoregressions centres each equation on a random walk. That means a coefficient of one on the variable's own first lag and zero elsewhere. Its prior variances shrink as the lag length grows. They are also tighter on other variables' lags than on the variable's own lags.

Why it appears here. This result underlies §5.6 ("a penalty is a prior, and a prior is a penalty"), the Bayesian VARs mentioned there, and the regularised predictors of §10.4.

Deeper. Litterman (1986); Tibshirani (1996).

A.10 Stein's paradox and empirical Bayes

The idea. Consider estimating the skill of 2,000 fund managers from their track records. The top of the ranking contains managers with skill and managers with luck. Shrinking every estimate toward the group average removes some of the luck and lowers the total error, even though it biases each individual estimate. **Stein's paradox** is the surprising theorem that this shrinkage beats the unshrunk estimates for any true values, once an analyst estimates three or more together. **Empirical Bayes** chooses the amount of shrinkage from the data. The spread of the estimates, compared with their noise, indicates how much real variation exists.

Formally. Let $\hat{\theta}_i \sim N(\theta_i, s^2)$ independently for $i=1,\dots,p$ with $p\geq 3$. The James–Stein estimator $\tilde{\theta}_i=(1-(p-2)s^2/\sum_j\hat{\theta}_j^2)\hat{\theta}_i$ has lower total mean squared error than $\hat{\theta}_i$ for every value of the θ_i. In the empirical Bayes version, the true values come from $N(\mu, A)$. The posterior mean is

$$\mu+\frac{A}{A+s^2}\,(\hat{\theta}_i-\mu),$$

Here the cross-section supplies the estimates of μ and A, with A estimated as the variance of the estimates minus s^2. Shrinkage is stronger when the estimates are noisier or the true dispersion is smaller.

Why it appears here. §5.6 states Stein's paradox and recommends shrinking manager and analyst estimates. The same logic underlies the forecast-combination puzzle of §8.8 and the winner's curse of §4.7.

Deeper. Efron & Morris (1977).

A.11 The truncated normal and the inverse Mills ratio

The idea. Suppose an outcome is observed only when an unobserved index is high enough. A wage is observed only when someone chose to work. A fund's return is observed only when the fund survived. The observed errors then have a non-zero mean, because they belong to the units that selection favoured. The shift in their average depends on how likely selection was. For normal errors the shift has a closed form, the **inverse Mills ratio**. Heckman's selection correction adds it to the regression as the omitted variable.

Formally. For $v\sim N(0,1)$, $\mathbb{E}[v\mid v>-a]=\varphi(a)/\Phi(a)\equiv\lambda(a)$, which is large when a is small and near zero when a is large. Selection is rare in the first case, and only large v get through. In Heckman's model, $Y_i=X_i'\beta+u_i$ is observed only if $W_i'\gamma_s+v_i>0$, with (u_i,v_i) jointly normal, $\mathrm{Var}(v_i)=1$, and correlation ρ_s. Then

$$\mathbb{E}[Y_i\mid X_i,\text{selected}]=X_i'\beta+\rho_s\sigma_u\,\lambda(W_i'\gamma_s),$$

so a probit for selection gives $\hat{\lambda}$, which the regression adds as a regressor. Because λ is nearly linear over much of its range, credible identification needs a variable in W_i that affects selection but not Y_i. The letters W_i, γ_s, and ρ_s are local to this entry.

Why it appears here. §6.4 describes selection as an omitted variable. The equivalences table in §12.2 names this term.

Deeper. Heckman (1979); Wooldridge (2010), the chapter on sample selection.

Part III — Testing and resampling. This part covers standard errors computed by resampling, tests that are exact by design, and corrections for testing many hypotheses at once.

A.12 The jackknife, the percentile-t bootstrap, and block bootstraps

The idea. The **jackknife** recomputes an estimate n times, leaving out one observation each time. It estimates the variance from how much the estimate moves. The bootstrap (§4.9) instead resamples with replacement. Two refinements matter. The first refinement, the **percentile-t** method, resamples the t-statistic instead of the estimate. It gives more accurate intervals, because the distribution of a t-statistic depends less on unknown parameters. The second refinement applies to dependent data. Resampling **blocks** of consecutive observations preserves the dependence inside each block.

Formally. The jackknife variance is $\frac{n-1}{n}\sum_i \left(\hat{\theta}_{(-i)} - \bar{\theta}_{(\cdot)}\right)^2$, where $\hat{\theta}_{(-i)}$ omits observation i and $\bar{\theta}_{(\cdot)}$ averages them. A percentile-t interval bootstraps $t^* = (\hat{\theta}^* - \hat{\theta})/\,\mathrm{se}^*$ and reports $[\hat{\theta} - t^*_{0.975}\,\mathrm{se},\ \hat{\theta} - t^*_{0.025}\,\mathrm{se}]$. The **moving block bootstrap** resamples overlapping blocks of fixed length L_b. The length must grow with the sample, and a common rate is $T^{1/3}$. The **stationary bootstrap** draws blocks of random, geometrically distributed length with mean L_b. This makes the resampled series itself stationary.

Why it appears here. §4.3 notes that HC3 is close to the jackknife. The table and refinements in §4.9 rest on these constructions.

Deeper. Efron (1979); Künsch (1989); Politis & Romano (1994).

A.13 The wild and wild cluster bootstrap

The idea. Keep the regressors fixed and create new artificial samples by multiplying each residual by a random sign. Each observation keeps its own error variance, so the samples preserve heteroskedasticity without modelling it. The **cluster** version flips the sign of a whole cluster's residuals at once. This also preserves any correlation inside the cluster. The procedure uses the actual cluster structure instead of an approximation that assumes many clusters. It therefore stays accurate with far fewer clusters than the cluster-robust t-test needs.

Formally. Estimate the model with the null imposed (the **restricted** version), and keep the residuals $\tilde{u}_i$. For $b = 1, \ldots, B$, draw a weight w_g for each cluster, either ± 1 with equal probability (Rademacher) or from the six-point Webb distribution $\{\pm\sqrt{1/2}, \pm 1, \pm\sqrt{3/2}\}$. Then form $Y_i^* = X_i'\tilde{\beta} + w_{g(i)}\tilde{u}_i$, re-estimate, and compute the cluster-robust t_b^*. The p-value is the share of $|t_b^*|$ at least as large as the observed $|t|$. With G clusters, Rademacher weights generate only 2^G distinct samples. Researchers therefore use Webb weights when G is small. When treatment reaches only one or two clusters, sign flips of their residual sums cannot reproduce the treatment's noise, and the procedure becomes very conservative (§4.4).

Why it appears here. This is the remedy for few clusters in §4.4. That section's figure simulates it.

Deeper. Cameron, Gelbach & Miller (2008); MacKinnon, Nielsen & Webb (2023).

A.14 Randomisation inference

The idea. In an experiment, the uncertainty comes from the random assignment of treatment. Under the null that treatment did nothing to anyone, each unit's outcome would be the same under any assignment. The outcomes under every alternative assignment are therefore known. Recompute the test statistic for every assignment the design could have produced, and compare the actual statistic with them. The resulting p-value is exact at any sample size and needs no distributional assumption.

Formally. Test the **sharp null** $Y_i(1) = Y_i(0)$ for all i. Take a statistic $T(\mathbf{d}, \mathbf{Y})$, such as the difference in means under assignment vector $\mathbf{d}$. The p-value is the share of assignments $\mathbf{d}$ that the randomisation design allows with $|T(\mathbf{d}, \mathbf{Y})| \geq |T(\mathbf{D}, \mathbf{Y})|$, where $\mathbf{D}$ is the realised assignment. When the assignments are too numerous to enumerate, a large random sample of them suffices. The test concerns the sharp null of no effect for anyone. This null is stronger than a zero *average* effect.

Why it appears here. §4.4 and §7.2 recommend it for small experiments and single treated units. The placebo inference of synthetic control in §7.7 is a version of it.

Deeper. Imbens & Rubin (2015), the chapter on Fisher's exact p-values.

A.15 Family-wise error and false discovery rates

The idea. When a study tests many hypotheses, "5%" can mean two different things. The **family-wise error rate** is the probability of making *any* false discovery. Controlling it is strict. It suits cases where a single false claim is costly. The **false**

discovery rate is the expected *share* of discoveries that are false. Controlling it allows many discoveries while bounding the share that are false, which suits screening.

Formally. With K tests and p-values sorted as $p_{(1)} \leq ... \leq p_{(K)}$:

- **Bonferroni** rejects H_j when $p_j \leq \alpha/K$. The family-wise error rate is then at most α.
- **Holm** rejects $H_{(1)}, H_{(2)}, ...$ in order while $p_{(j)} \leq \alpha/(K-j+1)$. It controls the same rate and is at least as powerful.
- **Benjamini–Hochberg** finds the largest j with $p_{(j)} \leq j\alpha/K$ and rejects $H_{(1)}, ..., H_{(j)}$. The false discovery rate is then at most α for independent or positively dependent tests.

With $K = 100$ and $\alpha = 5\%$, Bonferroni requires $p \leq 0.0005$, a two-sided $|t|$ above about 3.48.

Why it appears here. §4.8 tabulates these corrections. §9.6 discusses the $t > 3$ hurdle for new return predictors.

Deeper. Benjamini & Hochberg (1995); Harvey, Liu & Zhu (2016).

A.16 Testing the best of many: reality check, SPA, and stepdown

The idea. After a search over many strategies or forecasting models, the test asks whether *any* of them beats a benchmark. The natural statistic is the performance of the best candidate. Its distribution under the null must reflect both the size of the search and the correlation among the candidates, which share the same data. White's reality check obtains that distribution by bootstrapping all the candidates together over the same resampled dates.

Formally. For candidate k at date t, let $d_{k,t}$ be its performance relative to the benchmark, such as a loss difference or a return difference, with sample mean $\bar{d}_k$. The null is $\max_k \mathbb{E}[d_{k,t}] \leq 0$, and the statistic is $V = \max_k \sqrt{T}\,\bar{d}_k$. A stationary bootstrap (A.12) resamples the same dates for every candidate. The p-value is the share of bootstrap values $\max_k \sqrt{T}(\bar{d}^*_k - \bar{d}_k)$ that exceed V. Hansen's **SPA** test studentises each candidate and stops very poor candidates from inflating the null distribution. This makes it more powerful. The **Romano–Wolf stepdown** procedure goes further. It identifies *which* candidates beat the benchmark while controlling the family-wise error rate.

Why it appears here. These tests appear in the table of §4.8 and in the trading-rule evidence of §9.6.

Deeper. White (2000); Hansen (2005); Romano & Wolf (2005).

A.17 The Hausman test

The idea. Compare two estimators. One is consistent whether or not an assumption holds, but imprecise. The other is more efficient, but consistent only if the assumption holds. If the assumption is true, both converge to the same value, and their difference is just noise. If the difference is too large to be noise, reject the assumption.

Formally. With $\hat{\theta}_C$ consistent under both hypotheses and $\hat{\theta}_E$ efficient under the null,

$$H = (\hat{\theta}_C - \hat{\theta}_E)' [\hat{V}_C - \hat{V}_E]^{-1} (\hat{\theta}_C - \hat{\theta}_E) \xrightarrow{d} \chi^2_q,$$

where q is the number of coefficients compared. The simple variance of the difference, $\hat{V}_C - \hat{V}_E$, is valid only because $\hat{\theta}_E$ is efficient under the null. In panels, the fixed-effects estimator is $\hat{\theta}_C$ and the random-effects estimator is $\hat{\theta}_E$. The null is that the unit effects are uncorrelated with the regressors. With clustered or heteroskedastic errors, random effects is no longer efficient. The usual robust alternative is the Mundlak regression of §7.4 with a cluster-robust Wald test on the unit means. The same logic compares OLS with IV to test for endogeneity.

Why it appears here. §7.4 uses it to choose between fixed and random effects.

Deeper. Hausman (1978); Wooldridge (2010), the chapter on linear panel models.

Part IV — Identification tools. This part covers the formal tools behind the research designs of §6, §7, and §10.5.

A.18 d-separation and the back-door criterion

The idea. A causal graph shows which variables can pass statistical association to which others. Association flows along any path between two variables, whatever the direction of its arrows, unless the path is blocked. Conditioning on a variable

in the middle of a chain or a fork blocks the path. Conditioning on a collider, where two arrows meet, *opens* a path that was blocked. The back-door criterion uses these rules to decide which controls identify a causal effect.

Formally. A set of variables S **blocks** a path between D and Y in either of two cases. The path contains a chain $A \to B \to C$ or a fork $A \leftarrow B \to C$ with B in S. Or the path contains a collider $A \to B \leftarrow C$ with neither B nor any of its descendants in S. If every path is blocked, D and Y are **d-separated** given S. They are then independent conditional on S in every distribution the graph allows. A set S satisfies the **back-door criterion** for the effect of D on Y if two conditions hold. No member of S is a descendant of D, and S blocks every path from D to Y that begins with an arrow into D. The adjustment formula then identifies the effect:

$$\mathbb{E}\big[Y \mid \mathsf{do}(D = d)\big] = \sum_{s} \mathbb{E}[Y \mid D = d, S = s]\ \Pr(S = s),$$

where $\mathsf{do}(D = d)$ denotes setting D by intervention rather than observing it.

Why it appears here. §6.5 gives rules for confounders, mediators, and colliders. It also warns that some pre-treatment controls are colliders.

Deeper. Pearl (2009); Cinelli, Forney & Pearl (2024).

A.19 Sensitivity analysis: Oster's δ and the robustness value

The idea. Analysts cannot test for unobserved confounding directly. They can measure how strong a confounder would have to be to explain the estimate away. Both methods here calibrate that strength against something observed. Oster's δ compares it with the selection implied by the available controls. The robustness value expresses it as the share of variance that the confounder would need to explain.

Formally. *Oster.* Let $\mathring{\beta}, \mathring{R}^2$ come from the regression without controls and $\tilde{\beta}, \tilde{R}^2$ from the regression with them. Suppose selection on unobservables is δ times selection on observables, and that a regression including every confounder would reach $R^2_{\max}$. The bias-adjusted coefficient is approximately

$$\beta^* \approx \tilde{\beta} - \delta\,(\mathring{\beta} - \tilde{\beta})\,\frac{R^2_{\max} - \tilde{R}^2}{\tilde{R}^2 - \mathring{R}^2}.$$

Report the δ that sets $\beta^* = 0$, or report β^* at $\delta = 1$ with $R^2_{\max} = 1.3\,\tilde{R}^2$. *Cinelli and Hazlett.* Let t_D be the treatment's t-statistic, let df be the residual degrees of freedom, and let $f = |t_D|/\sqrt{df}$. The **robustness value** $\frac{1}{2}(\sqrt{f^4 + 4f^2} - f^2)$ is the smallest partial R^2 with both the treatment and the outcome that a confounder would need to reduce the estimate to zero.

Why it appears here. §6.7 recommends reporting one of these whenever an estimate rests on selection on observables.

Deeper. Oster (2019); Cinelli & Hazlett (2020).

A.20 Weak-instrument tools: Anderson–Rubin, LIML, and the effective F

The idea. When instruments are weak, the 2SLS t-statistic is far from normal (§7.3). Three responses help. Use a test whose validity does not depend on instrument strength (Anderson–Rubin). Use an estimator that is less biased toward OLS (LIML). Measure strength with a statistic suited to the dependence in the data (the effective F).

Formally. *Anderson–Rubin.* To test $\beta = \beta_0$, regress $Y_i - X_i'\beta_0$ on the instruments and any exogenous controls. Test that the instrument coefficients are zero, using a robust Wald statistic compared to χ^2_ℓ. Under the null the dependent variable is just the structural error, so the test is valid however weak the instruments are. The β_0 values that the test does not reject form a confidence set. This set can be very wide or unbounded when the instruments are weak, which reflects how little they reveal. *LIML* belongs to the family of k-class estimators, which includes OLS ($k = 0$) and 2SLS ($k = 1$). It chooses k from the data, slightly above one. It equals 2SLS in just-identified models. With many instruments it is approximately median-unbiased, at the cost of fatter tails. *The effective* F of Montiel Olea and Pflueger rescales the first-stage F-statistic to account for heteroskedasticity, clustering, or serial correlation. It compares the rescaled statistic with critical values for the bias of 2SLS. In just-identified models it equals the robust first-stage F.

Why it appears here. These tools support the inference and diagnostics for IV in §7.3.

Deeper. Anderson & Rubin (1949); Montiel Olea & Pflueger (2013); Andrews, Stock & Sun (2019).

A.21 Dynamic panel GMM

The idea. With a lagged dependent variable and unit fixed effects, demeaning biases the estimate (the Nickell bias of §7.4). First-differencing removes the fixed effect but creates a new problem. The differenced lag is correlated with the differenced error, because both contain last period's shock. Levels of the dependent variable from two or more periods back are correlated with the differenced lag. If the shocks are serially uncorrelated, those levels are not correlated with the differenced error, so they can serve as instruments.

Formally. In $Y_{it} = \phi Y_{i,t-1} + X_{it}'\beta + \alpha_i + u_{it}$, differencing gives $\Delta Y_{it} = \phi\,\Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta u_{it}$, where $\Delta Y_{i,t-1}$ and Δu_{it} both contain $u_{i,t-1}$. If the u_{it} are serially uncorrelated, the moment conditions $\mathbb{E}[Y_{i,t-s}\,\Delta u_{it}] = 0$ for $s \geq 2$ identify ϕ by GMM (§5.3). Two diagnostics are standard. The differenced residuals should show first-order but no second-order autocorrelation, and the J-test should not reject. The number of instruments grows with the square of T. Too many instruments bias the estimate toward OLS. When ϕ is near one, lagged levels predict differences poorly and the instruments are weak. The usual response is "system GMM", which adds equations in levels.

Why it appears here. §7.4 names it as the fix for the Nickell bias.

Deeper. Arellano & Bond (1991); Wooldridge (2010), the chapter on dynamic panel models.

A.22 Local polynomial regression and bandwidth choice

The idea. To estimate a regression function at one point without assuming its shape, fit a straight line using only the data near that point. Weight the nearest observations most. The **bandwidth** defines how near is "near." A wide bandwidth uses more data. This lowers variance but lets curvature further away bias the fit. A narrow bandwidth does the reverse. Regression discontinuity needs these fits at the edge of the data on each side of the cutoff. Global polynomials perform worst at those edges.

Formally. On the right of a cutoff c, with running variable R_i and kernel K, solve

$$\min_{a,\,b} \sum_{i:\,R_i \geq c} K\Big(\frac{R_i - c}{h}\Big)\big(Y_i - a - b(R_i - c)\big)^2,$$

do the same on the left, and estimate the jump as $\hat{a}_+ - \hat{a}_-$. The triangular kernel $K(z) = \max(0, 1 - |z|)$ is standard. The bias of a local linear fit is of order h^2, and the bandwidth that minimises mean squared error shrinks at rate $n^{-1/5}$. At that bandwidth the bias is of the same order as the standard error, so a conventional confidence interval is not centred on the true value. **Robust bias correction** estimates the bias with a higher-order fit and subtracts it. It also widens the standard error to account for the correction.

Why it appears here. §7.6 uses these methods for estimation and inference in regression discontinuity.

Deeper. Calonico, Cattaneo & Titiunik (2014); Cattaneo, Idrobo & Titiunik (2020).

A.23 Approximate sparsity

The idea. With hundreds of candidate controls, estimation is feasible only if most of them barely matter. **Sparsity** means that only a few controls have non-zero coefficients. **Approximate sparsity** relaxes this to a more realistic condition. A few controls matter a lot. The many remaining small effects add up to an error smaller than the sampling noise. Under that condition, a selection method that finds the important few loses little.

Formally. Take a conditional mean $\mathbb{E}[Y \mid X] = \sum_{j=1}^{p} \beta_j X_j$, possibly with $p > n$. It is approximately sparse if some s of the regressors approximate it with an approximation error whose root mean square is of order $\sqrt{s/n}$. Here s is small relative to n. Inference results require s to grow slowly relative to n, up to logarithmic factors in p. The LASSO then selects a model whose prediction error is close to what an oracle that knew the s important controls would achieve.

Why it appears here. §10.5 states that post-double-selection inference is valid under approximate sparsity.

Deeper. Belloni, Chernozhukov & Hansen (2014); Chernozhukov, Hansen, Kallus, Spindler & Syrgkanis (2024).

A.24 Neyman orthogonality and cross-fitting

The idea. Regularisation deliberately biases machine-learning predictions. If a causal estimate depends on them directly, that bias carries into the estimate. The fix is to build the estimating equation so that small errors in the auxiliary predictions

do not move the causal estimate *to first order*. Their effect then enters only as the product of two small errors, which is negligible. **Cross-fitting** adds a second safeguard. The predictions for each observation come from models that never saw that observation, so overfitting cannot bias the estimate.

Formally. A moment function $\psi(W;\theta,\eta)$ with nuisance functions η is **Neyman orthogonal** if its expectation has zero derivative in every direction of η at the truth:

$$\frac{\partial}{\partial r}\,\mathbb{E}\big[\psi(W;\theta_0,\,\eta_0+r(\eta-\eta_0))\big]\Big|_{r=0}=0.$$

For §10.5's partialling-out moment, $\psi=\big(Y-\ell(X)-\theta(D-\pi(X))\big)\big(D-\pi(X)\big)$ with $\eta=(\ell,\pi)$. Perturbing ℓ changes the expectation by a multiple of $\mathbb{E}[D-\pi(X)\mid X]=0$, and perturbing π by a multiple of $\mathbb{E}[u\mid X]=0$, so both derivatives vanish. The remaining bias is of order $\|\hat{\ell}-\ell\|\cdot\|\hat{\pi}-\pi\|$, which is small enough for $\sqrt{n}$-inference if each prediction error shrinks faster than $n^{-1/4}$. **Cross-fitting** splits the sample into folds, fits $\hat{\ell}$ and $\hat{\pi}$ on all folds but one, and computes residuals on the held-out fold. It rotates the held-out fold and estimates θ from the pooled residuals. Here W collects the data, and r is a scalar.

Why it appears here. This entry covers the two "ingredients" of double machine learning in §10.5.

Deeper. Chernozhukov et al. (2018).

Part V — Time series and volatility.

A.25 The Kalman filter

The idea. Many economic quantities are never observed directly. Examples are the true level of a trend, a factor that drives the yield curve, and an expected return. The **Kalman filter** estimates such an unobserved state from noisy observations. Each period it predicts the state and the next observation, compares the prediction with the actual observation, and corrects the predicted state in proportion to the difference. The correction is larger when the prediction was uncertain and smaller when the observation is noisy. The prediction errors also give the likelihood of the model, so maximum likelihood can estimate its parameters.

Formally. With state $\xi_t=F\xi_{t-1}+\eta_t$ and observation $y_t=H\xi_t+w_t$, where $\mathrm{Var}(\eta_t)=\Sigma_\eta$ and $\mathrm{Var}(w_t)=\Sigma_w$:

$$\begin{aligned}
\text{predict:}\quad & \hat{\xi}_{t|t-1}=F\hat{\xi}_{t-1|t-1}, \qquad P_{t|t-1}=FP_{t-1|t-1}F'+\Sigma_\eta,\\
\text{update:}\quad & \mathcal{G}_t=P_{t|t-1}H'\big(HP_{t|t-1}H'+\Sigma_w\big)^{-1},\\
& \hat{\xi}_{t|t}=\hat{\xi}_{t|t-1}+\mathcal{G}_t(y_t-H\hat{\xi}_{t|t-1}), \qquad P_{t|t}=(I-\mathcal{G}_tH)P_{t|t-1}.
\end{aligned}$$

P is the variance of the state estimate and $\mathcal{G}_t$ the **gain**. The letters in this entry are local to it.

Why it appears here. §7.8 lists term-structure models estimated by the Kalman filter among structural models in finance.

Deeper. Hamilton (1994), chapter 13. Section 6.6 of Trend-Following in Financial Markets applies it to trend estimation.

A.26 The Granger representation theorem

The idea. If several non-stationary series are cointegrated, the gaps between them are stationary, so the series must adjust toward each other. The **Granger representation theorem** says that this adjustment must appear in the short-run dynamics. At least one of the series must respond to last period's deviation from the long-run relationship. Cointegration and error correction are two descriptions of the same system.

Let $\mathbf{y}_t$ be a vector of n_y integrated series. They are cointegrated with rank r if r linearly independent vectors exist, the columns of an $n_y\times r$ matrix B, such that $\mathsf{B}'\mathbf{y}_t$ is stationary. The vector autoregression then takes the form of the **vector error-correction model**

$$\Delta\mathbf{y}_t=\mathsf{A}\,\mathsf{B}'\mathbf{y}_{t-1}+\sum_{j=1}^{p-1}\Gamma_j\,\Delta\mathbf{y}_{t-j}+\mathbf{e}_t,$$

where the $n_y \times r$ matrix A holds the speeds of adjustment. The rank of AB' is r. With $r = 0$ the model is a VAR in differences, and with $r = n_y$ the series are stationary in levels. The Johansen procedure estimates r by testing the rank of this matrix. The letters A, B, Γ_j, and n_y are local to this entry.

Why it appears here. This theorem underlies the error-correction model of §8.4 and the equivalence listed in §12.2.

Deeper. Engle & Granger (1987); Johansen (1991).

A.27 Market microstructure noise

The idea. Transaction prices differ from the "efficient" price. They bounce between the bid and the ask, move in discrete ticks, and carry temporary price impact. Each deviation is tiny. Real price moves shrink as the sampling interval shrinks, but these deviations stay the same size. At high enough frequency the noise dominates squared returns, and realised variance measures the noise instead of the volatility.

Formally. Write the observed log price as $p_t^{\text{obs}} = p_t^* + \upsilon_t$, with υ_t independent noise of variance σ_υ^2. An observed return over a short interval is then the efficient return plus $\upsilon_t - \upsilon_{t-1}$. With M intraday returns, the expectation of realised variance is about the integrated variance plus $2M\sigma_\upsilon^2$. Its bias therefore grows linearly with the sampling frequency. The same differenced noise gives observed returns a negative first-order autocorrelation. The standard responses are sampling at moderate frequencies, such as every five minutes, and using noise-robust estimators such as realised kernels and pre-averaging.

Why it appears here. §8.6 names noise as the limit on realised variance's precision.

Deeper. Andersen, Bollerslev, Diebold & Labys (2003).

A.28 Loss functions for volatility forecasts: MSE and QLIKE

The idea. Variance is never observed, so evaluations score volatility forecasts against a noisy proxy such as a squared return. Some loss functions rank competing forecasts in the same order whether the score uses the true variance or an unbiased noisy proxy. Others can prefer the wrong forecast. Mean squared error and QLIKE belong to the first group.

Formally. Let $\widehat{h}_t$ be a variance forecast and $\widehat{\sigma}_t^2$ a proxy with $\mathbb{E}[\widehat{\sigma}_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$. The two standard robust losses are

$$\text{MSE}_t = (\widehat{\sigma}_t^2 - \widehat{h}_t)^2, \qquad \text{QLIKE}_t = \frac{\widehat{\sigma}_t^2}{\widehat{h}_t} - \log \frac{\widehat{\sigma}_t^2}{\widehat{h}_t} - 1.$$

Both losses have their minimum expected value at $\widehat{h}_t = \sigma_t^2$. QLIKE depends only on the ratio of proxy to forecast, so it is scale-free and less dominated by a few high-variance days than MSE. Losses such as the mean squared error of volatilities or of log variances are not robust in this sense.

Why it appears here. §8.6 warns about evaluating volatility forecasts.

Deeper. Patton (2011); Hansen & Lunde (2005).

A.29 IVX

The idea. A predictor that is nearly a random walk gives the predictive regression a non-standard sampling distribution. The right correction depends on exactly how persistent the predictor is, and the sample estimates that persistence poorly (§9.2). **IVX** avoids this problem. It builds an instrument from the predictor's own past changes. The filter deliberately makes the instrument less persistent than the predictor. The method then uses that instrument in an IV regression. The resulting test statistic is approximately normal whatever the predictor's persistence.

Formally. For $r_{t+1} = \alpha + \beta x_t + u_{t+1}$, construct $\widetilde{z}_t = \sum_{j=0}^{t-1} \varrho^j \, \Delta x_{t-j}$ with $\varrho = 1 - c_z/T^a$, for a constant $c_z > 0$ and $a \in (0, 1)$. This makes $\widetilde{z}_t$ "mildly integrated". It is less persistent than a unit root but more persistent than a stationary series. Using $\widetilde{z}_t$ as an instrument for x_t gives a Wald statistic with a χ^2 limit whether x_t is stationary, local to a unit root, or exactly integrated. The cost is some loss of power when the predictor is in fact stationary and exogenous. The symbols ϱ, c_z, and a are local to this entry.

Why it appears here. §9.2 lists IVX among the inference methods for persistent predictors.

Deeper. Kostakis, Magdalinos & Stamatogiannis (2015).

Part VI — Finance. This part covers the finance concepts and methods used in §3.8, §4.7, and §9.

A.30 Excess returns and the Sharpe ratio

The idea. An investment's **excess return** is what it earns above a risk-free rate, the reward for bearing risk. The **Sharpe ratio** divides the average excess return by its volatility, giving reward per unit of risk. Leverage does not change it, because doubling a position doubles both the numerator and the denominator. The Sharpe ratio is also, almost exactly, the signal-to-noise ratio of a strategy's mean return, which determines how hard that mean is to estimate.

Formally. With excess return r_t^e, $\mathrm{SR} = \mathbb{E}[r_t^e]/\operatorname{sd}(r_t^e)$. For independent returns, a per-period Sharpe ratio scales to annual units by $\sqrt{A}$, with A periods per year. The t-statistic of the mean excess return over T periods is approximately $\mathrm{SR}_{\text{period}}\sqrt{T}$, which equals the annual Sharpe ratio times the square root of the number of years. For independent normal returns the standard error of an estimated per-period Sharpe ratio is approximately $\sqrt{(1+\mathrm{SR}^2/2)/T}$ (§2.6).

Why it appears here. The Sharpe ratio appears throughout the chapter: in the precision of means (§2.5–§2.6), the value of predictability (§3.8), the winner's curse (§4.7), and backtests (§9.1, §9.7).

Deeper. Lo (2002); Appendix A.32 and A.45 of Portfolio Construction and the Covariance Matrix.

A.31 Risk aversion, rational expectations, and the stochastic discount factor

The idea. An investor values a payoff by how much an extra dollar is worth in the state of the world where the payoff arrives. A dollar in a recession, when consumption is low, is worth more than a dollar in a boom. The **stochastic discount factor** is that state-by-state valuation. The price of every asset is its expected payoff weighted by the discount factor. **Risk aversion** sets how steeply the valuation rises in bad states. **Rational expectations** is the assumption that investors' expectations equal the true conditional expectations. This assumption lets an econometrician test the resulting pricing equations with realised data.

Formally. With power (constant relative risk aversion) utility $u(C) = C^{1-\eta}/(1-\eta)$, relative risk aversion $-Cu''(C)/u'(C)$ equals η. An investor who can buy an asset with gross return R_{t+1} is at an optimum only if $u'(C_t) = d\,\mathbb{E}_t[u'(C_{t+1})R_{t+1}]$, which rearranges to

$$\mathbb{E}_t\big[M_{t+1}R_{t+1}\big] = 1, \qquad M_{t+1} = d\,(C_{t+1}/C_t)^{-\eta},$$

and for an excess return, $\mathbb{E}_t[M_{t+1}r_{t+1}^e] = 0$. Under rational expectations $\mathbb{E}_t$ is the mathematical expectation given $\mathcal{F}_t$ (A.1). So any variable known at t is a valid instrument. The mean-variance investor of §3.8 maximises expected return minus $\gamma_r/2$ times variance. This investor corresponds to quadratic utility, whose marginal utility is linear. So the investor's discount factor is linear in the return on the investor's portfolio.

Why it appears here. The SDF concepts appear in the Euler-equation row of the table in §1.5 and in GMM estimation in §5.3. They also appear in the SDF approach to asset-pricing tests in §9.4, the rational-expectations critique in §13.3, and the optimal portfolio in §3.8.

Deeper. Cochrane (2005); Hansen & Singleton (1982).

A.32 Portfolio sorts, value weighting, and NYSE breakpoints

The idea. The simplest non-parametric test of whether a characteristic predicts returns ranks stocks on it every period, groups them into portfolios, and tracks each portfolio's subsequent return. The return of the top group minus the return of the bottom group is the characteristic's "factor" return. Two implementation choices decide whether the result describes the economy or only its smallest stocks. One is how the sort weights stocks inside each portfolio. The other is where the sort draws the group boundaries.

Formally. At each rebalancing date, compute percentile **breakpoints** of the characteristic and assign stocks to groups. Then compute next-period portfolio returns, either equal-weighted, $\frac{1}{N_p}\sum_i r_i$, or **value-weighted**, $\sum_i \frac{ME_i}{\sum_j ME_j} r_i$, where ME_i is market capitalisation at formation and N_p the number of stocks in the portfolio. **NYSE breakpoints** compute the percentiles from New York Stock Exchange stocks only, which are larger on average. The sort then assigns every stock to a group. Without NYSE breakpoints, the extreme groups fill with microcaps, which are about 60% of stocks and about 3% of market

value (§9.1). Evidence comes from the time series of the long–short return, with HAC standard errors, and from its alpha against existing factor models (§9.4).

Why it appears here. §1.4 lists portfolio sorts as a descriptive tool. The replication debate in §9.6 depends partly on weighting and breakpoints.

Deeper. Fama & French (2008); Hou, Xue & Zhang (2020).

A.33 Mean-variance efficiency, the tangency portfolio, and the GLS R^2

The idea. Among all combinations of a set of risky assets, one has the highest Sharpe ratio. This is the **tangency portfolio**. A factor model is exactly right when some combination of its factors reaches that maximum, so that adding the test assets cannot improve on the factors. So the time-series test of a factor model asks a question about Sharpe ratios. For the same reason, a goodness-of-fit measure for a factor model should weight pricing errors by how much they matter for the maximum Sharpe ratio, not equally.

Formally. For assets with mean excess returns μ and covariance matrix Σ, the maximum squared Sharpe ratio is $\mu'\Sigma^{-1}\mu$, attained by weights proportional to $\Sigma^{-1}\mu$. If α collects the test assets' intercepts in regressions on the factors, and Σ_e their residual covariance, then

$$\alpha'\Sigma_e^{-1}\alpha = \mathrm{SR}^2(\text{factors and test assets}) - \mathrm{SR}^2(\text{factors}),$$

which is what the GRS statistic scales and tests. A **GLS** R^2 in a cross-sectional regression weights pricing errors by the inverse covariance matrix. It therefore measures errors on the portfolio combinations that matter for the maximum Sharpe ratio. It also depends on how close the model's **factor-mimicking portfolio** comes to the tangency portfolio. The factor-mimicking portfolio is the combination of test assets most correlated with the factor. In this entry μ and α are vectors over test assets.

Why it appears here. These ideas underlie §9.4's reading of the GRS test and Lewellen, Nagel and Shanken's recommendation.

Deeper. Gibbons, Ross & Shanken (1989); Lewellen, Nagel & Shanken (2010); Cochrane (2005).

A.34 Errors in betas: the Shanken correction

The idea. Fama–MacBeth standard errors treat the first-pass betas as known. The first pass estimates the betas, so the second-pass risk premia carry extra uncertainty that the standard errors miss. Shanken derived the adjustment. It inflates the variance by a factor that grows with the squared Sharpe ratio of the factors. It also adds a term for the sampling variation of the factors themselves.

Formally. For an OLS second pass with K factors, estimated premia $\hat{\lambda}$, factor covariance matrix Σ_f, and Fama–MacBeth variance $\hat{V}_{\text{FM}}$, the corrected variance is approximately

$$\hat{V}_{\text{Shanken}} = (1 + \hat{\lambda}'\Sigma_f^{-1}\hat{\lambda})\,\hat{V}_{\text{FM}} + \frac{\Sigma_f}{T}.$$

For traded factors, $\lambda'\Sigma_f^{-1}\lambda$ is the factors' maximum squared Sharpe ratio per period. This is small in monthly data: a monthly Sharpe ratio of 0.15 gives a multiplier of about 1.02. The correction therefore matters most with several factors or high Sharpe ratios. Here K is the number of factors.

Why it appears here. §9.4 discusses errors in variables in two-pass tests.

Deeper. Shanken (1992); Cochrane (2005), the chapter on regression-based tests.

A.35 Driscoll–Kraay standard errors

The idea. Sum each date's moment contributions across all firms. This produces a single time series with one observation per date. Any correlation across firms at a date is inside that sum. Then apply Newey–West to the time series, which handles correlation across dates. The result is robust to arbitrary cross-sectional dependence and to serial dependence that fades, provided the sample has many dates.

Formally. With $h_t = \sum_i X_{it}\hat{e}_{it}$ and Bartlett weights $w_j = 1 - j/(L+1)$,

$$\hat{V}_{\mathrm{DK}} = (\mathbf{X}'\mathbf{X})^{-1}\Big[\sum_t h_t h_t' + \sum_{j=1}^{L} w_j \sum_t (h_t h_{t-j}' + h_{t-j} h_t')\Big](\mathbf{X}'\mathbf{X})^{-1}.$$

Consistency requires the number of dates to grow. The number of firms can be fixed or large.

Why it appears here. These standard errors appear in the standard-error table in §1.6 and the panel table in §9.5.

Deeper. Driscoll & Kraay (1998); Petersen (2009).

A.36 The deflated Sharpe ratio

The idea. The best of many backtests has a Sharpe ratio biased upward by the search that selected it. The **deflated Sharpe ratio** tests whether the best result beats what the best of that many unskilled trials would have produced by luck. It also accounts for the sample length and for non-normal returns. It converts a statement such as "the best configuration scored 1.4" into the probability that the configuration has real skill.

Formally. The expected maximum Sharpe ratio across K independent unskilled trials is approximately

$$\mathrm{SR}_0 = \sqrt{\mathrm{Var}(\widehat{\mathrm{SR}}_k)}\,\Big[(1-\gamma_E)\,\Phi^{-1}(1-\tfrac{1}{K}) + \gamma_E\,\Phi^{-1}(1-\tfrac{1}{Ke})\Big],$$

where $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant and $\mathrm{Var}(\widehat{\mathrm{SR}}_k)$ is the variance of the Sharpe ratios across trials. The deflated Sharpe ratio is then

$$\Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}_0)\sqrt{T-1}}{\sqrt{1 - \hat{s}_3\,\widehat{\mathrm{SR}} + \frac{\hat{s}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2}}\right),$$

with $\hat{s}_3$ and $\hat{s}_4$ the skewness and kurtosis of returns. The number of trials K should count *effectively independent* trials. That count is smaller than the raw count when configurations are correlated (§4.8). In this entry K is the number of trials.

Why it appears here. §4.8 and §9.7 list it among the adjustments for search. The research log in §11.5 exists to supply K.

Deeper. Bailey & López de Prado (2014); Appendix A.46 of Portfolio Construction and the Covariance Matrix.

Part VII — Machine learning.

A.37 Trees, forests, boosting, and neural-network regularisation

The idea. A **regression tree** splits the data into boxes using thresholds on the predictors, such as "book-to-market above 0.8 and size below the median." It predicts the average outcome in each box. It therefore captures non-linearities and interactions without any specification of them. A single deep tree overfits badly. A **random forest** averages many trees. Each tree grows on a bootstrap sample and splits only on a random subset of predictors, which reduces variance. **Boosting** grows a sequence of small trees. Each tree fits what the ensemble so far has failed to explain, and the ensemble adds it with a small weight. This reduces bias gradually. **Neural networks** stack layers of non-linear transformations. Four devices keep them from overfitting: weight penalties, **early stopping** (halting training when validation loss stops improving), **dropout** (randomly silencing units during training), and averaging several networks. A **causal forest** grows trees whose splits seek differences in treatment effects instead of differences in outcomes. It uses separate subsamples to choose the splits and to estimate the effects within them. This separation is what makes its confidence intervals valid.

A tree predicts $\hat{f}(x) = \sum_m c_m\,\mathbb{1}\{x \in \mathcal{R}_m\}$ for boxes $\mathcal{R}_m$ chosen greedily to reduce squared error, with c_m the mean outcome in box m. A forest averages B trees, $\bar{f}(x) = \frac{1}{B}\sum_b \hat{f}_b(x)$. Boosting builds $\hat{f}_M(x) = \sum_{m=1}^{M} \varsigma\,\hat{t}_m(x)$, where the algorithm fits each small tree $\hat{t}_m$ to the current residuals (for squared loss) and ς is a small learning rate. Validation chooses the tuning parameters: depth, number of trees, learning rate, and penalties (§10.6). The symbols in this entry are local to it.

Why it appears here. These methods appear in §10.4's comparison of return-prediction methods and in the causal forests of §10.5.

Deeper. Hastie, Tibshirani & Friedman (2009), chapters 9, 10, and 15; Wager & Athey (2018); Gu, Kelly & Xiu (2020).

A.38 Mallows's C_p and the optimism of in-sample fit

The idea. A model's error on its fitting data understates its error on new data. The understatement grows with the number of parameters, because each parameter lets the fit absorb some of the noise. **Mallows's C_p** adds the expected understatement back to the in-sample error. This lets an analyst compare models of different sizes on an estimate of their out-of-sample error without holding out data.

Formally. Take linear regression with k coefficients, n observations, and error variance σ^2. The expected in-sample mean squared error falls short of the expected out-of-sample error at the same regressor values by $2\sigma^2 k/n$ (§10.2). One common form of the criterion is

$$C_p = \frac{1}{n}\sum_{i=1}^{n} \hat{e}_i^2 + \frac{2k\hat{\sigma}^2}{n},$$

with $\hat{\sigma}^2$ estimated from a large model. Choosing the model with the smallest C_p approximately minimises expected prediction error. With normal errors and known σ^2, ranking models by C_p is equivalent to ranking them by the AIC.

Why it appears here. §10.2 derives the optimism and names C_p as the criterion built on it.

Deeper. Hastie, Tibshirani & Friedman (2009), chapter 7.

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